

Polyphenol extraction from black tea by membrane filtration

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Polyphenol Extraction
from Black Tea
by Membrane Filtration

Lin Sun

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ABSTRACT

Black tea is one of the most widely consumed beverages which contains a high level of antioxidant tea polyphenols. Polyphenols have beneficial function in lowering the risk of heart disease and radical-scavenging abilities. Nanofiltration (NF) has the potential to be used to recover polyphenols from tea solution.

In this study, an investigation of extraction methods of polyphenols on pre-treatment from black tea was done in conjunction of nanofiltration fouling studies. Extraction temperatures, infusion time, tea concentration and solvent constituent were studied. UV-Vis spectrophotometric method was applied to measure the total polyphenols extracted in different conditions.

Two types of pre-treatments (vacuum filtration and ultra-filtration) were conducted on the original black tea feed to attempt to reduce subsequent fouling in nanofiltration. For ultra-filtration process, three different UF membranes, regenerated cellulose membrane with molecular weight cut off (MWCO) of 1K Da and 5K Da, and polyethersulfone membrane with MWCO of 5K Da were used. Both permeate flux and rejection of polyphenols were studied in this part. In addition, backwash and chemical cleaning were both conducted to the fouled UF membrane to observe their flux recovery.

In the nanofiltration process, membrane with molecular weight cut off of 200-400 and 700 Daltons were compared. Effects of trans-membrane pressure, filtration time, stirring rate were evaluated on the membrane performance. UV-Vis spectrophotometer and GC/MS were used to analyze the total amount of

polyphenols and composition of tea samples separately in nanofiltration. Streaming potential of virgin and fouled NF membranes were measured to assess changes to surface chemistry. After filtration, chemical cleaning was applied to the fouled NF membrane to study the flux recovery and rejection after each cleaning cycles, to explain the fouling mechanism and membrane performance.

Keywords: Polyphenols, Extraction, Ultra-filtration, Nanofiltration, UV-Vis spectrophotometer, CE, GC/MS, Chemical cleaning.

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Chapter 1 INTRODUCTION

1.1 Background and objectives

As reported by Nawaz et al. [1], polyphenols was found to be the most dominant metabolites in plants. Polyphenols have an important research value due to its health related functions. With their antioxidant property, they can help human body in cellular process, keep the high-density lipids from oxidizing and clean low-density lipids. Besides, polyphenols also play a part in fighting against ulceration, carcinogen and mutagen because of its ability of absorbing free radicals [1].

Nowadays, human being is exposed to more and more severe pollution, such as car exhaust, ultra violet rays, and radiation from mobile phones. People are desire to improve their health through food and beverage consumption, which provides a vital important trigger to develop the food and beverage process [2].

Concentration (or enrichment) of antioxidant elements or separation of different phenolic compounds from natural sources including fruits, vegetables and other plants would enable quantitative evaluation of specific benefit of selected compounds. The concentrated beneficial compounds can also be used as supplementary food additive.

Tea is the most popular beverage in the world. Green tea and Oolong tea are consumed mostly in China and Japan, whereas black tea is preferred in the United states and Europe [3]. Tea is a major source of polyphenols, with the major constituents of flavonols, flavones, and phenolic acids.

Conventionally, polyphenols in tea and other sources are extracted in organic solvent,

such as methanol, ethanol and hexane. However, this method always leads to chemical residual which are toxic and dangerous for human intake [4]. Solid phase extraction (SPE) is another method currently being used for large scale. Although SPE is safe and efficient, the cost of setting up is quite high. Besides, during extraction, high temperature is required in order to increase the extraction rate, which may denature the polyphenols [4].

Membrane filtration is an alternative choice over other conventional extraction methods. First, it is toxic solvent free. Second, membrane filtration provides easy automation [5] and scale up [1], shorter process time [5], lower labour and energy costs [6], less waste disposal [6], no additive and mild operation conditions [1].

However, the membrane fouling in filtration is the main disadvantage which results in the decline of permeate flux. Usually the fouling is caused by deposition and accumulation of solute or colloidal particles at the membrane surface or inside the pores [6].

The objectives of this research are:

- 1) To evaluate the extraction methods for tea polyphenols in order to establish a consistent source of polyphenols for concentration efficiency evaluation and membrane separation efficiency
- 2) To evaluate pre-treatment process to the polyphenol solution extracted from black tea leaves using different techniques in order to evaluate the benefit of the pre-treatment on the nanofiltration(NF)membrane performance
- 3) To compare different NF membranes with different molecular weight cut off and material in polyphenols concentration performance in terms of membrane flux trends and polyphenols retention
- 4) To evaluate the effect of nanofiltration operational condition on the membrane performance (flux, degree of fouling, polyphenols retention and membrane

cleanability)

1.2 Structure of the thesis

Chapter 1 and 2:

This thesis consists of 6 chapters. Chapter 1 presents the short introduction and the objectives of this study. Followed by chapter 2, that provides literature review of polyphenols and compositions of black tea, background information of fouling and cleaning in membrane systems and the analytic methods applicable in this research.

Chapter 3:

Chapter 3 describes the detailed information and the characterization of all the materials used in this study as well as the equipment and the analytical methods.

Chapter 4:

Chapter 4 presents the investigation of extraction methods of total polyphenols in black tea and pre-treatment for black tea feed. Different extraction conditions were tested in this chapter, including extraction temperature, infusion time, tea concentration and the solvent constituent. For pre-treatment determination, both vacuum filtration and ultra-filtration were examined. In UF pre-treatment, three UF membranes (regenerated cellulose and polyethersulfone membranes) were tested and compared via the permeate flux and rejection. Both backwash and chemical cleaning were applied on the fouled UF membranes and evaluated.

Chapter 5:

Chapter 5 contains the nanofiltration process and evaluation of polyphenols concentration from black tea feed, the evaluation of chemical cleaning protocols for NF membranes and analysis for both NF membranes and composition of black tea feed. Membranes NF 270 and NTR-7450 were used to carry out the research in this

chapter. Varied nanofiltration operation conditions were examined and evaluated, including trans-membrane pressure, filtration time and stirring rate. In cleaning phase, NaOH solution was applied to the fouled membranes in multiple cleaning cycles, and evaluated by the flux recovery and rejection of polyphenols. To analyze the results, UV-Vis spectrophotometer, GC-MS, and capillary electrophoresis were applied to investigate the tea samples in NF process and cleaning cycles.

Chapter 6:

The study concludes with chapter 6, which presents overall conclusions and recommendations.

Chapter 2 LITERATURE REVIEW

2.1 Introduction

This chapter aims to review the relevant background information and current research status in separation of polyphenols using membrane technologies. The first part of this chapter includes the information regarding polyphenols and compositions of black tea; the physical and chemical properties of polyphenols while the source of polyphenols and polyphenols in tea were also discussed. Due to the material used in this research is black tea, the differences of green tea, Oolong tea and black tea is also reviewed.

The second part of this chapter covers the review of ultrafiltration and nanofiltration processes that are most relevant to this research. In the review of membrane technology, different membrane materials and membrane filtration are reviewed following the introduction of membrane processing. Then, the previous research of membrane technology in application of polyphenols extraction is also reviewed. Ultrafiltration and Nanofiltration are the two processes that have been identified to be applicable to in this research, in this case, previous studies in this area are elaborated and evaluated. Because fouling is a severe and unavoidable problem in membrane processing, fouling characterization, types of fouling, fouling mechanisms, and fouling prevention & cleaning are intensively reviewed in this section to achieve better membrane performance in the following research.

In the third section of literature review, different analytic methods applicable in this research are introduced. This analysis of polyphenols in solution which includes estimation of concentration of total polyphenols and analysis of polyphenol

composition are introduced and applications reviewed. Analytical processes applicable for membrane fouling characterization are also reviewed.

2.2 Polyphenols

2.2.1 Introduction of polyphenols

Phenolic compounds or polyphenols constitute one of the most numerous and widely distributed groups of substances in the plant kingdom, with more than 8000 phenolic structures currently found [7]. Polyphenols are products of the secondary metabolism of plants. They arise biogenetically from two main synthetic pathways: the shikimate pathway and the acetate pathway [8].

Natural polyphenols can range from simple molecules, such as phenolic acids, to highly polymerized compounds, such as tannins. They occur primarily in conjugated form, with one or more sugar residues linked to hydroxyl groups, although direct linkages of the sugar unit to an aromatic carbon atom also exist.

Interest for polyphenolic compounds is initiated with their deleterious effects caused by the ability of certain polyphenols to bind and precipitate macromolecules, such as dietary protein, carbohydrate, and digestive enzymes, thereby reducing food digestibility[9].Recent interest in antioxidant and free radical-scavenging abilities associated with some phenolics and their potential effects on human health. Polyphenols play an important role in cellular processes within the body by acting against the oxidation of low-density lipoprotein (LDL) [10]. It is also suggested that polyphenols have the ability to possess antiulcer [11], anticarcinogenic[12], and antimutagenic activities [13]. All of those functions of polyphenols can be contributed to its strong antioxidant power in quenching free radicals [14].

2.2.2 Polyphenols in plants

Polyphenolic compounds as secondary metabolites are ubiquitous in all plant organs in response to stress conditions such as infection, wounding, and UV radiation [14]. Phenolics in plant include simple phenols, phenolic acid (both benzoic and cinnamic acid derivatives), coumarins, Flavonoids, stilbenes, hydrolysable and condensed tannins, lignans, and lignins [14]. These polyphenols are widely presented in various kinds of food, including cereals and legumes (barley, corn, nuts, oats, rice, sorghum, wheat, beans and pulses), oilseeds (rapeseed, canola, flaxseed, and olive seeds), fruits and vegetables and beverages (fruit juices, tea, coffee, cocoa, beer, and wine) [4].

Plant polyphenolics are not only an excellent source of antioxidants exhibiting health-related function, but also play an important role in the quality of fruit and vegetables with enzyme compositions. In addition, some specific polyphenols are found uniquely in some plants, such as cinnamic esters of tartaric acid in grapes, phloridzin in apples and flavanone glucosides in citrus fruits [14].

2.2.3 Physical properties of polyphenols

The physical properties of polyphenols including molecular weight, melting point, absorption of light, and optical rotation presented in Table 2.1.1 are extracted from Wei's thesis [39].

Physical properties of polyphenols	
Molecular weight	catechin 293 Epicatechin 294, epicatechin-(3-O)-gallate 445, procyanidin dimer 580, trimer, 870 tetramer 1160.
Solubility	catechins (lipid soluble) procyanidins (water soluble)
Melting point	catechin, 174° C epicatechin, 236° C epicatechin-3-gallate 236° C
Absorption of light	λ 210-280 nm.
Optical rotation (in ethanol)	[α]D of catechin, 0 Epicatechin, 58.3, epicatechin-3-gallate, 188°

Table 2.1.1 Physical properties of polyphenols

2.2.4 Chemical properties

The chemical and biochemical properties of polyphenols presented in Table 2.1.2 are also extracted from references.

Chemical properties of polyphenols	
Hydrogen donation	Multiple hydroxyl groups (-OH) that are hydrogen-donating antioxidants and singlet oxygen quenchers. Potent metal chelating agents, which can trap and quench free radicals and break the chain reaction.
Stability	Sensitive to oxygen, light, acid and alkaline, less sensitive to heat. Catechin and epicatechin are easily oxidized. Polyphenols (catechins) are relatively stable in pH 3.6-6.5. To deactivate polyphenol oxidase, short exposure to a temperature between 70 and 90°C can be applied (Macheix et al., 1990).
Coloring reactions	In alkaline solutions such as sodium carbonate, polyphenols can react and produce a blue color, which is based on the Folin-Denis and Folin-Ciocalteu methods for measuring total polyphenols.

Table 2.1.2 Chemical properties of polyphenols [4]

Polyphenols can be divided into at least 10 different classes depending on their basic chemical structure. Figure 2.1.1 [15] illustrates four of the basic chemical structures of the main polyphenolic compounds. Flavonoids, which constitute the most important single group, can be further subdivided into 13 classes, with more than 5000 compounds [16].

The structure of catechins (monomers) and some procyanidins are shown in Figure 2.1.1. Catechin and epicatechin are the two proanthocyanidin monomers. Dimers, trimers, tetramers, and so on are created when each of these two monomers bind at the alpha or beta position on their molecular structures. In addition, catechin and epicatechin can combine to create esters such as catechin/epicatechingallate. They can also bond with sugars and proteins to create glycosides and polyphenolic proteins [4].

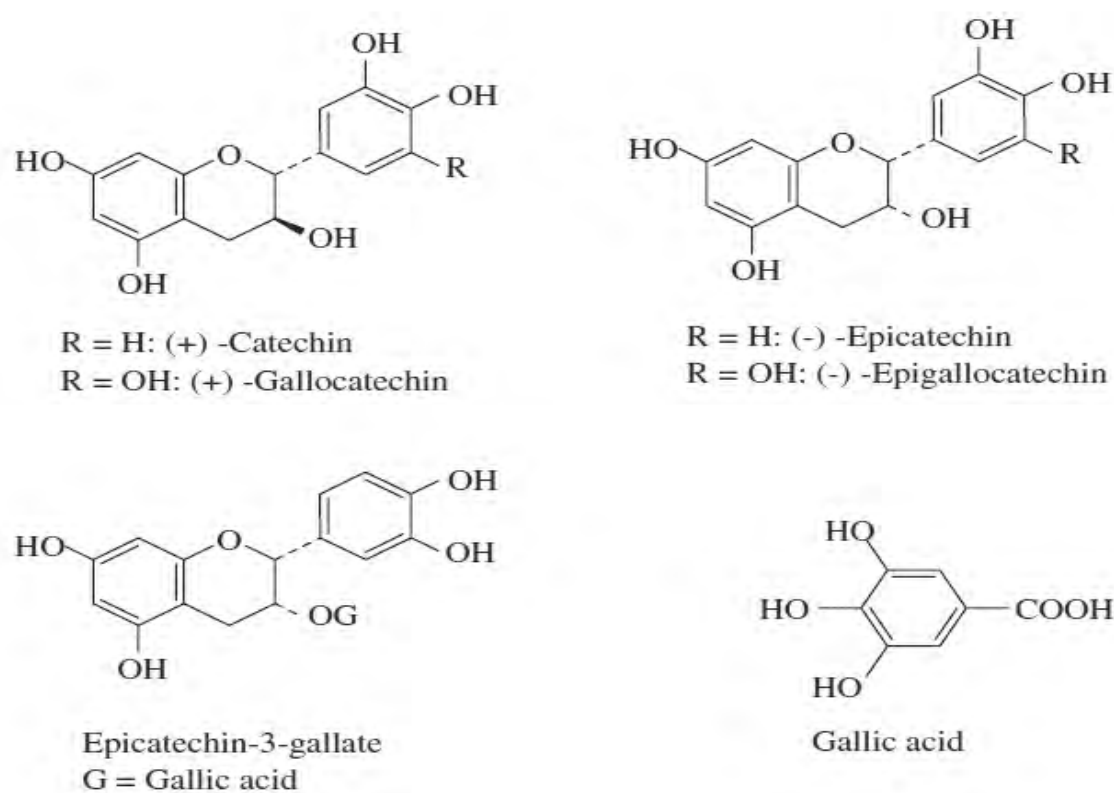


Figure 2.1.1: Structures of major polyphenols identified in tea[4]

2.3 Polyphenols in tea

In all of its various forms (green, black and Oolong) tea is a ubiquitous beverage, one of the most widely consumed beverages in the world other than water and is largely characterised by its high polyphenol content and their attendant antioxidant properties [17]. Green tea and Oolong tea are consumed mostly in China and Japan, whereas black tea is preferred in the United States and Europe [3]. There have been numerous studies linking drinking tea with reduction of risk of health problems, cancer in humans and even certain effects caused by alcohol intoxication and smoking although the evidence is still insufficient to allow for any clear statements to be made [18]. It is found that low-molecular weight tea components (caffeine and polyphenols, including the catechin group) have various effects at both individual and the cellular level, with beneficial physiological effects in antioxidation, antiobesity, acceleration of lipid metabolism, anti-carcinoma activity, and antimutagenicity [3]. These proposed health benefits have increased the popularity of tea beverages throughout the world.

2.3.1 Differences among green tea, black tea and Oolong tea

Depending on the treatment of harvest leaf, three main types of tea obtained.

1) Green tea:

The leaves are dried in the sun or on a tray in a stream of hot air as soon as chopped. In this case, the enzyme (for polyphenol oxidases) does not activate properly, and little polyphenols are oxidized [3].

The primary catechins in green tea are (-) epicatechin (EC), (-) epicatechin-3-gallate (ECG), (-) epigallocatechin (EGC), and (-) epigallocatechin-3-gallate (EGCG). In addition, caffeine, theobromine, theophylline, and phenolic acids, such as gallic acid,

are also present as minor constituents of green tea [19].

2) Oolong tea:

The leaves are allowed to stand for less than an hour before the enzyme is inactivated by heating.

3) Black tea:

Black tea is obtained from green tea by fermentation. The rolled chopped leaves are allowed to stand for 3-6 hours, the colourless catechins are oxidized by the enzyme to a complex mixture of orange-yellow to red-brown compounds, other types of polyphenols, mainly theaflavinegallate and thearubigins [20]. About 15% of tea catechins remained unchanged whereas approximately 10% is oxidised to bisflavanols, theaflavins and other oligomers which have molecular weight of 500-3000 Da. The remaining 75% of the catechins are converted to a poorly defined and intensively coloured complex group of substances known as thearubigins whose molecular weight ranges from 700 to 40000 Da [18].

2.3.2 Compositions of tea polyphenols

Tea is a major source of polyphenols, with the major constituents of flavanols [(+)-catechin, (-)-epicatechin, and (-)-epicatechingallate], flavonols (quercetin, kaempferol, and their glycosides), flavones (vitexin, isovitexin), and phenolic acids (gallic acid, chlorogenic acid) [4]. Flavanols are water soluble and colorless, and give an astringent taste. They oxidize readily in an alkaline environment. Tea beverage quality is positively correlated with fresh-leaf flavanol concentration. Green tea is pale yellow-green, and slightly more astringent than black tea, having a brothy characteristic imparted by theanine and other amino acids [21]. In Table 2.2.1, it shows that most polyphenolic compounds found in black tea are between 100-200 Da.

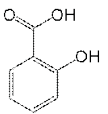
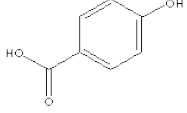
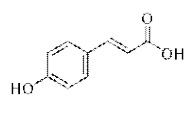
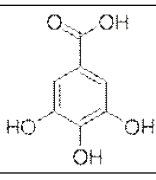
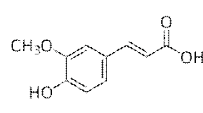
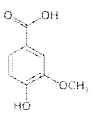
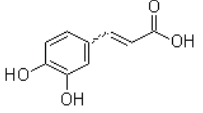
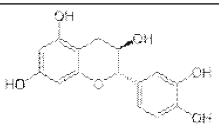
Components	Formula	Molecular Formula	Molecular weight (g/mol)	Usage
Salicylic acid (phenolic acid)		$C_7H_6O_3$	138	Medical and cosmetic uses
P.H.B.A (benzoic acid)		HOC_6H_4COOH	138	Preparation of biocides, antiseptics and bacteriostatic agents. It is used as a chemical intermediate for synthetic drugs, pharmaceuticals, dyes and plasticizers
Coumaric acid (phenolic acid)		$C_9H_8O_3$	164	Antioxidant properties (cancer)
Gallic acid (phenolic acid)		$C_7H_6O_5$	170	Antioxidant and helps to protect the cells against oxidative damage (and cancer...)
Ferulic acid (organic compound that is an abundant phenolic phytochemical)		$C_{10}H_{10}O_4$	194.1	Antioxidant in the sense that it is reactive toward free radicals such as reactive oxygen species
Vanillic acid (benzoic acid)		$C_8H_8O_4$	168	Flavouring agent
Caffeic acid		$C_9H_8O_4$	180	Pharmaceutical uses (carcinogenic)
Catechin (flavonoids)				Antioxidant (antibiotic properties.)

Table 2.2.1 Detailed properties of black tea polyphenols [19]

As a major source of polyphenols, tea polyphenols are constituted of flavanols, flavonols, flavones, and phenolic acids, which occupies approximately 30% of green tea dry weight [22], and 9-10% of black tea dry weight, while the percentage in Oolong tea is between green tea and black tea [23].

2.3.3 Composition of black tea

The composition of tea essentially varies with its species, season, climate, age of leaf, part of plant and manufacturing procedure. The approximate amounts of polyphenols in fresh leaves, green and black tea are in the range of 30-35%, 10-25% and 8-21%. The other constituents are amino acids, proteins, fibres, ash and carbohydrates.

Theaflavin and thearubigin are soluble in hot water and represent 20-35% of dry tea. The composition of black tea by percentages reported by Luczaj and Skrzydlewska [18] are shown in Table 2.2.2

Catechins	3-10
Theaflavins	3-6
Thearubigins	12-18
Flavonols	6-8
Phenolic acid and depsides	10-12
Amino	13-15
Methylxanthines	8-11
Carbohydrates	15
Protein	1
Mineral matter	10
Volatiles	<0.1

Table 2.2.2: Major components of black tea beverage in wt% of extract solids [18]

2.4 Membrane separation technology

Generally, membrane is defined as a selective barrier between two phases, and the term “selective” is inherent to a membrane or a membrane process. A membrane can be thick or thin, its structure can be homogeneous or heterogeneous, transport can be active or passive, passive transport can be driven by a pressure, concentration or a temperature difference. Besides, membranes can be natural or synthetic, neutral or charged [24].

2.4.1 Types of membrane process

Every membrane separation process is characterized by the use of a membrane to accomplish a particular separation. Figure 2.3.1 summarizes the application range of the various membrane processes. Solid and porous membranes (polymeric or in some cases ceramic or glass) are described here.

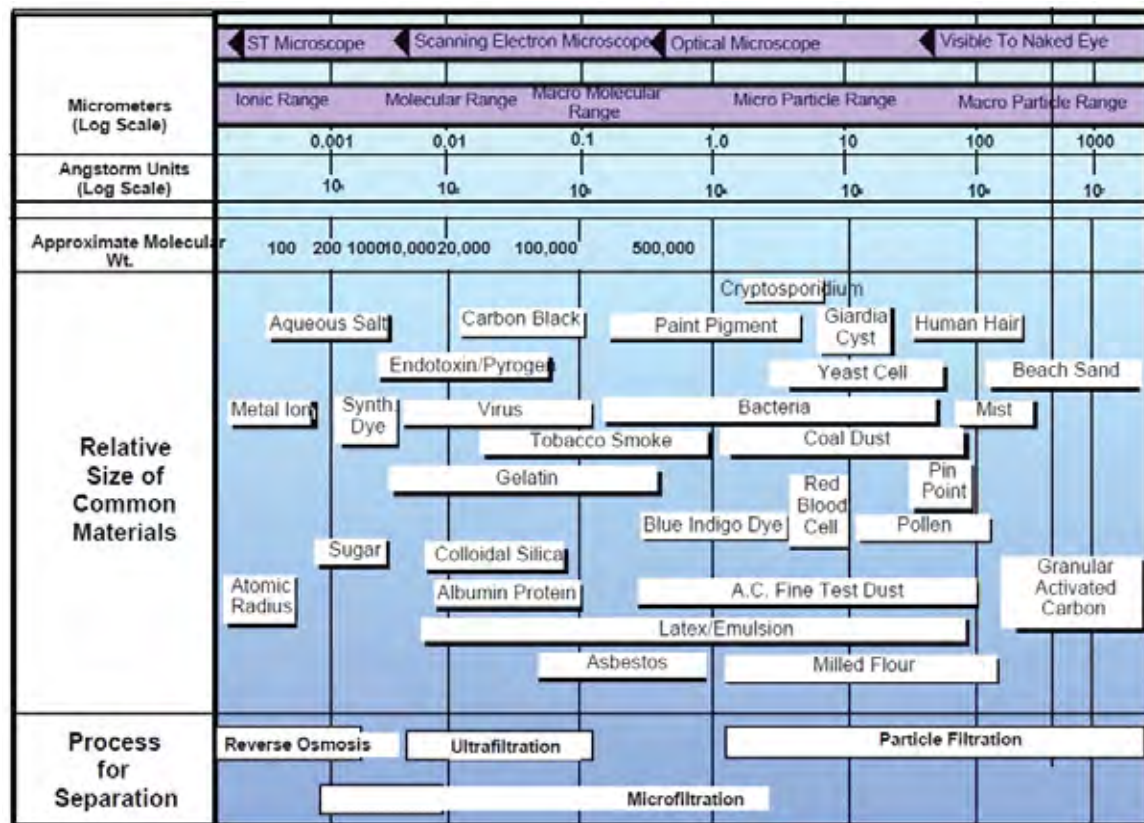


Figure 2.3.1: Types of membrane filtration process based on pore size [25]

Four types of membrane process are classified by membrane molecular weight cut off: Microfiltration (MF), Ultrafiltration (UF), Nanofiltration (NF) and Reverse Osmosis (RO). In Table 2.3.1, all the four membrane separation range investigated, compared and summarized [26].

	Microfiltration	Ultrafiltration	Nanofiltration	Reverse Osmosis
Pore size	4-0.02 μm	0.2-0.02 μm	<0.002 μm	0.002 μm
Components being rejected	Particles, clay, bacteria	Macro molecules, proteins, polysaccharides, vira	mono-, di- and oligosaccharides, negative ions	sodium chloride, glucose, amino acid
Operating pressure	<2 bar	1-5 bar	5-20 bar	10-100bar
Application (eg)	Water and wastewater treatment	Enrichment of trace chemicals	Removal of disinfection, BVPs, softening of water	Drinking water, desalination

Table 2.3.1: Comparisons of the four membrane processes [26]

2.4.2 Membrane materials

Synthetic membranes can be divided into organic (polymeric) and inorganic membranes, and the most important class of membrane materials being organic, i.e. polymers or macromolecules.

Based on different applications and chemical compatibility, varied organic membrane materials have been utilized in industry. In table 2.3.2, membrane materials and structures used in different processes are depicted respectively [27].

Membrane types	Method of preparation	Structure	Membrane processes
Asymmetric single layer	Casting and phase-inversion (cellulosics, polyamides, polysulphone)	Dense or finely-porous skin grading to (macro) microporous substructure	MF, UF, NF, RO, gas separation
	Dip coating	Finely porous layer on macroporous structure	MF, UF
Asymmetric composite	Film formation microporous support (dissimilar material)	Dense skin on microporous support	NF, RO, gas separation, pervaporation
Dynamic	Deposition of fine precoated on microporous barrier	Thin (removable) layer on microporous membrane	MF, UF, NF, RO
Symmetric-microporous	Stretching (PTFE, polypropylene)	Random network (0.02 to 10 μ m)	MF
	Irradiation and track etching (polycarbonate, polyester)	Parallel pores, 0.3 to 8 μ m	MF
	Casting and phase inversion (cellulose esters, nylon)	Random pores, 0.1 to 1 μ m	MF
	Molding and sintering	Random pores, 0.02 to 20 μ m	UF

Table 2.3.2: Membrane materials and structures [27]

Inorganic membranes generally possess superior chemical and thermal stability relative to polymeric membranes but are relatively costly. Nowadays, all kinds of applications of inorganic material are found in the field of microfiltration and ultrafiltration. Four types of inorganic material are frequently used: ceramic membranes; glass membranes; metallic membranes (including carbon); zeolitic membranes. These materials have very narrow pore sizes and can be applied in gas separation and pervaporation [24].

2.4.3 Membrane module design and configurations

In addition to the membrane, the filtration system also includes pumps, feed vessels, tubing, associated connections, necessary monitors and controllers for pressure, temperature, flow rate and membrane module. This system is a physical unit which

houses membrane in an appropriate configuration and satisfy mechanical, [26]

In order to avoid severe concentration polarization, mass transfer in the feed is required to be sufficiently rapid. Concentration polarization is often the dominant factor that controls the membrane performance, which refers to accumulation of highly concentrated boundary layer of retained solutes at the membrane surface [26].

Dead-end filtration test rig is the simplest and most generally applied setup in the laboratory. In this set-up, the permeate passes through the membrane while the concentration of the particles or large molecules that could not pass through the membrane increases, causing particle deposition or formation of fouling layer, leading the increase of filtration resistance and reduction of permeate flux. The deposit is eventually removed by backwashing (reversing permeate flow) or by other physical/chemical cleaning.

Crossflow is a process where the feed flows parallel to the membrane. The crossflow or tangential flow of the feed stream on the membrane surface is forced through the membrane and could partially prevent the build-up of particle deposition or fouling layer. The larger particulates which cannot pass through the membrane are swept along the upstream by the tangential force, which can be used as self cleaning effect. The clean-keeping efficiency of the flowing liquid increases with its velocity [26, 28].

Usually, in most sterile filtration of clean streams, clarifying prefiltration, MF sterilization cartridges in the pharmaceutical industry, dead-end flow is applied. When the retained material is likely to plug the membrane, such as UF, cross-flow filtration is the better choice [29].

In order to design a suitable filtration set, general requirements should be considered and followed [29]:

- ✧ Thin active layer of the membrane;
- ✧ Stable membrane with long service life;
- ✧ Mechanically strong;
- ✧ Large membrane surface area with a small footprint;
- ✧ Controlled concentration polarization;
- ✧ Easy cleaning;
- ✧ Inexpensive building cost;
- ✧ Low operating costs.

In this research, considering the concentration of black tea feed is not high, a stirred cell was tested, which needs less feed and is faster to set up than cross-flow to access some of the operating conditions.

2.4.4 Advantages and disadvantages of membrane process

Membrane technology is an emerging technology and because of its multi-disciplinary character it can be used in a large number of separation processes. The advantages of membrane separation processes in industrial applications include [24]:

- ✧ Separation can be carried out continuously
- ✧ Energy consumption is generally low
- ✧ Membrane process can easily be combined with other separation process (hybrid processing)
- ✧ Separation can be carried out under mild conditions
- ✧ Up-scaling is easy
- ✧ Membrane properties are variable and can be adjusted
- ✧ No additives are required

Pressure-driven processes such as ultrafiltration, nanofiltration, and microfiltration

are already established and various applications have been commercialized in the fields of pharmaceutical and biotechnology [1]. The combination of molecular separation with a chemical reaction, or membrane reactors, offers important new opportunities for improving the production efficiency in biotechnology and in the chemical industry.

However, the drawbacks of membrane technology should be mentioned as well [24]:

- ✧ Concentration polarization/membrane fouling;
- ✧ Low membrane lifetime if aggressive cleaning is used;
- ✧ Low selectivity or flux for excessively fouling feed conditions;
- ✧ Up-scaling factor is more or less linear (less economy of scale for very large installations)

2.5 The applications of membrane technology in polyphenols extraction

In this section, literature studies on membrane filtration of polyphenols were summarized.

2.5.1 Total tea polyphenol extraction

Usually, extraction is performed in three methods: organic solvent extraction, ultrafiltration and solid phase extraction (SPE). Mixtures with water and other solvents, such as methanol, ethanol, acetone, ethyl acetate are used in organic solvent extraction [30]. In this method, the mixture is constantly stirred until uniform slurry is obtained. Then the final extract can be obtained by filtering out the organic solvent. To increase the extraction rate, it can be achieved by increasing concentration gradient, a larger diffusion coefficient or smaller particle size. Although

this method is efficient, the final extraction compounds are not safe due to the chemical residual [4].

As another option, solid phase extraction is currently used in large scale production. This procedure of SPE separation is to condition a column with an appropriate solvent and then force the sample through the packing material (here, chrome was used) [39]. The next step is to wash out the impurities, leaving the compounds of interest bound to the absorbent. In the final step, the compounds are eluted from the solid phase using a strong solvent.

In this method, high pressure is required to extract the desirable compounds. The advantages of SPE are user friendly, economical process in long run, safe products. But the high expense of setting up system, high running temperature which may denature polyphenols cannot be ignored.

Ultrafiltration is another common means to separate and concentrate target compounds. The driving force of ultrafiltration is trans-membrane pressure, which is usually chosen at 2-10 bar. With ultrafiltration, the concentration of diluted polyphenolics solution can be increased. However, like other membrane filtration, the retention and permeate flux of ultrafiltration are also affected by membrane fouling, which is caused by accumulation of particles on the membrane surface and inside of membrane pores [4].

2.5.2 Ultrafiltration, Nanofiltration and Reverse Osmosis of polyphenols

Most phenols and flavonoids are low molecular weight phenolic compounds. The molecular weight of most polyphenols found in black tea (shown in Table 2.2.1) is

less than 300 Da.

As shown in Table 2.3.1, the pore size of Ultrafiltration is between 0.2 to 0.02 μm while nanofiltration and reverse osmosis are less than 0.002 μm . Nanofiltration can be applied in food industry for concentration and demineralization of whey, concentration of sugar and juice, and also concentration of polyphenols in juice and tea. Besides, nanofiltration are also used to concentrate the textile dye effluents in the pulp and paper industry [31].

It would be more appropriate to choose a nanofiltration in order to concentrate the polyphenols, although membranes with pore size larger than the polyphenol molecular weight could also retain the polyphenols which has been observed with filtration of other feed streams [21].

Therefore this study will investigate the filtration of black tea solution with NF membranes as well UF membranes (molecular weight cut off 1k Da and 5k Da, regenerated cellulose membrane) and the combination of both in order to achieve better separation factor while minimize membrane fouling and reduce membrane cleaning frequency. Furthermore, membrane made of regenerated cellulose possesses good solvent resistance with both aqueous and organic solvent, and compatible to a wide range of pH [32]. As a result, it shows low fouling characteristics and it is easy to clean.

Evans et al. [33] applied UF membrane on the clarification of black tea by using polymeric regenerated cellulose (RC) and fluoropolymer (FP) membranes and evaluated the separation efficiency and final product quality. In order to obtain the desired temperature, the black tea powder was initially dissolved in reverse osmosis (RO) water at 80°C and then mixed with RO water at ambient temperature to make up to final temperature of 50°C. With the conditions of 1.0 bar trans-membrane

pressure, 0.44 m/s cross flow velocity and 50°C feed temperature, different membranes with varied materials and molecular mass cut-off values (FP=10, 30, 100 kg mol⁻¹ molar mass cut-off (MMCO), RC=10, 30, 100 kg mol⁻¹MMCO) were observed and it's been found that increased fouling was present on rougher, more hydrophobic FP surfaces. For RC membranes, when roughness, charge and hydrophobicity were similar, it showed that no significant influence to the filtration properties has been found due to variation in pore size. All RC and FP 100 kg mol⁻¹ were cleaned relatively easily by sodium hydroxide (NaOH) solution. Generally all the RC membranes indicated superior performance to FP membranes in longer fouling runs due to their demonstrations of high flux with the least fouling. HPLC was used to measure the theaflavin and caffeine content in feed, and permeate. It is found the concentration of theaflavins in the permeate tea solids did not show obviously different to the initial feed solution after filtration of RC membranes with values of 99-101% recorded. However, the results by FP membranes showed a substantial variation in transmission of total theaflavins from 64 to 108% [33].

Zhang and Matsuura [21] investigated on reverse osmosis concentration of green tea juice. In their research, membranes of different pore sizes from different polymeric materials (cellulose acetate, polyethersulfone, aromatic polyamide and polyvinyl alcohol) were tested. Green tea powder was weighed and dissolved in distilled water at 90°C, then centrifuged at 6000 rpm to remove the sediment. Analysis of caffeine, polyphenols and amino acids was performed by spectrophotometry. The concentration of polyphenols was determined by ultraviolet spectrometry at 540 nm after adding ferrous tartarate into the sample solution and the diluting with buffer solution. Based on data, the concentration of green tea juice can be performed effectively by reverse osmosis membranes prepared from cellulose acetate and aromatic polyamide materials. In addition, the cellulose acetate membranes could achieve the fractionation of caffeine (in the permeate) and other tea components (in the retentate) at low feed concentrations (1.09×10 kg/m³, 1%), however, the

fractionation becomes less effective at high tea juice concentration (2.83 and $4.71 \times 10 \text{ kg/m}^3$, 3% and 5%). And the optimum pH for both caffeine separation and the product rate was observed as pH 5 [21].

2.6 Membrane fouling

During membrane filtration at constant pressure, the flux at a finite time is always less than the original value. At the beginning of filtration, this phenomenon is caused by concentration polarization, and when steady state conditions have been attained, a continuous decline in flux can often be observed. Such continuous flux decline is the result of membrane fouling, which may be defined as the (ir)reversible deposition of retained particles, colloids, emulsions, suspensions, macromolecules, salts on or in the membrane. Almost all feed components can foul on the membrane to a certain extent, which is strongly influenced by the physio-chemical nature of the membrane and solute(s) [35].

2.6.1 Flux decline in filtration

There are several reasons for the decreasing flux in a real-world feed stream.

- ✧ Changes in membrane properties: these can occur as a result of physical (high pressure) or chemical deterioration (pH, temperature, and other environmental factors). Besides, in food and biological applications, the necessary wash and frequent cleaning regimes contribute to the decreasing of membrane lifetime as well.
- ✧ Change in feed properties: this factor can be easily predicted due to the increasing feed stream's viscosity and density and diffusivity changes as solids levels accumulated, leading to lower flux than that of the initial feed.
- ✧ Concentration polarization: this is frequently confused with membrane fouling in

flux-lowering phenomena. Theoretically, with decreasing transmembrane pressure, lower feed concentration or increasing cross-flow velocity, the flux depressing effects caused by concentration polarization should be reversible. If this can be done, the real cause of decreasing flux should be due to polarization and not true fouling [36].

2.6.2 Types of fouling

The phenomenon of fouling is very complex and difficult to describe theoretically. Even for a given solution, fouling will depend on physical and chemical parameters such as concentration, temperature, pH, ionic strength and specific interactions. Generally, fouling can be divided into the types below:

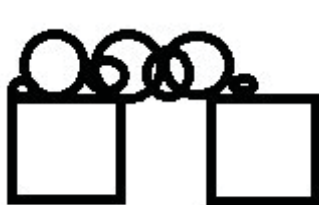
Colloidal: Clays, flocs;

Biological: Bacteria, fungi, and EPS;

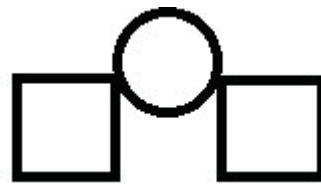
Organic: Humics, oils, polyelectrolytes;

Scale: Mineral precipitates (CaCO_3 , SiO_2 ,...)

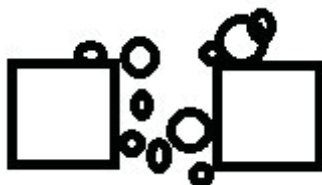
In addition, the four fouling mechanisms are shown in the figure and table below.



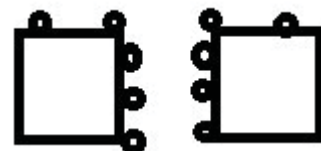
Cake filtration



Complete blocking



Intermediate blocking



Standard blocking

Figure 2.5.1: Diagrams of different blocking

Law	Physical cause	Description	Equation
Cake filtration	Boundary layer resistance	Deposit of particles larger than the membrane pore size onto the membrane surface	$t/V=AV+B$
Complete blocking	Pore blocking	Occlusion of pores by particles with no particles superimposition	$-\ln(J/J_0)=At+B$
Intermediate blocking	Long term adsorption	Occlusion of pores by particles with particle superimposition	$1/J=At+B$
Standard blocking	Direct adsorption	Deposit of particles smaller than the membrane pore size onto the pore walls, reducing the pore size	$t/V=At+B$

Table 2.5.1: Introduction of different types of fouling mechanisms (J:flux, J_0 :initial flux, V:the cumulative volume of permeate at time t, A and B:model parameters)

2.6.3 Fouling prevention and cleaning

Strategies to reduce fouling can include reducing flux, careful membrane material selection and pre-treatment (coagulation, adsorption) [37].

✧ Pre-treatment of the feed solution:

As a process stream to either remove or modify potential foulants, pre-treatment is an effective method for reduction of flux decline and is usually easy to implement where membrane filtration is retrofitted into an existing conventional treatment plant [37]. Pre-treatment can remove larger particles or reduce the fouling potential of remaining solutes. Pre-treatment methods employed included: heat treatment, pH adjustment, addition of complexing agents (EDTA etc.), chlorination, adsorption onto

active carbon, chemical clarification, pre-microfiltration and pre-ultrafiltration [24]. In NF normally frequent cleaning is avoided by using different types of pretreatment. MF and UF can be effective in removing large particulates, but small colloidal matter may still permeate, which can result in a very low (less than 3) silt density index (SDI) [38]. SDI is a measure of fouling properties commonly used in the water treatment industries.

✧ Membrane properties

A change of membrane properties can reduce fouling. Fouling with porous membranes (microfiltration, ultrafiltration) is generally much more severe than with dense membrane (pervaporation, reverse osmosis). Furthermore, a narrow pore size distribution can reduce fouling (although this effect should not be overestimated). Besides, hydrophilic membranes can help reducing fouling than hydrophobic membrane by reducing adsorption. In addition, charged membranes can also help, especially in the presence of (negatively) charged colloids in the feed [24].

✧ Module and process conditions

Fouling phenomena diminish as concentration polarization decreases. Concentration polarization can be reduced by increasing the mass transfer coefficient (high flow velocities) and using low(er) flux membranes. Also the use of various kinds of turbulence promoters will reduce fouling [24].

✧ Cleaning

Physical cleaning methods:

- Backflush/Forward flush/Reverse Flush
- Scrubbing (e.g. using foam balls for tubular modules)
- Air sparge
- Vibrations
- Sonication

Chemical cleaning methods:

Chemical cleaning is the most important method for reducing fouling, with a number of chemicals being used separately or in combination. Chemical cleaning relies on chemical reactions to break bonds and cohesion forces between membranes and foulants. Some important chemicals are:

- Acids (strong such as H_3PO_4 , or weak such as citric acid)
- Alkali (NaOH)
- Detergents (alkaline, non-ionic)
- Enzymes (proteases, amylases, glucanases)
- Complexing agents (EDTA, polyacrylates, sodium hexametaphosphate)
- Disinfectants (H_2O_2 and NaOCl)
- Steam and gas (ethylene oxide) sterilization

According to the information presented by Diqiao Wei [39], the cleaning of ultrafiltration membrane was performed by 0.1 M NaOH at 35 °C for 30-60 minutes. In Li and Elimelech's paper [40], the model chemical cleaning solution used NaOH (pH11) as an alkaline solution.

2.7 Analysis of polyphenols

In this section, analytical techniques for both total polyphenols and specific composition of polyphenols in black tea are reviewed. UV-Vis spectrophotometry is easy to operate and reliable to measure the total polyphenols with Folin-Ciocaltea method which is applied [41]. In order to evaluate the detailed polyphenolic compounds in black tea, capillary electrophoresis (CE) and GC/MS have also been applied previously.

Total phenol content determination: The Folin-Ciocalteau (FC) method is routinely used for colorimetric assay of soluble phenols in wines and fruit juices. It is adapted

for plant extracts and liquid nutrient media too. UV-Vis spectrophotometer is a widely used analysis instrument to determine the total phenol content after Folin-Ciocalteu reaction which is easy to operate.

Composition analysis of polyphenols: In order to determine the specific solution composition, many analysis methods can be applied. GC, GC/MS, HPLC, CE are all reported to be used in research to investigate the compositions of polyphenols [17, 42]. In this study, GC/MS and CE are chosen to analysis the specific polyphenolic compounds in black tea samples solution.

2.7.1 UV-Vis spectrophotometer

UV-Vis spectrophotometry is a mature analytical technique, from which numerous procedures have been developed. Absorptiometry is the main tool, and a few methods are based on the study of the absorbance spectrum. UV absorptiometry is still used as a detection technique for liquid chromatography and a coloured reaction is always needed for visible colorimetric methods [46].

Analysis with UV-Vis spectrophotometer were used to measure total polyphenols in many previous papers [47]. Ultraviolet-visible spectroscopy or ultraviolet-visible spectrophotometry (UV-Vis or UV/Vis) refers to absorption spectroscopy in the UV-visible spectral region. This means it uses light in the visible and adjacent (near-UV and near-infrared (NIR)) ranges. The absorption in the visible range directly affects the perceived color of the chemicals involved. In this region of the electromagnetic spectrum, molecules undergo electronic transitions. This technique is complementary to fluorescence spectroscopy, in that fluorescence deals with transitions from the excited state to the ground state, while absorption measures transitions from the ground state to the excited state [48].

From Jac et al. review [49], it's proved that most polyphenols were absorbed between 200 and 300 nm (Table 2.6.1). Based on the information in Table 2.6.1, 210 nm and 250 nm were widely used to analyse polyphenols in previous research. Our own tests of scan of wavelength also shown a better absorptions at 210 nm and 250 nm. In this research, 210 nm and 250 nm were chosen to measure the concentration of black tea in feed, permeate and retention.

Analytes/matrix	Detection	Reference
Catechin, epicatechin,	UV 214 nm	[50]
Gallic acid, catechin, catechin-3-gallate, epigallocatechin, epigallocatechin-3-gallate, epicatechingallate, gallocatechin, gallocatechingallate	UV 200 nm	[51]
Caffeic acid	UV 250 nm	[52]
Caffeine, catechin, epicatechin, catechingallate, epicatechingallate, epigallocatechin, epigallocatechingallate (tea beverage)	UV 210 nm	[53]
Caffeic, vanillic acid, catechin, epicatechin (red wine)	UV 254 nm	[54]
Caffeine, catechin, epicatechin, epicatechingallate, epigallocatechin, epigallocatechingallate (black/green tea)	UV 200 nm, 266 nm	[55]

Table 2.6.1 General characteristics of the reviewed methods

2.7.2 Gas chromatography and mass spectrometry (GC/MS)

GC/MS is the combination of gas chromatography (GC) for separation and mass spectrometry (MS) for detection and identification of the components of a mixture of compounds, which has become a definitive analytical tool in the research and commercial analytical laboratory [56].

The GC/MS is composed of two steps, the gas chromatograph (step 1) and mass spectrometer (step 2). The gas chromatograph is characterized by the column's dimensions (diameter, length) as well as the phase properties. The difference in the

chemicals properties present in the sample will separate all molecules over the length of the column. Difference in molecular weight causes difference in times to travel through the GC column (the retention time). Then, the mass spectrometer captures and detects different molecules by breaking them into ionized fragments and detecting these fragments using their mass to charge ratio [56].

In the research by Wu et al., GC/MS was used to determine the phenolic acids and 2,4-dihydroxy-7-methoxy-1,4-benzoxazin-3-one in wheat [57]. In the paper by Gherman et al., both GC and GC/MS were applied to analyze the different types of *Menthapiperita* L. (Labiatae family) after rapid extraction, and the main volatile compounds identified by the gas chromatography–mass spectrometric analysis [42]. In addition, GC/MS was also used by Milou to determine the components in black tea before and after ultra-/ nano-filtration [58].

2.7.3 Capillary electrophoresis (CE)

Capillary electrophoresis (CE) has proved to be a rapid and versatile analytical technique that combines simplicity with high reproducibility. The narrow diameter (normally between 20 and 100 μm) of the silica glass capillaries allows the application of high voltages and ensures rapid heat dissipation. CE can be utilized to analyze a wide variety of charged and uncharged species. Size range from that of small analytes such as metal ions and low molecular weight alcohols to larger molecules, oligosaccharides, proteins, and nucleic acids [59].

CE presents the potential of producing high-resolution separations of biopolymers, as well as smaller pharmaceutical agents, with minimum use of both sample and reagents. HP³D CE of Agilent Technologies will be used in this work. Table 2.6.2 gives us the method to analysis polyphenols samples provided by HP³D CE of Agilent Technologies.

Instrument	Column	Buffer	Rinse	Reference
P/ACE 5500 CE (Beckman)	57cmx75µm I.Dx375µm O.D	0.15 M H ₃ BO ₃ at pH 8.5 (adjusted with NaOH)	Consecutive washing with water (5min), 0.1 M sodium hydroxide (5min), and water (2min) followed by the running buffer (15min)	[60]
P/ACE 5000 (Beckman)	77cmx50µm I.D	20 mM borax buffer, and adjusted to the appropriate pH with hydrochloric acid	After each separation, the a\capillary was rinsed sequentially with distilled water, 0.1 M HCl, distilled water, 0.1 M NaOH and buffer.	[61]
³ DCE(Hewlett-Packard)	56 cm x50 µm I.D fused-silica 'bubble cell'	Borate buffer (32 mM, pH 9.2), SDS (4.5 mM)	Acetonitrile for 5 min, 1 M NaOH for 10 min and deionized water for 20 min. also flushed with 0.1 M NaOH for 5 min and corresponding buffer for 10 min between runs	[62]
4-tray CZE instrument (Lauerlabs)	50µm idx85 cm with the detection window located 15 cm from the outlet side	4 mM sodium tetraborate, 12 mM potassium hydrogenphosphate, and 40 mM SDS (pH7).	Flushed with 1M NaOH for 10 min, then extensively rinsed with water, and finally conditioned with the separation buffer solution for 15 min	[55]
³ DCE (Hewlett-Packard)	40cmx50µm ID	800 µl of 500 mM boric acid (pH7.2), 200µl of 100 mM KH ₂ PO ₄ , 450 µl of 20 mM β-CD and 550 µl of acetonitrile	Prior to sample introduction, the capillary column was preconditioned with the buffer solution for 3 min.	[17]
P/ACE 5000	57cmx360µm ODx75µm ID		0.1 M hydrochloric acid, 0.1 M sodium hydroxide and then the electrolyte.	[63]
HP ³ DCE from Bio-Rad (Scientific Group)	51cmx50µm ID	50 mM K ₂ SO ₄ and 600 mM boric acid titrated to pH 7 using 1.0 M NaOH	Rinsing with running buffer for 2 min	[64]

Table 2.6.2 Different instruments & methods for CE

In the paper of Arce et al.[60], polyphenols presented in green tea were determined

by capillary electrophoresis. A Beckman P/ACE 5500 CE unit equipped with a diode array detector was used to separate and quantify the analytes. The running buffer used was a solution of 0.15 M H_3BO_3 at pH 8.5 (adjusted with NaOH). The tea samples (1.75 g) were extracted with 100 ml of boiling water.

Tea catechins and theaflavins were also analyzed by capillary electrophoresis, comparing to HPLC [17]. In this paper, Hewlett-Packard^{3D}CE was applied. The buffer solution was freshly prepared by mixing 800 μl of 500 mM boric acid, 200 μl of 100 mM potassium dihydrogenphosphate, 450 μl of 20 mM β -CD and 550 μl of acetonitrile together, and mixed uniform.

Separation mechanism and determination of flavanones were also performed by capillary electrophoresis by Wang et al.[65]. All their CE experiments were performed on an Agilent Hewlett-Packard^{3D} CE instrument. Uncoated fused-silica capillary was used. Running buffer consisted of borate buffer (32 mM, pH 9.2), SDS (4.5 mM) and with applied voltage of 15 kV.

2.8 Analysis of membrane surface charge

Stream potential is also tested in order to investigate the change of membrane surface charge before and after filtration. Since the interaction of organic and inorganic colloidal substances with membrane surfaces in aqueous media is dependent on the charge of the membrane surface [43, 44], determination of the stream potential of the membrane surface is critical to membrane fouling research. This property has generally been overlooked in efforts to select optimum operating conditions for various membrane separation processes. The results of streaming potential measurements of virgin and fouled membranes can be used to characterize the membrane surface cleanliness and permeability. In addition, the charge property of foulant can also be studied.

Zeta potential can be determined from several electrokinetic measurements: streaming potential, sedimentation potential, electro-osmosis, or electrophoresis. Among these techniques, streaming potential is the most widely used for a flat membrane surface [44].

In Elimelech's research, streaming potential analyzer was performed to determine the zeta potential of RO membranes. In their work, the origin of charge of polymeric surfaces in aqueous solutions and basic principles of streaming potential measurements were discussed [44]. In Nystrom's study, streaming potential measurements were investigated on NF membranes about the fouling and retention, in which zeta potential was calculated as a function of pH and ionic strength [45].

Although some research on streaming potential has been investigated, more membrane types are still waiting to be considered.

2.9 Summary

Based on the literature review of the chemistry and functions of polyphenols, especially the specific compounds existing in black tea, the extraction and filtration process with membrane has great potential. The membrane processes applicable (either UF or NF or combination of UF and NF) should be explored for the specific application and the membrane operation conditions can be optimized for optimized membrane performance such as the separation/concentration factor and membrane fouling management.

Research opportunity on different extraction methods, different membrane processes, the permeate flux and the polyphenols content in both permeate and retentate streams at different pressure and retention time exists and this work will focus on the following aspects.

The extraction methods are very important process in further investigation of membrane process on tea polyphenols. In this case, varied parameters (e.g. temperature, concentration, extraction time, solvent constituent) were tested and compared. Not only is it important to collect and measure the total polyphenols, also the stability of the majority of polyphenols in black tea cannot be ignored.

After the initial extraction, the tea solution should be pretreated to remove the particulates existed. In this phase, two different pretreatments (vacuum filtration with Whatman 42 ashless filter paper and ultra-filtration with MWCO of 1 K Da and 5 K Da) was applied to the raw black tea solution. The permeate flux of nanofiltration and polyphenols content were measured and compared after different pretreatment.

During the nanofiltration phase, varied parameters (e.g. TMP, retention time, stirring rate) were studied, in order to obtain the influence of different parameters and decide a more proper and sustainable conditions to concentrate polyphenols in black tea solution. Considering the components discovered in black tea, two nanofiltration membranes were with different molecular weight cut off were applied and compared.

In order to analyze the results and effects of different conditions, several analysis technologies were applied. UV-Vis spectrophotometer was used to measure the amount of total polyphenols, while GC-MS and Capillary Electrophoresis were used to characterize the detailed components in black tea feed, permeate and retentate. Besides, surface charge of virgin and fouled membranes were tested by Zeta Potential.

Chapter 3 EXPERIMENTAL METHODS

3.1 Membrane filtration apparatus

Nanofiltration with stirred cell (Sterlitech, US) was conducted to separate polyphenolics and caffeine from black tea solution (see in Figure 3.1). In this stirred-cell rig, feed pressure was maintained constant and monitored by a pressure indicator. The retentate was recycled back to the feed. The permeate was collected in a separate beaker which was placed on a balance for instantaneous measurement of the weight of permeate collected. The balance was connected to a computer, and temperature was consistent with the room temperature.

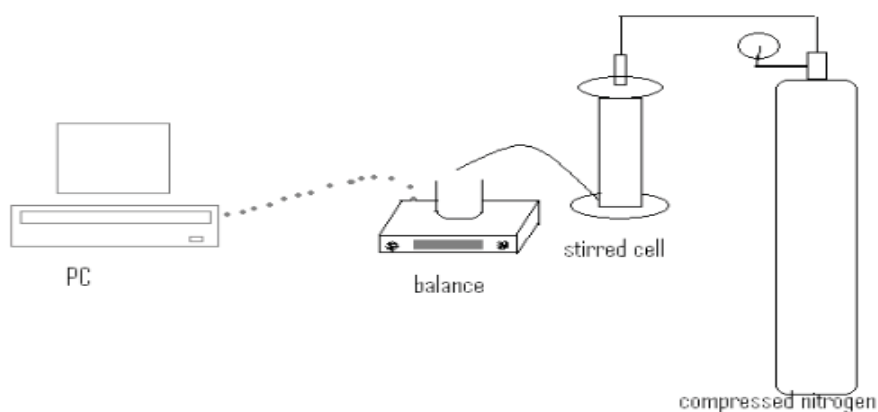


Figure 3.1: Schematic of the stirred-cell rig



Figure 3.2 Photo of experiment set-up

In this research, NF and UF membranes were selected. NF membrane with molecular weight cut off of 270 (NF 270, polyamide TFC) and 700 (NTR-7450) which are close to the molecular weight cut off (MWCO) of polyphenols in black tea were tested for flux and rejection of polyphenols.

The molecular weights cut off of most polyphenols are below 300 Da, in light of this, 270 membrane was applied to the nano-filtration. Membranes with higher MWCONF membrane and UF membranes will also be tested.

3.1.1 NF 270 membrane

NF 270 membrane is made of polyamide which is hydrophilic and with the permeability of 19.45 L/h.m².bar. The MWCO is considered to be 200-400, maximum operating temperature is 45°C and the maximum pressure tolerance is 41 bar. The pH range for continuous operation is from 3 to 10 (cleaning pH from 1 to 12) [66].

3.1.2 NTR 7450

NTR-7450 is made of polymide with a MWCO of 700 Da with a permeability of 15.67 L/h.m².bar. Its maximum operating temperature is 45°C and the maximum pressure tolerance is 40 bar. The pH range is from 3 to 10 [34].

Ohaus balance (Adventurer™) was set for data logging. A computer program was used to calculate the flux with permeate weight and time:

$$\text{Flux}(j) = \frac{\text{volume}_{\text{permeate}}(j) - \text{volume}_{\text{permeate}}(j-1)}{[\text{time}(j) - \text{time}(j-1)] * \text{area}_{\text{membrane}}}$$

NUOVAII stirrer was used in this research. Three stirring rates were applied in the following experiments and compared: 250 rpm, 450 rpm, and 650 rpm. Compressed

nitrogen 3.0 (8.1 m³) was used to apply the pressure for Nanofiltration.

3.2 Materials

Lipton black tea (100 bags, 180 g in total) was purchased from a local supermarket. Sodium carbonate, methanol, sodium acetate trihydrate, acetic acid, potassium dihydrogenphosphate, boric acid and sodium hydroxide were purchased from Ajax Chemicals. Folin-Ciocalteu's phenol reagent (2N), β -CD were from Sigma. Organic acid standard and buffer were from Agilent technology. Acetonitrile was from BDH.

3.3 The effect of operating parameters

The effects of different feed concentration, temperature, extraction solution, transmembrane pressure, stirring speed, filtration time, and cleaning protocol have been compared and studied to determine the optimum operating conditions.

Feed concentrations were tested between 0.33% g to 1.4% (black tea wt%) to extract polyphenols. In the extraction tests, temperature of 70 and 100 ° C were compared while the extraction solution were tried with Milli-Q water and 6% methanol. The transmembrane pressures were examined between 3 bar to 8 bar. The pressure was controlled via the pressure gauge mounted on the stirred-cell rig. The stirring speed was studied between 250 rpm, 450 rpm, 650 rpm. The filtration time was tested between 2 hours to 6 hours. Different cleaning protocols with varied concentration of cleaning solutions, cleaning time were also investigated.

The experiments were performed as follows:

- ✧ Feed concentration: 0.33%, 1.4% (black tea wt%)
- ✧ Extraction solution: Milli-Q water, 6% methanol in Milli-Q water

- ✧ Temperatures: 70° C, 100° C
- ✧ Transmembrane pressure: 3 bar, 5 bar, 8 bar
- ✧ Stirring speed: gear 2, 4, 6 by the scale of the stirrer
- ✧ Filtration time: 2 hours, 4 hours, 6 hours
- ✧ Cleaning protocol:
 - Cleaning solution: 0.01 M NaOH, 0.1 M NaOH
 - Membrane handling: cleaning separately, cleaning in rig without taking out

3.4 Experimental Procedure

This investigation began with vacuum filtration of black tea solution, then followed by nanofiltration with the applied pressure, and finally analysis feed, retentate and permeate by varied instruments. Total polyphenols in feed, retentate and permeate was analyzed by UV-Spectrophotometry with Folin-Ciocalteu's phenol reagent. The polyphenols composition in feed and retentate of Nanofiltration process was also analyzed by GC-MS. The streaming potential of virgin membrane and fouled membrane was investigated by Zeta-Potential.

3.4.1 Extraction methods

There has been various reports in the literature as how the tea solution was obtained from the tea leaves and the information was summarized in Table 3.1. The first two methods were from literature and used as reference for this research, the rest of the conditions were conducted in this work.

Weight of tea (g)	Volume of water (ml)	30% Methanol content (%)	Infusion temperature (° C)	Infusion time (min)
0.2	100	0	100	5 [62]
0.5	80	6	70	15 [41]
1.8	300	0	100	5
1.8	300	0	100	10
3.6	300	0	80	5
3.6	300	0	100	5
1	300	0	100	5
1	300	0	70	5

Table 3.1: Infusion methods in literature and tested

After cooling the mixture to room temperature, the solution was filtered with a vacuum filter using Whatman ashless filter paper 42 with particle retention of 2.5 µm. This solution was regarded as basic sample solution.

In consideration of avoiding oxidation factor, extraction temperature of 70 ° C was chosen which can reserve more polyphenols (theaflavin) due to most theaflavin would be degraded at a temperature above 70 ° C. Besides, the diffusion time was chosen as 5 minutes to reduce oxidation time.

3.4.2 Pretreatment of feed

✧ Vacuum filtration

After extraction and cooling, the black tea solution was vacuum filtered (Figure 3.3) using Whatman ashless filter paper 42 (see in Table 3.2). The flow rate was not measured during the filtration process.

Whatman grade No.42 quantitative filter paper, ashless	
Diameter (cm)	5.5
Whatman No.	1442-055
Porosity	Fine
Flow rate	Slow
Particle retention	2.5 µm

Table 3.2: Properties of Whatman grade No.42 quantitative filter paper



Figure 3.3: Photo of vacuum filter



Figure 3.4: Photo of used vacuum filter paper

✧ Ultrafiltration

After vacuum filtration, Ultra-filtration was also applied before Nanofiltration. In this procedure, UF membranes with MWCO of 1 K Da and 5 K Da were tested (in Table 3.3):

MWCO of UF membranes	Material
1 K Da	Regenarated Cellulose, PLAC 15005
5 K Da	Regenarated Cellulose, PLCC 15005
5 K Da	Polyethersulfone (by Nitto Denko)

Table 3.3 Ultra-filtration membranes tested

3.4.3 Filtration procedures

✧ Initial water soaking

Before filtration of feed solution, membrane was soaked in Milli-Q water for 1 day in order to remove the storage coating in the membrane.

✧ Initial water flux

Pure water flux was performed on the virgin membrane at feed pressure of 3 bar for 30 minutes in order to test the membrane permeability before fouling.

✧ Filtration

The feed was pressed through the membrane with stirred cell. Trans-membrane pressure was controlled by the gauge mounted to nitrogen cylinder to give a pressure of 3 to 8 bar. The permeate flow was measured by measuring the weight of permeate every two minutes. The balance measuring permeate weight was connected to computer which can calculate the flux of permeate by a program in it. Samples were collected from initial black tea feed solution, finial permeate and finial feed solution (retentate). Permeate samples were also collected at each half hour during filtration process in one test.

✧ Final water flushing

Final water flushing is the same as initial water flushing. The proper execution is important to flush out any sample solution (fouling) left on the membrane surface. Insufficient or improper flushing may cause entrapment of foulants in the membrane structure, which result in faster fouling that cannot be

recovered. After the final water flushing, pure water flux was also measured and recorded.

✧ Cleaning

The cleaning was performed by 0.1 M and 0.01 M NaOH solutions at room temperature (21 ° C) for 30 minutes at stirring rate of 450 rpm. After cleaning, membrane was stored in pure Milli-Q water in the fridge.

3.5 Analysis methods

3.5.1 Spectrophotometric method

In this study, 210 nm was chosen to measure the concentration of black tea in feed, permeate and retention. A scan was conducted in the wavelength range from 200 nm to 800 nm and the absorption was given in Figure 3.5.

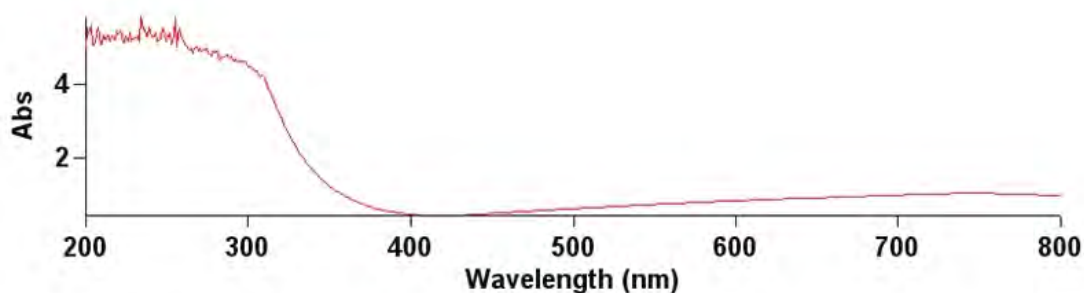


Figure 3.5: Total polyphenols absorption in wavelength from 200 to 800 nm

Analytes/matrix	Detection	reference
Catechin, epicatechin,	UV 214 nm	[50]
Gallic acid, catechin, catechin-3-gallate, epigallocatechin, epigallocatechin-3-gallate, epicatechingallate, gallocatechin, gallocatechingallate	UV 200 nm	[51]
Caffeic acid	UV 250 nm	[52]
Caffeine, catechin, epicatechin, catechingallate, epicatechingallate, epigallocatechin, epigallocatechingallate (tea beverage)	UV 210 nm	[53]
Caffeic, vanillic acid, catechin, epicatechin (red wine)	UV 254 nm	[54]
Caffeine, catechin, epicatechin, epicatechingallate, epigallocatechin, epigallocatechingallate (black/green tea)	UV 200 nm, 266 nm	[60]

Table 3.4: Analytes and wavelength used for detection using UV spectrophotometer found in literature

The total content of polyphenolic was determined by using Folin-Ciocalteu Reagent. This assay is based on the absorbance of colored molybdate and tungstate complexes formed by phenolic groups under alkaline conditions. The results were presented as catechin or gallic acid equivalents.

✧ Instrument: Cary100 UV-Vis Spectrophotometer (see in Figure 3.3)

✧ Reagent: Folin-Ciocalteu reagent (Sigma)

33% sodium carbonate solution

Acetate buffer: 1.16 g/100ml sodium acetate trihydrate

0.34 ml/100ml acetic acid

✧ Method:

- Prepare Folin-Ciocalteu reagent, 33% sodium carbonate solution and acetate buffer as above.
- Pipette 1 ml samples (feed, permeate or retentate) and dilute with 4 ml Milli-Q water and 5 ml acetate buffer
- Pipette 1 ml diluted sample into a 10 ml flask and add 0.5 ml Folin-Ciocalteu reagent. Make up volume to 10 ml with 33% sodium carbonate solution.

- Wait for the color changing to be stable.
- Blank sample was made by 0.5 ml Folin-Ciocalteu reagent in 10 ml flask, making up with 33% sodium carbonate solution
 - Measure the absorbance of samples in Cary-100 UV-Spectrophotometer at 210 nm
- ✧ Calculation: Before measuring the sample, the measurement was zeroed by blank sample which can eliminate the interference of the color of the Folin-Ciocalteu reagent itself.



Figure 3.6: Photo of Cary 100 UV-Vis Spectrophotometer

3.5.2 Gas Chromatography-Mass Spectrometry (GC-MS)

- ✧ Instrument: GC-MS analysis was performed using a Thermo Electron Corporation (Trace DSQ) Mass spectrometer, equipped with a TR-5MS SQC capillary non polar column (length: 15m; ID: 250 μ m; film thickness: 2.5 μ m).
 - ✧ Reagent: acetate ethyl used as mobile phase
- 5-Methoxy Salicylic acid as internal standard
- ✧ Method:
 - The inlet temperature was set at 220°C while the oven temperature was

initially at 80°C (held for 1min) and then increased to 260°C at 25°C/min (held for 3min)

- The mass spectrometer was operated in the positive electron ionization (EI) mode with ionization energy of 70 eV. The ion source temperature was 260°C.
- 1µl of each derivatised sample was injected into a capillary column [60]
- All the components in the sample were identified using mass spectral library provided by the GC-MS supplier



Figure 3.7: Photo of GC-MS machine

3.5.3 Capillary electrophoresis

- ✧ Instrument: Agilent CE (see in Figure 3.4)
- ✧ Columns:
 - AGIG1600-61239, PVA CE-cap. Ext. light path 50 µ m 56 cm
 - AGIG1600-62211, Capillary Standard HPCE 50 µm id 72 cm
 - AGIG1600-61232, Ext. Light Path CE cap. 50µm 56 cm
- ✧ Sample preparation: The sample was prepared by extracting 3g of Lipton black tea from 300 ml Milli-Q water and incubating in 70 °C for 5 min. After vacuum filtration with Whatman ashless 42, an aliquot of 0.5 ml of the tea extract was

mixed with 1 ml 20% (v/v) acetonitrile

✧ Buffer preparation:

- The buffer solution was freshly prepared by mixing 800 μl of 500 mM boric acid, 200 μl of 100 mM potassium dihydrogenphosphate, 450 μl of 20 mM β -CD and 550 μl of acetonitrile together, and mixed uniform
- 0.15 M H_3BO_3 (PH 8.5), adjusted with NaOH



Figure 3.8: Photo of Agilent CE

3.5.4 Streaming potential

- ✧ Instrument: Zeta Potential Analyser (Sure Pass, Anton Parr, see in Figure 3.5)
- ✧ Reagent: 0.001 potassium chloride solution (background electrolyte), 0.25 M hydrochloric acid, and 0.1 M potassium hydroxide solution
- ✧ Method:
 - Sample preparation: soak the membranes to be analysed in Milli-Q water for 24 hours and prepare 2 litres of background electrolyte, 0.25 M HCl, and 0.1 M KOH for the titration
 - Sample mounting: mount in the following order (from bottom to top)---

small transparent block, spacer without connection, membrane with hole facing up, spacer with connection, membrane without hole facing the other membrane and big transparent polymer block

- Pre-treatment: conduct a flow check and pH titration

Measurement: empty and rinse the equipment, and then conduct the measurement of the surface charge of target membrane



Figure 3.9: Photo of Zeta Potential analyser

Chapter 4 POLYPHENOLS EXTRACTION AND BLACK TEA PRE-TREATMENT

4.1 Introduction

The focus of some previous research is to study the feasibility of using membrane filtration process for concentration and separation of polyphenols. Giving the high content of polyphenols in tea leaves and the absence of sugar, the polyphenolic compounds used in this study are extracted from black tea in purchased tea bags (Lipton, purchased from Coles Supermarket) which is largely consumed by local market comparing to loose tea leaves. All of the tea bags are from a large package to maintain the consistency of the tea property.

The colourless catechins which are the main polyphenols can be oxidized by the enzyme to a complex mixture compounds with the colours of orange-yellow to red-brown. The main tea catechins have been identified as epicatechin, which includes(–)-epigallocatechin (EGC), (–)-epigallocatechingallate (EGCG), (–)-epicatechingallate (ECG), and (–)-epicatechin (EC). In those epicatechins, (–)-EGCG has the strongest free radical scavenging ability in singlet oxygen-induced system [65]. It is also the most abundant catechin in green tea. In addition, it is a unique compound only occurring naturally in tea species so it is considered as the quality indicator of tea products. In this case, catechins are the main polyphenolic compounds that should be measured and identified in the following research.

In a work by Wang et al. [65], the stability of catechins in an aqueous system was studied and a mathematical model accounting for the simultaneous degradation and epimerization was established with (–)-EGCG as an indicator. The mathematical

model was validated by a reaction at 70°C. It revealed that (–)-EGCG decreased with increasing temperature and time while (–)-GCG increased. The higher the temperature, the faster the concentration of (–)-EGCG decreased (40°C to 100°C) and the quicker the concentration of (–)-GCG increased.

In the work reported by Zhang [21], the theaflavins extracted in 80% ethanol reached peak absorption when the brew temperature was 75°C. With temperatures higher than 75°C, absorption dropped due to the oxidation of theaflavins. In this case, 70°C was considered as a better brew temperature since it can preserve the diversity of polyphenols obtained. In the research presented by Shrikande et al. [67], polyphenols in grapes were extracted by hot water. Water temperatures of 140---210°F (60---100°C) were employed and he proved that a temperature range of 160---180°F (70---80°C) is more preferable.

The extraction of polyphenols from the dry tea is to provide polyphenol solution for the subsequent membrane filtration process evaluation. The extraction process and parameters are firstly evaluated to provide information of the amount of polyphenols in the solution for subsequent filtration. The standard extract process for this study will be established prior filtration in order to maintain the consistent feed supply for the membrane filtration process. The analytic method for evaluating the amount of polyphenols in the solution will also be evaluated.

While the extraction process aims to get the maximum amount of polyphenols in the solution, consistency of the extracted solution, simpler procedure are also considered important in evaluation of the extraction method. The main extraction method applied in this study is using hot water, while parameters such as temperature, infusion time, concentration of tea powder are evaluated to compare the amount of polyphenols extracted. Solvent constitute in the extraction solution was also evaluated and compared with the non-solvent solution.

In order to remove particulates that may be present in the solution, a pre-treatment procedure was applied by vacuum filtering the solution (cooled to the room temperature) using Whatman No.42 ashless filter paper with 2.5µm particle retention from the solution prior to the membrane filtration tests.

4.2 Extraction of polyphenols from black tea

Hot water infusion is the main method for polyphenol extraction from tea leaves for consumption, and various studies also applied similar method for extracting polyphenols. Jurisic et al. extracted polyphenols from plants at 70°C with application of solvent [41], while Largera et al. extracted tea polyphenols with 100°C boiling water [62]. Hot water of 80°C and 100°C were also used to extract polyphenols from black tea by Milou [58] which resulted in similar amount of polyphenols. This study evaluated both extraction with boiling water (100°C) and hot water at 70°C).

The infusion time is another factor that could affect the quantity and quality of polyphenols from tea leave. Astill et al. [68] reported the effect of brewing time of tea leaves on the amount of polyphenol extracted and observed that the amount of catechins extracted from tea bags increased with the infusion time from 30 seconds, 1, 2 and 5 minutes. Longer extraction time may result in more polyphenol extracted; however, oxidation of some constitute such as “the colourless catechins” are at the risk of being oxidized by the enzyme to a complex mixture of orange-yellow to red-brown compounds. Milou [58] compared the infusion time of 5 minutes and 10 minutes for the amount of polyphenols extracted and observed that higher polyphenols were detected in the 5 minutes infusion time.

Various studies have applied mixture of hot water with solvent to achieve higher extraction efficiency, for example, Naczek et al. used mixtures of water and solvents

such as methanol, ethanol, acetone to extract polyphenols from target materials and obtain higher extraction efficiency than pure water [30]. However, chemical residue can be a safety issue when solvents are involved in the extraction process [4].

4.2.1 Effect of extraction temperature

In these experiments, 1 g black tea was placed in 300 ml Milli-Q water of 70°C and 100°C (boiling water) for 5 minutes infusion time. The 70°C infusion temperature was maintained relatively constant by placing the beaker (containing tea solution) in a larger container filled with 70°C water. While the infusion with boiling water was relatively less constant since the infusion was exposed in room temperature.

The concentration of total polyphenol in the solution was analyzed with Folin-Ciocalteu method with the following procedure. 1 ml tea feed was mixed with 5 ml acetate buffer and 4 ml Milli-Q water. With 0.5 ml FC reagent, 33% Na₂CO₃ solution was added into 1 ml tea feed mixture until 10 ml obtained in the test tube (the sample preparation was specially explained in chapter 3.5.1).

The tea samples for UV-Vis spectrophotometry were made in 4 concentrations with 0.5, 1, 1.5 and 2 ml tea feed in the sample mixtures. The total polyphenol content in the black tea feed was examined by UV-Vis Spectrophotometer at 210 nm.

The average absorption with 3 readings and standard deviation (STDEV) calculated as error bar was used to compare the temperature effect on extraction. Figures 4.1.1 and 4.1.2 shows that the absorbance values in varied concentration units were linear with repeated tests (test 1 and test 2 are two separate solutions extracted with same condition) at 70°C and 100°C. By calculation, the total amount of polyphenols extracted at 100°C was approximately 20% higher than at 70°C, indicating that total polyphenols extraction increased with temperature. In Milou's report [58], the

amount of polyphenols (e.g. gallic acid) obtained at 80°C was around 35 (the internal standard was marked as 100) in the GC-MS peak height, and the results observed at 100°C was approximately 40. In this case, the amount of gallic acid extracted at 100°C was about 14% higher than at 80°C. In the study by Shrikhande et al., extraction temperatures was also investigated and it reported that temperatures over 100°C are detrimental to polyphenols, which can cause denaturation [67]. While more polyphenols could be extracted at 100°C, due to the possible oxidation of polyphenols and the relatively unstable temperature, it is still more practical to conduct the extraction at 70°C.

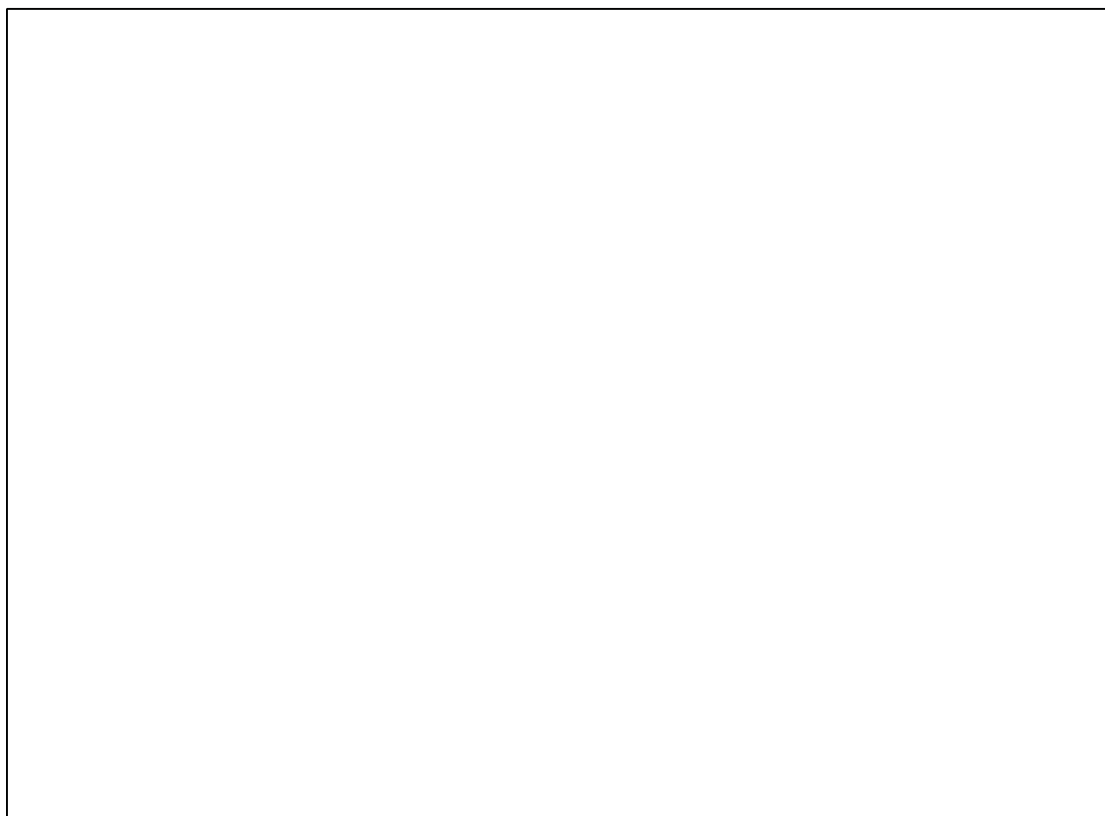


Figure 4.1.1: Absorbance values at 70°C versus concentration units.

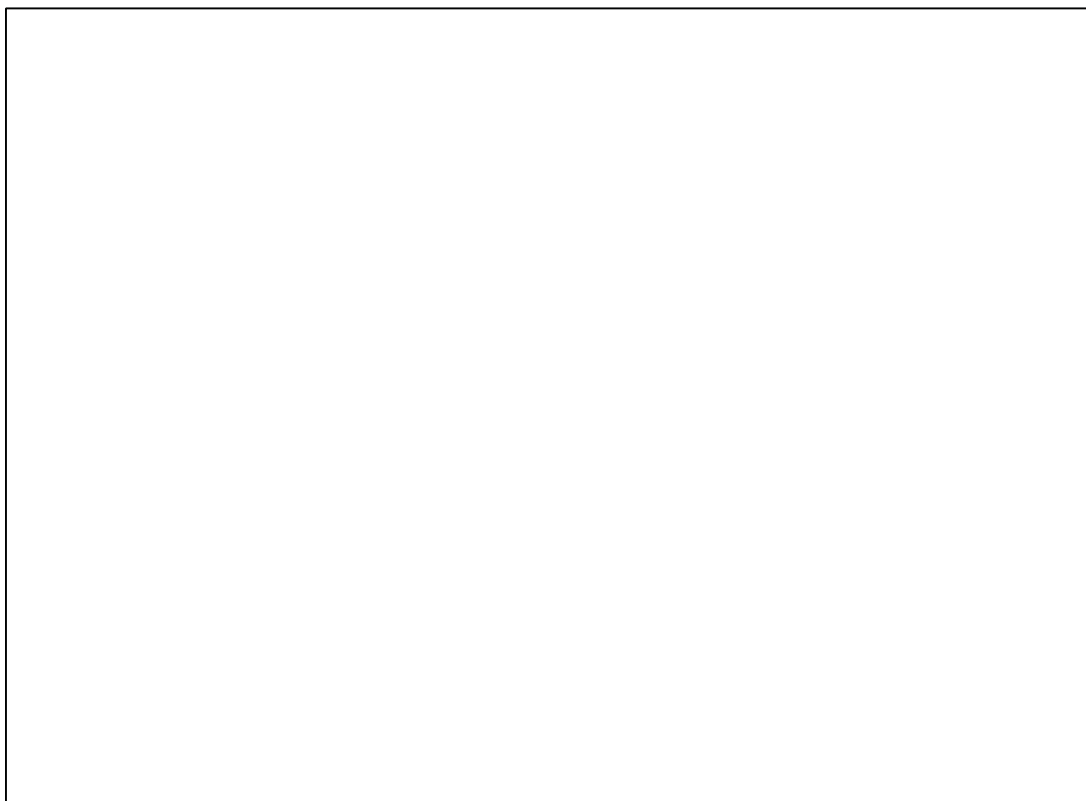


Figure 4.1.2: Absorbance values at 100°C versus concentration units.

4.2.2 Effect of infusion time

A previous colleague, Bouchra Milou [69], conducted 5 and 10 minute infusion tests and the compositions were analyzed by GC-MS. Following her protocol, 1 g black tea was infused in 70°C 300 ml Milli-Q water for 5 min and 10 min respectively before left to cool to the room temperature. The solution was vacuum filtered at the same condition as described previously. Following the same sample preparation procedure (described in Chapter 3, section 3.5.1), the total polyphenols in the solution was estimated.

The tea samples for UV-Vis spectrophotometry were made in 4 concentrations with 0.5, 1, 1.5 and 2 ml tea feed in the sample mixtures. Average readings were also calculated to compare the differences at two extraction times.

Total polyphenols were approximately 30% higher when infusion time doubled (the UV absorption of total polyphenols increased from 0.4242 to 0.5519 as the infusion time increased from 5 minutes to 10 minutes). In longer infusion time, more polyphenols such as flavanoids and phenolic acids can be soluted. However, oxidization can be much more severe after long exposure in a high temperature environment which results in the disproportionate increase in total polyphenols. Considering keeping the variety of tea polyphenols in feed, and 5 minutes was chosen as the optimum infusion time in order to minimize the oxidization of polyphenols as much as possible.

Astill et al. [68] reported the effect of brewing time of tea leaves on polyphenol extracts. The amount of catechins extracted from tea bags after 30, 60, 120, 300 second infusion were 22.1, 36.5, 62.9, and 90.5 mg/L respectively. When the brewing time increased from 2 minutes to 5 minutes (2.5 times), the catechins extracted increased by 48.9%. By calculation, if the brewing time doubled, the catechins amount would rise approximately 35.1%. This is close to the results gained from the effect of infusion time on total polyphenols extraction above (approx. 30% total polyphenol increase when the brewing time doubled from 5 minutes to 10 minutes).

In Milou's results [69] the chromatography of polyphenolic compounds infused in 5 minutes appeared more distinct than after 10 minutes. This result may be due to the fact that smaller amount of components of very low quantities no longer appeared. However, in this study, the analysis was focused on the total amount of polyphenols. In a heating system, degradation and epimerization will occur simultaneously on tea polyphenols, and produce catechins. For example, (-)-EGCG will epimerize to (-)-GCG, and transform from epi-structure to non-epi-structure [70]. In this case, the species of polyphenolic compounds may change by degradation and epimerization, but the higher extraction temperature can still obtain more polyphenols in total.

4.2.3 Effect of tea concentration

Previous study in evaluation of concentration of tea powder to extract polyphenols indicated that, the polyphenols analysis with the chromatography was clearer with a lower concentration (1 tea bag approx. 1.8 g compared to 2 tea bags approx. 3.6 g) [69]. Following the previous observation [10], two quantities of 1.6 g and 3.6 g of black tea were blended in 250 ml Milli-Q water at 70°C for 5 min.

The total polyphenols extraction tests were repeated two times and the UV absorption results are shown in Figure 4.1.3, below.



Figure 4.1.3: Effect of tea concentration in extracting polyphenols from black tea in repeated tests

The results indicated that the total polyphenols concentration was consistent with tea weight. In the three groups of tests, the absorption with 3.2 g of tea was approximately twice compared to the results with 1.6 g of tea.

From Milou's observation [58] of polyphenols in black tea sample by GC-MS, the gallic acid extracted from 2 tea bags was roughly 35 (the internal standard was

marked as 100), and the result obtained from 1 tea bag was approximately 30. The total amount of gallic acid just increased about 17% as the tea doubled. However, in the GC-MS graph, it showed that the number of peaks detected from the tea sample extracted from 2 tea bags were more than the peaks after 1 tea bag infusion. This result indicated that the amount measured for a specific polyphenolic compound may not rise significantly, however the species of polyphenols detected were increased and lead to a total amount of polyphenols increasing [69]. In this case, Milou's study can still support the results showed in Figure 4.1.3.

4.2.4 Effect of solvent constituent

Due to the organic nature of polyphenolic compounds, the black tea polyphenols are more soluble in organic solvents, such as ethanol and methanol, than in pure water. However, considering the simplicity and consistency of operation, pure Milli-Q water was compared with the solvent extraction method.

Comparing the amount of polyphenols extracted with pure water and 30% ethanol in water mixture at 70° C for 5 minutes, the polyphenols extracted with 30% ethanol mixed in extraction solvent was about 30% higher than pure Milli-Q water as shown in Table 4.1 (on page 73).

While the results indicate that extraction with organic solvent is higher than pure Milli-Q water, the hot water extraction process avoided the usage of solvent thus the avoided the solvent separation process. This study adopted the hot water extraction method.

Previous results showed that the amount of gallic acid extracted by the solvent with 50% ethanol was approximately 15% higher than 20% ethanol [58]. According to the review in Shi et al. paper, although extraction procedures of polyphenols with organic

solvents were efficient, further research diminished the use of organic solvents considering about the safety of human consumption [4].

4.2.5 Effect of vacuum filtration on the amount of total polyphenols extracted

After extracting 300ml of tea polyphenols at a temperature of 100°C with 5 mins of infusion, total polyphenols were measured for solutions before the vacuum filter and after filter process to compare the total polyphenols.

In Table 4.1 by calculation, it can be seen that the average UV absorption of extracted polyphenols after vacuum filtration was 8% higher than those without vacuum filtration. It may due to the impurities in feed were removed by vacuum filtration interfered with UV absorption. This result was averaged over three tests.

4.3 Pre-treatment of black tea feed

As fouling of membranes is a significant issue for the efficiency of filtration in membrane processes, pre-treatment to remove the particulates that may be present in the extracted solution and other large natural organic particulates has been treated as a polishing step before a more retentive membrane process such as nanofiltration [37]. In the process of preparing polyphenol solution extracted from black tea powder for nanofiltration process, pre-treatment was performed and studied with vacuum filter paper (Whatman, NO.42 filter paper, with 2.5 μ m particle retention) and ultra-filtration (UF) membranes before nanofiltration as a pre-treatment technique in this research.

4.3.1 Vacuum filtration as pre-treatment

The purpose of vacuum paper filtration was to remove suspended particulates that are present in the tea solution after extraction. Most of the dissolved matter passed easily through the filter paper. As shown in Figure 4.1.4, the vacuum filter paper used for filtering 300 ml of tea solution extracted from 1 g of tea powder with 70°C hot water was covered by a layer of brown impurities which would interfere the UV absorption results of tea feed and cause membrane fouling.

This brown substance consisted of large particles which existed in the original tea solution.



Figure 4.1.4: Photo of Whatman No.42 filter paper after vacuum filtration of 300 ml of tea solution extracted from 1 g of tea powder with 70°C hot water

4.3.2 Ultra-filtration as pre-treatment

Usually, ultra-filtration was considered as a proper method to separate and concentrate desirable compounds from solution and the UF membranes are able to retain compounds with the MWCO of 300-500,000 Da [4]. Ultra-filtration is commonly applied with a driving force of 2-10 bar [4]. In this study, to remove the

impurities and concentrate the polyphenols, ultra-filtration was used as a pre-treatment before nanofiltration.

In Evans et al. paper, both regenerated cellulose (RC) and fluoropolymer (FP) membranes were used to investigate on the clarification of black tea. RC membrane demonstrated less fouling tendency and stable theaflavins rejection [71].

In the extraction part, the amount of total polyphenols extracted in different conditions was analyzed by UV-Vis spectrophotometer at 210 nm. The absorption was used to study the concentration of polyphenols extracted.

The particle size retained by the vacuum filter paper used as pre-treatment is 2.5 μm , which is much larger than most of the polyphenolic compounds existing in black tea (MWCO of 200-1000). In this case, an ultra-filtration membrane with tighter MWCO was applied as the second step of pre-treatment to remove the other large sediment particles in black tea feed. The UF membrane also aims to remove dissolved high molecular weight molecules in the solution that has passed vacuum filter paper. Since those non-polyphenol large molecular weight molecules are more prone to foul the membrane and provide initiation sites for further fouling [72], the UF filtration process aims to reduce the membrane fouling and improve separation performance in the nanofiltration process.

Ultra-filtration as the second phase of pre-treatment after vacuum filtration

Three types of ultra-filtration membranes were selected and to be tested with vacuum filtered polyphenol solution. They were Regenerated Cellulose (RC) 1K Da, Regenerated Cellulose (RC) 5K Da, and Polyethersulfone (PES) 5K Da.

All the UF membranes were soaked in Milli-Q water overnight before filtration. The membrane initial water flux was measured before UF filtration of polyphenol

solution. In the UF filtration process, flux changed over time, and overall polyphenols retention by the UF filtration process was evaluated. Membrane fouling mechanisms on each membrane were also discussed.

✧ Flux change with different ultra-filtration membranes

Each UF membrane was pre-conditioned in Milli-Q water at 3 bar for 20 minutes and then the vacuum filtered black tea solution replaced pure water as feed with applied pressure at 5 bar for 2 hours. After the filtration with tea feed, backwash was applied on the fouled UF membrane at 5 bar, and then water flux of UF membrane was measured at 3 bar. The membrane initial water flux, flux at the end of UF filtration, membrane water flux after backwash and water flux after cleaning for all three UF membranes were summarized in Table 4.2.2. The UF flux variation over time is presented in Figure 4.2.1.

The total polyphenols in permeate of UF filtration with different membranes were estimated by using UV-spectrophotometer and presented in Table 4.2.4.

Selected membranes	UF 1K Da (RC)	UF 5K Da (RC)	UF 5K Da (PES)
Permeate of UF pre-treatment			Totally clear
Fouled membranes			

Table 4.2.1:Photos of permeate and fouled membrane after UF pre-treatment

Membrane L/m ² h	RC 1K Da			Avg $\pm$ STD	RC 5K Da			Avg $\pm$ STD	PES 5K Da		Avg $\pm$ STD
Initial water flux(3bar)	12.4	11.3	12.9	12.2 $\pm$ 0.58	32.5	28.3	32.6	31.1 $\pm$ 1.74	111.3	113.7	112.5 $\pm$ 1.73
UF filtration flux (5bar)	20.2	18.7	22.0	20.3 $\pm$ 1.17	41.2	41.5	43.2	42.0 $\pm$ 0.69	8.4	9.4	8.9 $\pm$ 0.71
Water flux after backwash(3bar)	10.3		10.8	10.6 $\pm$ 0.35	31.7	28.2	31.5	30.5 $\pm$ 1.39	22.5	22.5	22.5 $\pm$ 0
Water flux after chemical cleaning(3bar)		11.3							113.9	103.4	108.7 $\pm$ 7.43

Table 4.2.2: Flux change with different membranes in repeated tests



Figure 4.2.1: Flux change during UF pre-treatment with different membranes at 5 bar

	RC 1K Da	RC 5K Da	PES 5K Da	Tea feed
Absorption of total polyphenols in permeate (Abs) per	0.12	0.22	0.021	
	0.11	0.21	0.017	
	0.13	0.24	0.034	
			0.024	
Average	0.12 $\pm$ 0.01	0.22 $\pm$ 0.015	0.024 $\pm$ 0.007	0.38

Table 4.2.4: The absorption of total polyphenols (UV readings) in the pre-treatment

permeate with varied UF membranes

✧ Flux comparison of UF membranes

In the UF filtration tests, all the three UF membranes were applied at 5 bar and the permeate flux was recorded. RC 5K Da showed the highest permeate flux (41.2 L/m²h), and RC 1K Da was 20.2 L/m²h. Although PES 5K has the highest initial water flux (111 L/m²h), its flux during tea filtration was just 8.4 L/m²h (see Table 4.2.2).

The polyphenol concentration measurements in the permeates indicates that for RC membranes, the concentration in the permeate after UF filtration with larger molecular weight cut off (5K Da) membrane was approximately 1.83 times of the permeate after filtration with UF 1 K membrane. Furthermore, polyphenol concentration in the permeate of PES 5K Da membrane also showed quite high retention rates of polyphenols (0.024), which was just 10% of the concentration in the permeate with RC 5K Da and 6% of feed (see Table 4.2.4). The rejection of RC 1K Da and RC 5K Da were 68.4% and 42.1% respectively.

Integrating all the observations, the RC 5K Da membrane showed the most efficient filtration performance with the fastest flux rate. The PES membrane experienced the slowest flux which produced the least efficient pre-treatment. Besides, PES 5K Da experienced the highest polyphenols rejection of 93.7%, and in this case, the permeate (feed for nano-filtration) obtained few polyphenols to determine in the following experiments.

Considering the same molecular weight cut off of RC 5K Da and PES 5K Da, the significant difference in flux of tea filtration is due to the different membrane material. Some RC membranes have the larger pore distribution comparing to PES membrane which may contribute to the higher tea flux with RC 1K Da and 5K Da. According to Lee et al. paper, the permeability of RC membrane is three times of PES

membrane with the same molecular weight cut off [73].

✧ Investigation of UF membrane fouling by cleaning

In addition to reducing flux rates and selecting membrane material after giving close attention to coagulation and absorption characteristics, pre-treatment is characterized as a third effective strategy to reduce fouling [37]. As shown above in Table 4.2.2, all three types of UF membrane used for pre-treatment experienced flux decline during tea filtration in varying degrees. To investigate the mechanism of fouling on UF membranes, both physical and chemical cleaning tests were conducted on the fouled membranes.

✧ Backwashing UF membranes

Backwashing is an efficient physical cleaning method to remove the fouling cake on the membrane surface. In this study, the backwashing was conducted by pure Milli-Q water after filtration for approximately 20 minutes. After backwashing, the water flux of the cleaned membranes was measured again. If the fouled membranes regained most of the original flux rate, it means that the dominant fouling source is cake accumulation. If not, chemical cleaning will be applied next.

Membrane L/m ² h	RC 1K Da		RC 5K Da			PES 5K Da	
Initial water flux(3bar)	12.4	12.9	32.5	28.3	32.6	111.3	113.7
Water flux after backwash(3bar)	10.3	10.8	31.7	28.2	31.5	22.5	22.5
Recovery rate%	83.1	83.7	97.5	99.6	96.6	20.2	19.8

Table 4.2.5: Flux recovery after backwash

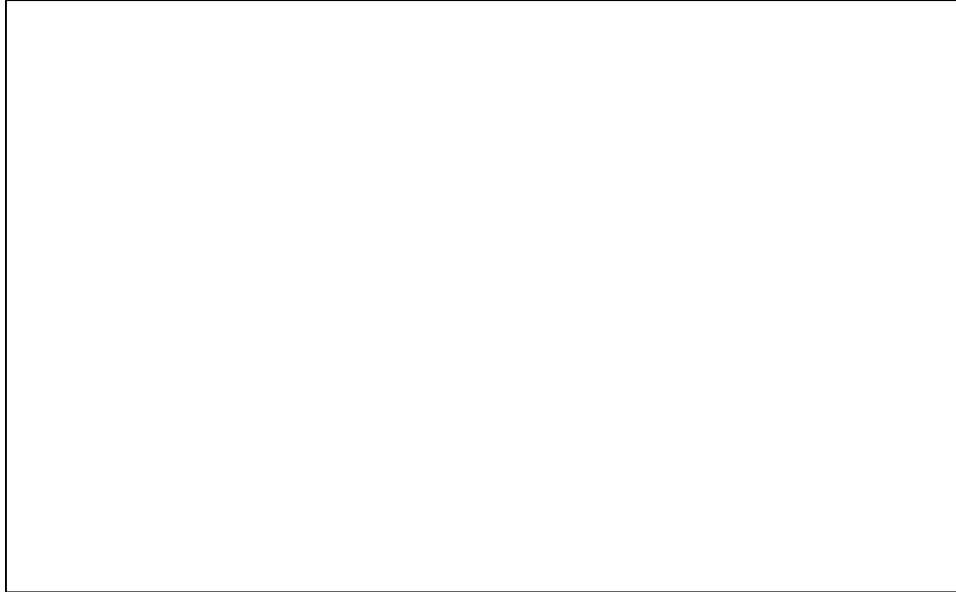


Figure 4.2.2: Water flux change after the backwash cleaning protocol

As shown in table 4.2.5 and figure 4.2.1, both of the RC membranes obtained high flux recoveries after backwash reaching approximately 83% (RC 5K Da) and 98% (RC 1K Da). However, the recovery of the PES membrane after this process was relatively poor (20%). After the backwashing, the membrane surface of PES 5K Da still showed significant cake fouling.

The fouling of the RC membranes was a result of cake/gel layer and standard blocking, because most of the major polyphenolic components in black tea have a MWCO less than 1000 Da which could be easily removed by backwashing. However, in the UF 5K Da (PES membrane), standard blocking was not the main reason for its low permeate flux (8.9 L/m²h) and low flux recovery (20%). The photo of the UF 5K Da membrane, taken after pre-treatment and backwashing, shows an obvious brown cake layer on the membrane surface. It indicates that the adhesion force of the foulant to the PES membrane was stronger on RC membrane [33]. The foulant layer which caused the flux decline on the UF 5K Da (PES) membrane may be due to concentration polarization while the standard blocking just contributing about 20% of the fouling on PES 5K Da membrane. This phenomenon may be caused by the membrane material (polyethersulphone) having more hydrophobic and adsorptive

for polyphenols in black tea.

In Lee et al. research, backwash experiments were applied on different UF membranes (RC membrane and PES membrane). After the backwash, RC membrane experienced the highest flux recovery, which can be attributed to the removal of the cake/gel foulant layer from both the membrane surface and inside of the pores. On the other hand, the foulant cannot be removed by backwash is due to strongly adherent pore blockage [73].

Weis et al.[74] have investigated the influence of different UF membranes fouled with spent sulphite liquor. Their research showed that both the PES and RC membranes showed decreased performance after fouling and cleaning which was indicated by the foulant peaks using the FTIR. However, the RC membrane experienced less significant change than the PES. The results reveal that the rougher, hydrophilic RC membrane had a greater tendency to resist adhesion than the smoother but more hydrophobic PES membrane. This tendency also proves the difficulty of cleaning the PES 5K Da membrane compared with the two RC membranes.

This trend is also shown in Puro et al. research [32] on the performance of RC and PES UF membranes in the filtration of pulp mill process waters. In their study, the two PES UF membranes, UH050P (MWCO of 50K Da) and UH030P (MWCO of 30K Da) experienced much more fouling than the RC UF membrane UC030 (MWCO of 30K Da).

✧ Effect of RC 5K Da membrane as pre-treatment for the nanofiltration process

Combining all the results obtained above, the RC 5 K membrane was chosen as the UF pre-treatment membrane due to its high efficiency and easy-to-clean properties. After ultra-filtration with the RC 5 K, the permeate was collected as feed for use in

the nanofiltration step, to investigate the effect of ultra-filtration in reducing the impurities in tea feed and on NF performance.

Figure 4.2.4 shows that the final permeate flux of tea feed with UF pre-treatment produced a similar result as the flux without UF which were both around 25 L/m²h. In Figure 4.2.5, the average and standard deviation of initial tea flux of NF 270 at 5 bar was 26.9 ± 6.4 L/m²h. In this case, it seems that the UF pre-treatment contributed little to reduce the fouling of the tea feed in the nanofiltration step. However, the initial permeate flux with UF pre-treatment was lower and the flux remain slightly more stable during the 2 hours filtration while the final flux of NF without UF was about 70% of the initial one. The steady permeate flux during nanofiltration indicates that the fouling did not accumulate significantly on the membrane surface, comparing to the obviously flux decline without UF pre-treatment. It means that the UF as pre-treatment improves NF performance by reducing membrane fouling.

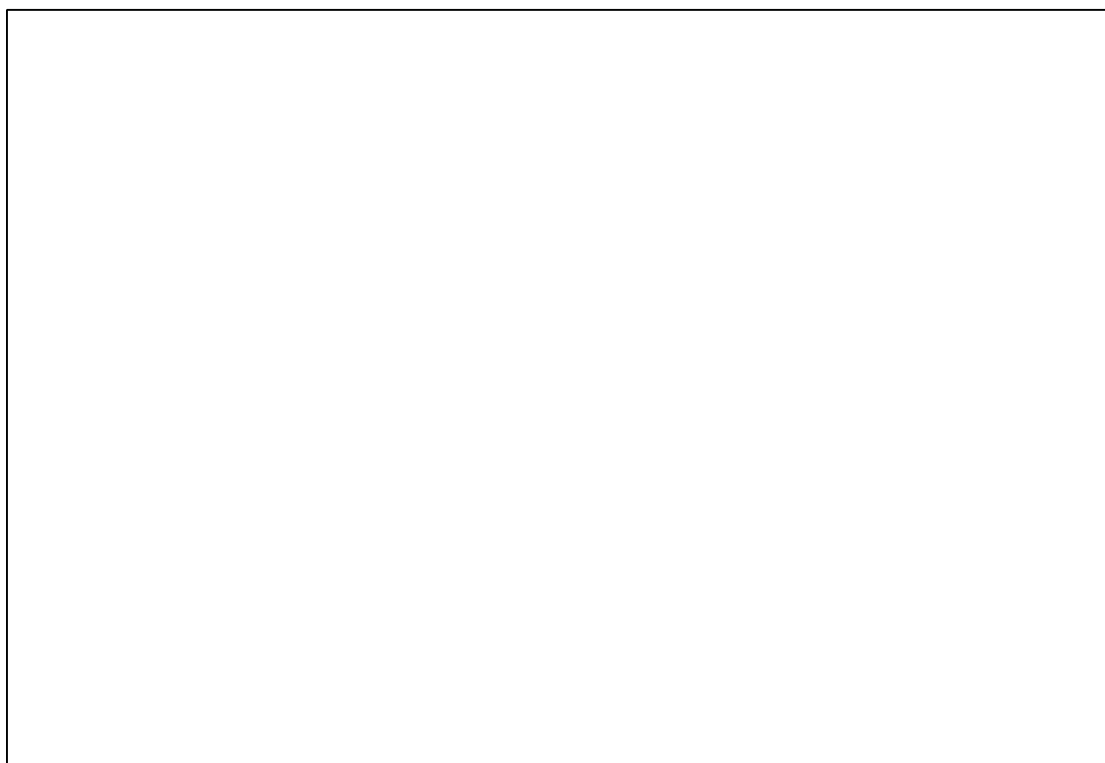


Figure 4.2.4: Comparison of NF flux with/without UF pretreatment



Figure 4.2.5: Initial flux of NF 270 with tea feed at 5 bar (without UF pre-treatment)

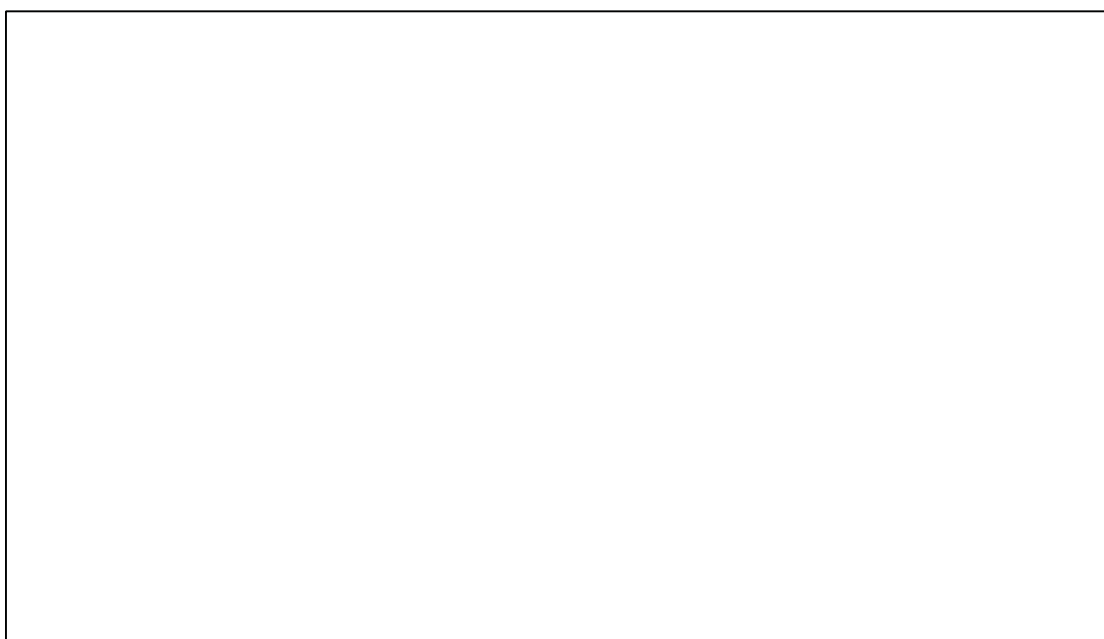


Figure 4.2.6: Polyphenols concentration ratio in UF 5K Da

In figure 4.2.6, the polyphenols concentrations were expressed as the relative concentration (concentration ratio) compared to the feed. The permeate concentration ratio was 0.74 (the UV reading of feed is defined as 1) while the retentate ratio was 1.19. The standard deviations of feed, permeate and retentate were 0.044, 0.045, 0.063 respectively.

Ultra-filtration as the sole pre-treatment before nanofiltration without vacuum filtration

In this part of the experiments, the UF membrane (RC 5K Da) was used as the only pre-treatment before nanofiltration to investigate its efficiency without vacuum filtration.

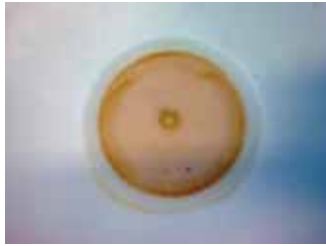


Figure 4.2.7: Photo of fouled UF membrane (RC 5K Da) after 2 hours filtration

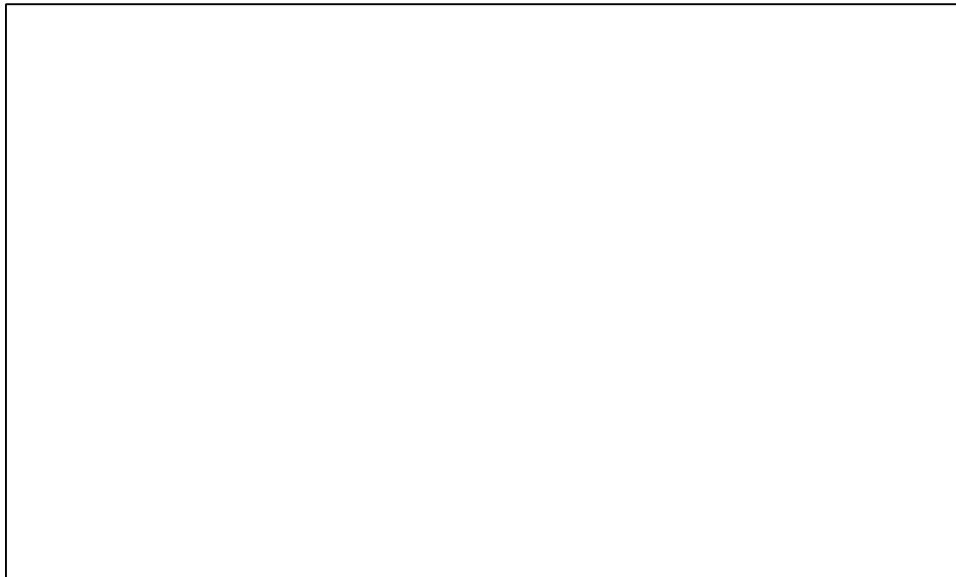


Figure 4.2.8: Flux changes during UF (RC 5K Da) at 5 bar pre-treatment in repeated tests



Figure 4.2.9: Comparison of permeate flux change in NF 270 after varied pre-treatments



Figure 4.2.10: Comparison of permeate flux of tea feed in UF with/without vacuum filtration

After 2 hours ultra-filtration, the permeate flux was around 33 L/m²h (figure 4.2.8). Figure 4.2.9 shows the flux changes in nanofiltration after varied pre-treatments. Although the final flux of UF pre-treatment was quite close to the results of vacuum and vacuum & UF pretreatment, Figure 4.2.9 indicates that the initial flux of UF pre-treatment alone was lower than the flux of vacuum pre-treatment (27 L/m²h compared to 35 L/m²h), which may be due to the membrane variations. However, in the 2 hours nanofiltration, the permeate flux after UF pre-treatment was steady while the flux of vacuum filtration alone experienced a significant drop from 35 to 25 L/m²h (approximately a 30% decrease). It means that UF pre-treatment has allowed the fouling accumulation on the membrane surface to be controlled. Considering flux change, applying UF alone for pre-treatment has been proven useful for removing impurities that may cause severe fouling of the NF membrane.

Also, the changes of permeate flux in ultra-filtration with/without vacuum filtration were measured. As can be seen in Figure 4.2.10, the initial permeate flux in both cases was around 50 L/m²h. After 2 hours ultra-filtration, the final flux with UF pre-treatment alone experienced more obvious decline to about 78% of the flux with

vacuum & UF pre-treatment.

Lastly, the absorbance value of total polyphenols was also measured by UV-Vis spectrophotometer, the polyphenols rejection being 74.7% calculated by mass balance.

4.3.3 Membrane cleaning

In the next phase, the PES membrane will be exposed to a chemical cleaning to remove the particulate fouling of the membrane pores.

1) Chemical cleaning for PES 5 K Da membrane

Weis et al. [74] described how two materials were applied to the fouled ultrafiltration PES membranes which were 0.1% wt Ultrasil 11 solution and 0.5% wt NaOH solution respectively. The membrane cleaned by NaOH produced a greater foulant peak height (by FTIR) than the one with Ultrasil 11. The result indicates that Ultrasil 11 also left a surfactant layer on the membrane surface, which affected the membrane performance.

Due to the low flux recovery after backwashing with Milli-Q water, chemical cleaning was applied to PES membrane to investigate the fouling mechanism more thoroughly. A 0.01 M NaOH solution ($\text{pH} \approx 12$) was applied to the fouled PES membrane with stirring but without additional pressure for approximately 30 minutes.

Permeate after chemical cleaning	UF 5K Da (PES) membrane after chemical cleaning
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Table 4.2.6: Permeate after chemical cleaning and cleaned UF 5K Da (PES) membrane



Figure 4.2.3: Water flux recovery with different cleaning protocols in UF 5K Da (PES) membrane

After chemical cleaning, the PES membrane was rinsed with Milli-Q water and was measured again after applying water flux at 3 bar pressure. As seen in Table 4.2.6, the color of permeate obtained during pure water filtration was light brown, which was attributed to the cake accumulated on the membrane surface despite sufficient cleaning with NaOH solution. Figure 4.2.3 shows that the water flux after chemical cleaning was 113.9 and 103.4 L/m²h, and the recovery rates were 102.3% and 90.9% separately. The data for backwash and chemical cleaning in Figure 4.2.3 are the average of 2 tests.

Although this UF membrane has a MWCO of 5K Da which is considerably larger than most of the polyphenolic compounds in black tea, it still shows significant fouling

from black tea. This result shows that the main reason for severe flux decline for the PES5K Da membrane may be attributed to concentration polarization on the membrane surface and pore blockage. The PES membrane showed a less roughness than RC membrane in AFM images [74], and on this basis we would expect PES should exhibit a higher flux and less flux decline [73]. However, the results obtained during black tea filtration in this study showed opposite observation. This phenomenon may due to the more hydrophobic membrane material and strong foulant adhesion force to the membrane surface. Polyethersulfone is relatively hydrophobic membrane material compared to regenerated cellulose, which means its lipophilic property is stronger. In addition, as mentioned in chapter 2, many polyphenolic compounds found in tea are aromatics with benzene rings, which means these compounds may be prone to adhere to PES membrane than RC membrane. In Lee et al. investigation, the feed is river/lake water which likely contains humic acid foulants in comparison to components in black tea [73]. In light of the factors mentioned above, sodium hydroxide solution with stronger cleaning ability than backwash is capable to remove the foulants adhered on PES 5K Da and achieve a high flux recovery.

4.4 Conclusion

4.4.1 Summary of extraction methods of tea polyphenols

The standard extraction conditions selected for subsequent filtration tests are: temperature of 70°C, infusion time of 5 min, solvent of pure Milli-Q water, tea weight of 1.0 g, and feed volume of 300 ml. However, the temperature of 70°C is not strict stable, and the flask with tea feed is surrounded by hot water bath, which can keep the temperature at 68-70°C.

Abs Reading NO.	Temperature/°C		Time/min		Tea weight/g		Solvent		Vacuum filtration	
	70	100	5	10	1	2	Water	30% ethanol	✓	×
1	0.41	0.54	0.41	0.57	0.41	0.38	0.41	0.63	0.54	0.42
2	0.47	0.52	0.47	0.54	0.47	0.44	0.47	0.55	0.52	0.46
3	0.40	0.50	0.40	0.55	0.40	0.41	0.44	0.58	0.50	0.48
Average	0.43	0.52	0.43	0.55	0.43	0.41	0.43	0.59	0.52	0.45

*Table 4.1: Summary of results and extraction parameters (UV readings)
(✓: with vacuum filtration, x: without vacuum filtration)*

In summary (Table 4.1), the comparison of the effects with various extraction methods were presented. A temperature of 70 °C and 5 minutes infusion time was chosen bearing in mind the likelihood of oxidation of the theaflavins. Because the polyphenols' extraction did not increase with increasing tea concentration, 1 g black tea was used for infusion in 300 ml Milli-Q water. Comparing solvents, although 30% ethanol was more effective for extracting polyphenols, pure Milli-Q water was chosen to avoid chemical residues. In addition, vacuum filtration was also proven to be an efficient way to remove the large particles in the tea solution. In this case, vacuum filtration was used as a pre-treatment before nano-filtration was performed.

4.4.2 Evaluation and summary of different pre-treatments

From the results obtained above, the research can conclude that ultra-filtration with/without vacuum filtration affords only slightly better membrane performance in nanofiltration with less fouling accumulation. However, the UF membrane can be cleaned and recycled by simple backwash to regain most of the membrane's performance, which can reduce flux drop in limited capital expense. Other factors also have to be included, for example, processing time and total polyphenols obtained.

The rate of tea feed processing by UF membrane is much slower than vacuum filtration. The 300 ml original tea solution can be filtered completely through vacuum filter paper in about 3 minutes (approximately 6000 L/m²h). However, it takes approximately 2 hours to obtain roughly 100 ml permeate (approximately 45 L/m²h) with a pressure of 5 bar in ultra-filtration which limits the productivity of the following nanofiltration and the process of concentrating polyphenols.

Chapter 5 SEPARATION PERFORMANCE AND CLEANING OF NANOFILTRATION MEMBRANE

5.1 Introduction

Membrane characterization by monitoring the flux declines and rejection changes is a traditional method applied in fouled membrane. This characterization reveals broad picture of the fouling degree on the membrane surface and in the membrane pores. As mentioned in chapter 2, most of the polyphenolic compounds found in black tea have a MWCO ranging from 100-1000 Da, and two types of nano-filtration membranes NF 270 and NTR-7450 were applied in this research to concentrate the polyphenols in black tea feed.

Nano-filtration has been used to separate and concentrate polyphenolic compounds in food industry, including juice, wine, and tea. Most of the previous concentration studies monitored flux and rejection changes to characterize the process and efficiency. Similar research about concentration of polyphenols from apple juice by nano-filtration membrane has reported the effects of varied operating parameters [75].

In Saleh et al. paper, 1K Da MWCO membrane was applied in the nano-filtration to separate and concentrate polyphenolics from apple juice. Variable operation conditions were tested in Saleh et al. study: temperatures from 30 to 50° C, pH from 2 to 5, trans-membrane pressures from 5 to 30 bar [75]. However, apple juice is different from tea juice because of the existence of pulp and sugar, which are not present in black tea. In addition, the membrane used in this previous study is spiral wound membrane in a pilot scale rig (cross flow). In this chapter, flat sheet

membranes (NF 270 and NTR-7450) were conducted in a stirred-cell rig to concentrate polyphenolic compounds from black tea feed. Thus, filtration time and stirring rate have been extensively investigated in the following experiments.

Membrane cleaning is another part of investigation in this chapter. Flux recovery and rejection after cleaning have been used primarily when evaluating cleaning efficiency [45]. Variable cleaning methods are reported in previous research, and chemical cleaning is an effective and the most common way to remove the foulants and maintain and restore most of the characteristics of NF membrane [45].

In previous studies, proprietary formulated alkaline cleaner or NaOH solution were carried out to remove the organic foulants on the polyamide NF membrane [76, 77]. NaOH solution at 0.05% (w/w) (pH=12) was used to clean the UF membrane fouled with kiwifruit juice in Cassano et al. paper [78], and more caustic NaOH solution (0.1 M) was carried out to remove the foulants on UF membrane according to Wei's research [39]. Cleaning solutions with pH ranging from 2 to 12.2 were tested on NF 255-400 and were evaluated by flux recoveries and membrane retention changes, and alkaline chelating cleaning agents were most efficient [45]. In addition to flux, streaming potential is another common method to apply to characterize the membrane properties before and after fouling [45].

In this study, two different concentrations of NaOH (0.1 M and 0.01 M) were applied to understand the mechanism of chemical cleaning on NF membranes (NF 270 and NTR-7450). Surface charges of virgin and fouled membranes were evaluated by Analyser (Sure Pass, Anton Parr, see in Figure 3.5).

To analyze the polyphenols in NF process, different analytical methods were carried out in previous study. UV-Vis spectrophotometric procedure with Folin-Ciocalteu's reagent is widely used to determine the amount of total polyphenols [21, 41, 47].

With the results obtained by UV-Vis spectrophotometer, mass balance was calculated the polyphenols distribution in NF process. To measure the specific composition of samples, GC-MS [58, 79] and CE (see Table 2.6.2) were applied.

5.2 Nanofiltration performance

In this part of the research, the process of polyphenols concentration from black tea solution was studied with different NF membranes and varied operational conditions. Regarding membrane processing, membrane fouling, membrane performance and the composition of the foulants were also investigated. Each nanofiltration membranes are pre-conditioned before tea filtration.

5.2.1 NF membrane selection

In the literature review (chapter 2), it was noted that the molecular weights of the main polyphenols in black tea were below 300 Da (Table 2.3 and 2.4). Traditional synthetic polymer membranes were used in this research due to their advantage compared to ceramic membranes.

Previously NF 270 has been proved to have both high retention and high permeability, and in addition, NF 270 can withstand alkaline and acidic cleaning and mechanical wear [66]. The fouling and retention of NTR-7450 also has been tested with FeCl_3 , vanillin and humic acid [80]. Membrane NTR-7450 showed also high retention to FeCl_3 at pH ranging from 4-10 [80].

Zhang et al. [21] investigation of optimum membranes and operating parameters to concentrate green tea suggested that polyamide membranes with molecular weight of 31,100 achieve very high sodium chloride separation (above 95%), and also

indicate high separation for all tea juice components. The average retention for polyphenols and caffeine are 98.18% and 95.58% respectively. In Zhang's study, the tea polyphenols separation using the PES membrane with molecular weight of 100,000 is lower at 57.78%. In Manttari et al. study, the pure water permeability of different NF membrane were tested and the NF 270 membrane demonstrated both relatively high retention (90%) as well as a high permeate flux (125 L/m²h) comparing to many other membranes [81].

Considering the membrane pore size in Table 2.6, a nanofiltration membrane was chosen to perform the filtration research. Both NF 270 membrane and NTR-7450 membrane were tested and compared. NF 270 membrane is a type of polyamide thin-film composite membrane, while NTR-7450 is a sulphonated polyethersulphon (SPES) membrane.

5.2.2 Effect of trans-membrane pressure (TMP) on filtration flux with nanofiltration membrane NF270

In Table 2.3.1 (chapter 2), the pressure tolerance for the nanofiltration membrane was established as ranging between 5 to 20 bar. In Milou's research on concentrating black tea, 9 bar pressure was applied [58]. In Saleh et al. results of the effect of trans-membrane pressure ranging from 5 bar to 30 bar, the rate of fouling deposition increased more significant than the rate of phenolics content in retentate when pressure raised from 20 bar to 30 bar, compared to the data obtained at 5 to 10 bar [75]. Based on Saleh's research, trans-membrane pressures below 10 bar shows acceptable retention and fouling deposits with less driving force. In this study, experiments at 3 bar, 5 bar, 8 bar, and 10 bar were conducted and then compared.

The vacuum filtered black tea solution was filtrated at 3 bar, 5 bar, 8 bar and 10 bar separately using NF 270 membrane, for approximately 2 hours.

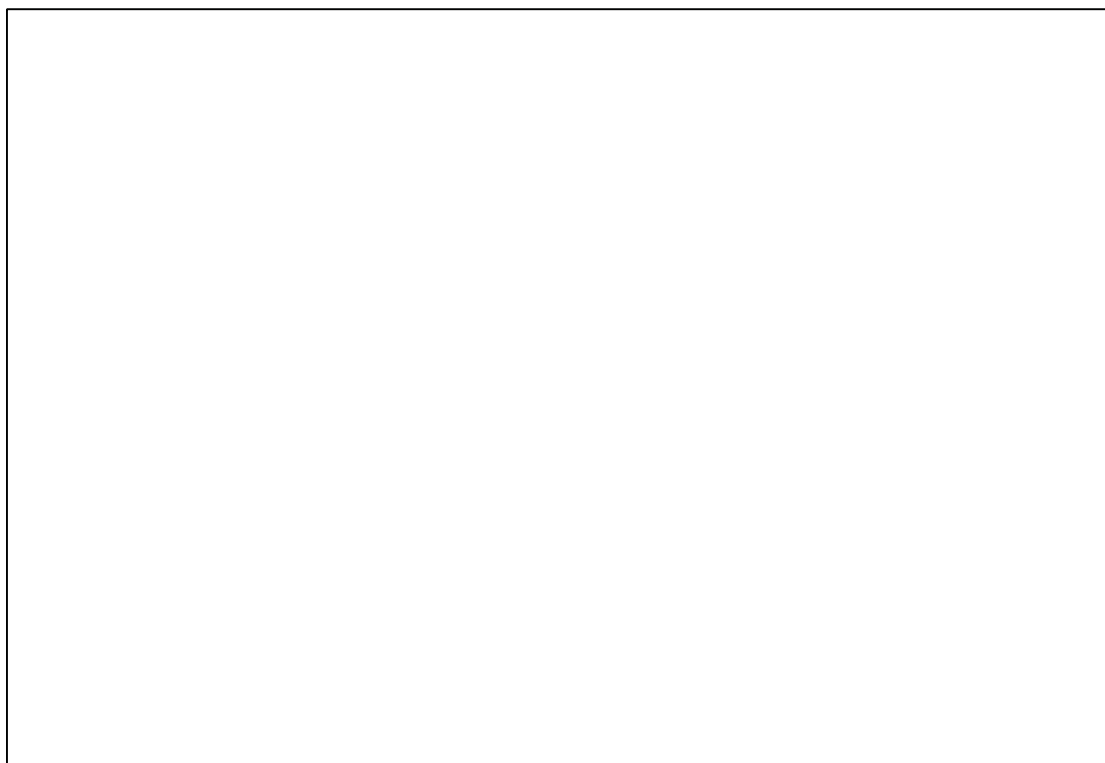


Figure 5.1.1: Comparison of flux at varying pressures

With the applied pressures ranging from 3 bar to 10 bar, the permeate collected were 52 ml, 90 ml, 118 ml, and 112 ml respectively. From Figure 5.1.1, it is clear that flux was increased as pressure was increased from 3 bar to 8 bar. However, the slopes of the flux at the beginning and end of the filtration were different at varying pressures. At 3 bar, the flux stayed steady at 16.8 to 14.5 L/m²h during the filtration. At 5 bar, the flux dropped from 34.9 L/m²h to 22.5 L/m²h, while at 8 bar, the initial flux was around 53.7 L/m²h and decreased gradually to approximately 28.1 L/m²h in 2 hours.

From 3 bar to 5 bar and then 8 bar, the percentage of pressure increasing was roughly 1.67, and the ratios of the initial flux at 3, 5, and 8 bar were 2.07 and 1.53. It is possible to say from these results that when pressure is increased from 5 bar to 8 bar, the flux did not increase proportionally, compared to the flux rising from 3 bar to

5 bar. It may be that the higher pressure of 8 bar intensified the degree of concentration polarization on the membrane surface, which slowed down the initial permeate flux of tea filtration and can be reversed by backwashing.

After 2 hours' filtration, the flux decline ratio for the final flux was 1.55 from 3 bar to 5 bar and 1.25 from 5 bar to 8 bar. Compared with the increasing ratio of the initial flux, this result indicated that the final flux level increased relatively consistently with rising pressure. It may be due to the layer of fouling which has a dominant effect on the permeate flux change during the 2 hours' tea filtration, the pressure issue being the lesser factor.

When the pressure rose to 10 bar, the permeate flux decreased initially by almost 29.2% for the first 20 to 40 minutes and did not show significant difference to 8 bar, which decreased from an initial 53.6 to final 38.1 L/m²h with a flux reduction of 29.1%. This result shows that when the pressure is too high there is little increase in permeate flux. On the contrary, the fouling polyphenols and other high molecular weight components in the vacuum filtered tea solution become markedly more severe, counterbalancing the potential increase of flux by raising the pressure. This observation of flux increase with increase in TMP to a certain level followed by flux reduction when TMP is further increased is also common in a number of UF and MF filtration processes [24].

In addition to the flux tests, a UV-Vis spectrophotometer was applied to measure the polyphenol concentrations of the permeate, retentate and feed to investigate membrane performance. It is measured by a membrane rejection figure. In addition, the error of mass balance is calculated as below:

	Rejection	Mass balance error
3bar x 2hr	0.9	29%
5bar x 2hr	0.86	8.50%
8bar x 2hr	0.86	15%

*Table 5.1.1: Mass balance of tea polyphenols at varied pressures
(*Mass balance is calculated as volume (ml) ×UV absorption.)*

Comparing the quantities of polyphenols in the feed, permeate and retentate, the difference between the expected amount and measured amount by UV-Vis spectrophotometer is shown in Table 5.1.1. The rejection rates of membrane NF 270 at 3 bar, 5 bar, and 8 bar were comparable at 90%, 86%, 86% respectively (Table 5.1.1). The data here suggests that over the span of 3-8 bar, membrane NF 270 continues to perform well in achieving a high rejection rate. The reasons of the mass balance errors may be caused by the measurement error or accumulation of polyphenols on the membrane surface. When the volumes of feed, retentate and permeate were measured, part of liquid may be retained in the stirred-cell or bleaker and affect the mass balance results. In addition, the absorbances obtained by UV-Vis spectrophotometer also experienced variation. In Milou's research, it was found that the accumulation of foulants on membrane can also cause the mass balance difference [58].

5.2.3 Effect of filtration time on filtration flux with nanofiltration membrane NF 270 at 3 and 5 bar

Given that the stirred-cell used in these experiments has a volume of 300 ml, the filtration time was set at 2 hours and 4 hours. Two different pressures (3 and 5 bar) were applied to the feed solution. Both the flux change and polyphenol concentration in feed, permeate and retentate were observed with tests of two

individual virgin NF 270 membranes.



Figure 5.1.2: Comparison of permeate flux of NF 270 at 2 hr and 4 hr at 3 bar

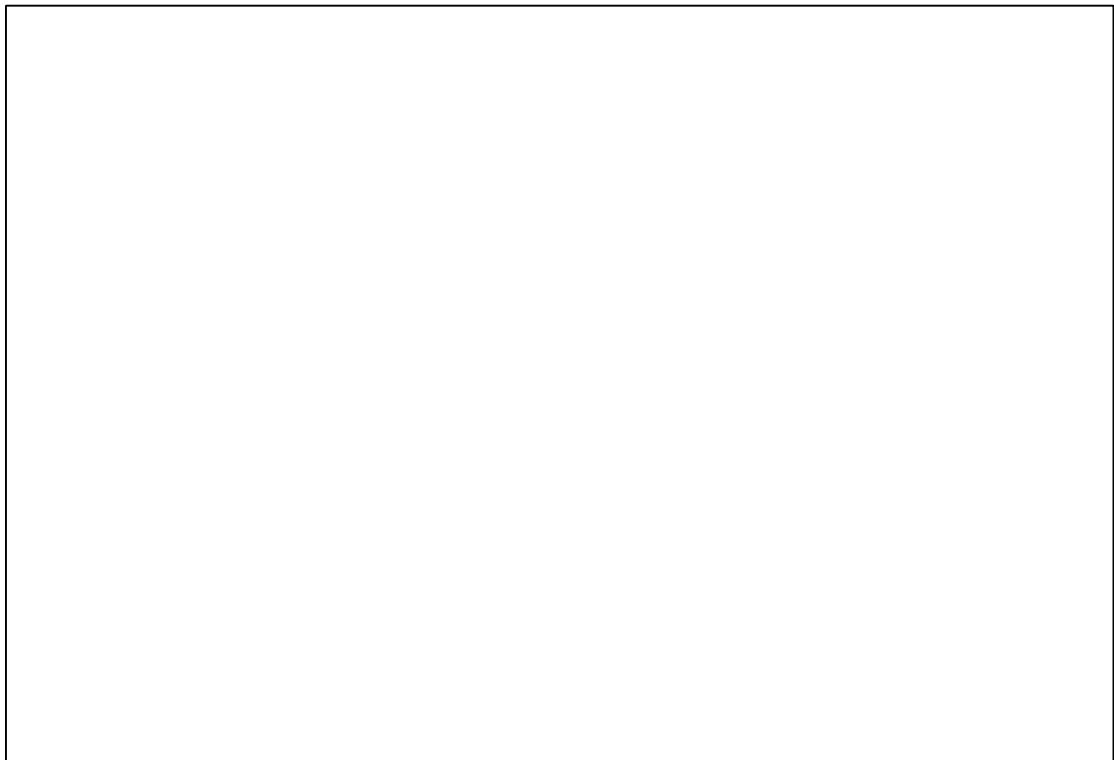


Figure 5.1.3: Comparison of permeate flux of NF 270 at 2 hr and 4 hr at 5 bar

At 3 bar, in the 2 hours' filtration, the permeate flux decreased from 17.2 L/m²h to

14.5 L/m²h (a 15.7% reduction) and in the 4 hour filtration, the flux rate dropped from 17.8 L/m²h to 11.5 L/m²h (a 35.4% reduction - see Figure 5.1.2).

From this analysis of 4 hours' filtration, it shows that in the first 2 hour filtration, the flux decreased by about 14.7% and in the next 2 hours, the flux rate dropped by another 20.7%. This phenomenon may be caused by the increasing concentration polarization on the membrane surface, because with longer processing time, the volume of tea solution in the stirred cell decreased and the concentration of the tea solution rose. Severe concentration polarization leads to reduce of permeate flux with time as suggested in previous studies [39].

At 5 bar, in the 2 hours' filtration, the permeate flux decreased from 35.0 L/m²h to 22.5 L/m²h (a 35.7% reduction) and in the 4 hour filtration, the flux rate dropped from 31.2 L/m²h to 16.9 L/m²h (a 45.8% reduction - see Figure 5.1.3). We can find that the flux dropped about 35.7% in the first 2 hours and 10% in the next 2 hours, which means that almost 80% flux decrease was occurred in the first 2 hours. This result was significantly different from the data obtained at 3 bar pressure. In this case, the issue of increasing pressure increasing may be the leading cause of the difference. At 5 bar, fouling accumulated quickly at the beginning of filtration due to higher pressure. It led to a fast decrease in the first 2 hours and then, fouling on the surface of the membrane achieved an equilibrium with the pressure which should lead to a higher permeate flux compared to 3 bar. The final flux of 5 bar is 1.47 times the value measured at 3 bar which is lower than the pressure increase ratio (1.67).

Similar effect of trans-membrane pressure was also found in Saleh et al. research, and the fouling increased more dramatic at higher pressure (20-30 bar) comparing to lower pressure (5-10 bar). This phenomenon can be explained that thicker and denser fouling layer of rejected solutes was compressed by higher pressure [75].

	Rejection	Mass balance error
5bar x 2hr	0.86	8.50%
5bar x 4hr	0.79	18.00%
5bar x 6hr	0.86	5%

*Table 5.1.2: Mass balance of tea polyphenols absorption at varied filtration time
(*Mass balance is calculated as volume (ml) ×UV absorption.)*

The UV absorbance of the feed, permeate and retentate in these two situations was also tested. Data for 4 hours filtration obtained a rejection rate of 79%, while the rejection rates of 2 hours and 6 hours were both 86% (Table 5.1.2). From the data above, the rejection rate did not show significant different with filtration time. It may due to the filtration time ranging from 2 hours to 6 hours did not affect the concentration polarization of feed severely and reached a equilibrium at the beginning of the filtration.

5.2.4 Effects of stirring rate on filtration flux with nanofiltration membrane NF 270 at 5 bar

Theoretically, permeate flux can be influenced by the stirring rate due to the turbulence caused by the stirrer on the membrane surface. With higher stirring rates, shear rate near the membrane surface should help to prevent membrane fouling thus achieving a higher flux. In this part of research, various stirring rates were tested on the NF 270 membrane at 5 bar to investigate the effect of turbulence on flux rates. Stirring rates of 250, 450, 650 rpm were tested and permeate flux observed.



Figure 5.1.4: Comparison of permeate flux of NF 270 at various stirring rates

As seen in Figure 5.1.4, the flux change over a 2-hour period under the three stirring rates (250, 450, 650 rpm) was from 29.3 to 22.5 L/m²h, from 35.0 to 22.5 L/m²h, and from 33.7 to 26.0 L/m²h respectively. The initial permeate flux at 250 rpm was the lowest value which was consistent with the lower mass transfer. The initial flux levels at 450 and 650 rpm were quite similar which may be explained by the variation of membrane samples (the virgin membrane used in tests were cut from the same column of membrane but different parts) but the final flux after 2 hours filtration at 650 rpm was somewhat higher (15.6%) than the value at 450 rpm. The analysis shows that the flux of tea filtration with the NF 270 membrane rises when the stirring rate increases from 250 to 650 rpm. The increase in turbulence here leads to increased mass transfer and enhances the flux. Not only does the mass transfer coefficient increased, and this reducing potential aggregation leaded to the wall concentration (or polarization modulus) become lower [82]. In Xu et al. study, in order to improve the permeate flux, a turbulence promoter to increase the stirring rate was applied in ceramic membrane reactor [83].

5.2.5 Comparison of filtration flux with nanofiltration membrane NF 270 and NTR-7450

Chapter 2 indicated that most polyphenolic compounds in tea have a MWCO below 1000 Da. In the research discussed above, the NF 270 membrane with a MWCO of 300 has been tested for flux change and rejection. To extend the study of the nanofiltration membrane's performance with a black tea solution, the performance of a looser NF membrane (NTR-7450) with a MWCO of 700 Da was compared. Both permeate flux and membrane rejection of total tea polyphenols were observed and compared with the denser membrane NF 270.

In order to compare the performance of the two types of membrane equally, identical pre-processing was applied on membrane NTR-7450: that is, overnight soaking in Milli-Q water and then pre-conditioning at 3 bar for 20 minutes before tea solution filtration. The water flux of NTR-7450 was approximately 30 L/m²h, which was roughly 9% lower than the water flux of NF 270 at 33 L/m²h.

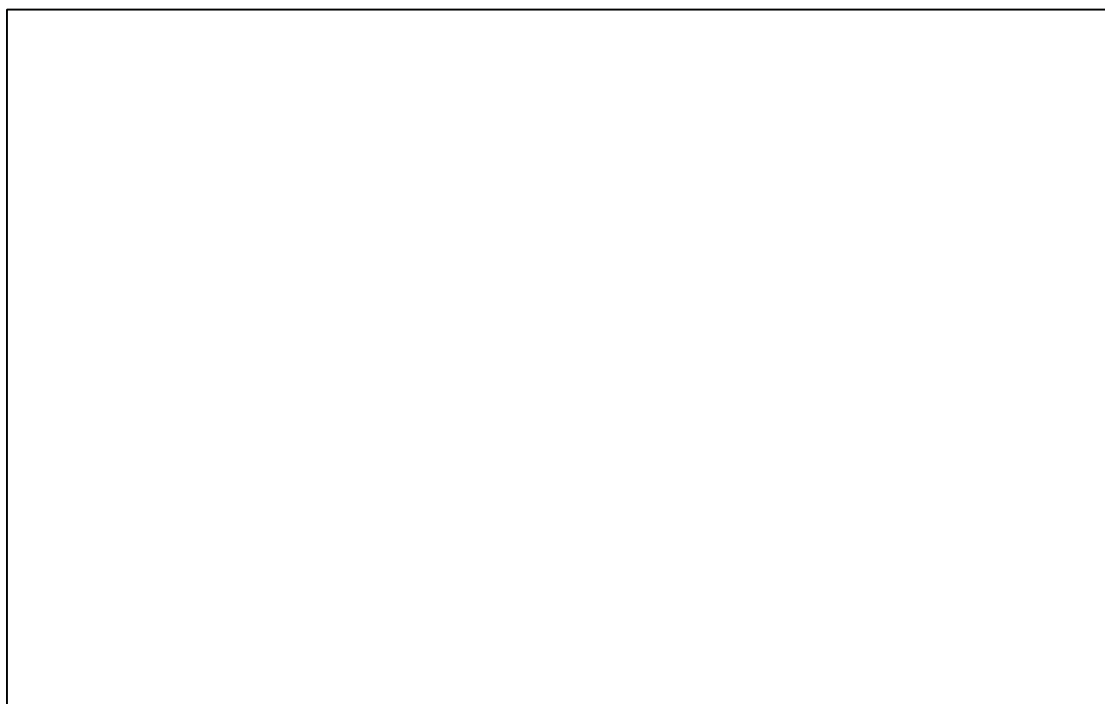


Figure 5.1.5: Comparison of permeate flux with membranes NF 270 and NTR-7450 at 5 bar for 2 hours

In the 2 hours of filtration, the permeate flux of NTR-7450 decreased from 25.8 to 17.7 L/m²h, and the flux of NF 270 dropped from 35.0 to 22.5 L/m²h. The initial permeate flux of NTR-7450 was approximately 26.3% lower than the initial value of membrane NF 270. In addition, the flux reduction of NTR-7450 was similar to NF 270. Based on the comparison above, membrane NF 270 achieved higher flux rates than NTR-7450 which means it is more efficient and productive in the membrane function.

In the research conducted by Liikanen et al. [84], varied nanofiltration membranes were used to try and improve the quality of chemically treated surface water. Those membranes included NF 70 (MWCO 60-200), NF 255 (MWCO 200-300), NTR-7450 (MWCO 700), NTR-7410 (MWCO 1100). Their results showed that membranes NF 70 and NTR-7450 had similar flux rates of 29.1 and 27.1 L/m²h respectively. The NF 255 membrane showed a flux of 39.1 L/m²h, which was about 44.4% higher than the flux of NTR-7450. Membrane NF 70 and NF 255 used in R. Liikanen's research were composed of polyamide [84], the same material as the NF 270 used in the current experiments. R. Liikanen et al. research results also identified a similar flux difference between NF 270 and NTR-7450 identified in the current experiments.

The NF 270 membrane as appears to be a better choice for processing tea filtration with better performance which is reflected in efficiency and productivity. NTR-7450 may suffer from intended fouling.

5.2.6 Change in Polyphenols concentration in permeate with nanofiltration membrane NF 270 and NTR-7450

In order to investigate the efficiency of total polyphenols rejection with NF 270 and NTR-7450, permeate samples were tested by UV-Vis spectrophotometer to estimate the relative concentration (ratio of concentration of polyphenols in the permeate to

the feed solution) at different times during nanofiltration process.



Figure 5.1.6: Concentration ratios of permeate collected at different times with NF 270 (1ml sample was collected from the permeate mixture at the time above, which is the overall permeate concentration at that time)

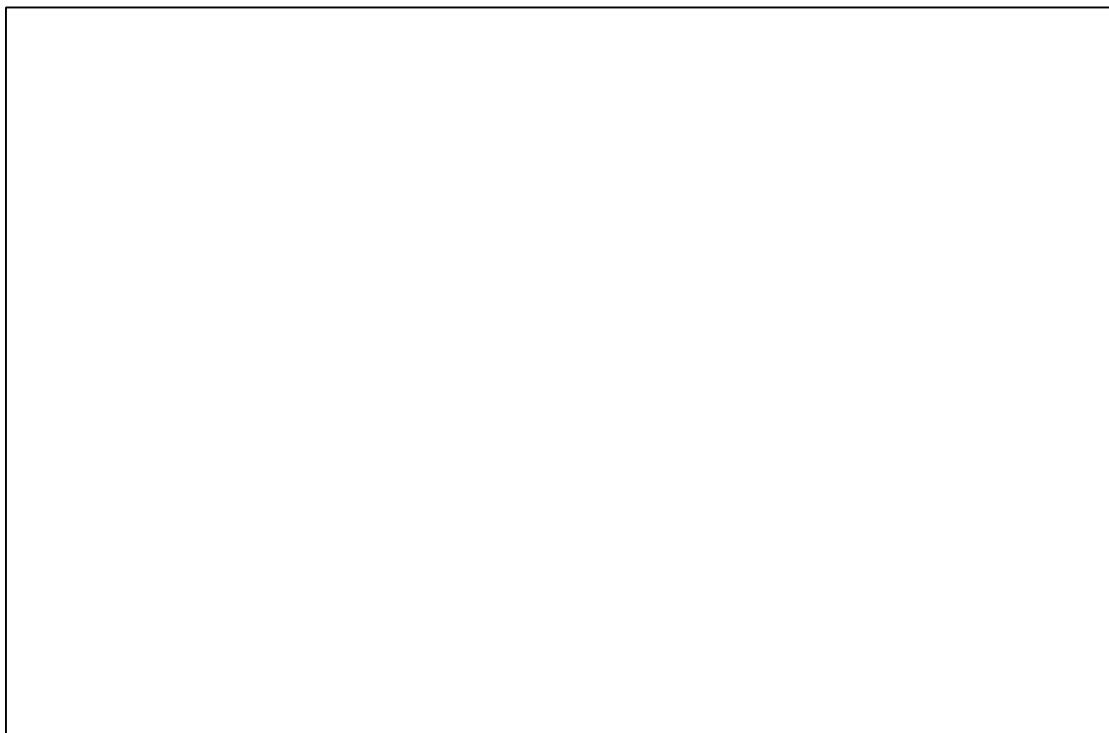


Figure 5.1.7: Concentration ratios of permeate collected at different times with NTR-7450

As shown in Figures 5.1.6 and 5.1.7, the absorbance values of permeate did not show obvious increase or decrease with the lapse of time. However, the permeate concentration obtained by the NTR-7450 membrane were higher (approximately 76%) than NF 270 (approximately 65%) which means that most of polyphenolic components are not being concentrated in retentate. This performance difference may due to the looser pore size of the NTR-7450 membrane which cannot retain most of the tea polyphenols with a distribution of MWCO from 200 to 1000 Da. In this case, judging by permeate absorbance, the NF 270 membrane shows better performance with rejection and concentration of polyphenols.

5.2.7 Chemical cleaning protocols for nanofiltration membranes

Due to the adsorption of molecules and the formation of a foulant layer, permeate flux decreases during filtration, and in a long run, the fouling layer on the membrane surface would affect the membrane property and performance. In this case, the cleaning of membrane after filtration is crucial step in membrane processing.

In order to remove the fouling on the surface of fouled membranes, chemical cleaning was applied. In Wei's thesis [39], it is presented that 0.1 M NaOH solution at 35° C for 30-60 minutes was used as cleaning method of ultra-filtration membrane. In Li et al. paper, NaOH solution (pH 11) is used as an alkaline solution to be the model chemical cleaning solution [40].

1) Cleaning solution

The 0.1 M NaOH solution was applied to cleaning the fouled membrane after 5 hours of filtration at 5 bar. After 30 minutes cleaning (stirring the cleaning solution for 30

minutes without any pressure), the pure water flux of this membrane was increased to 50 L/m²h from the initial flux of the clean membrane of 35 L/m²h. This may be because the cleaning solution was too strong for the membrane material and might have damaged the membrane structure.

A NaOH solution with a pH around 11-12 at concentration of 0.01 M was also tested. With this concentration, the cleaning solution did not damage the membrane, because the permeate flux did not change significantly compared to the result with a fresh membrane.

2) Cleaning protocols

Two sets of cleaning protocols are investigated in this research:

- ✧ Single stage filtration and cleaning evaluation
- ✧ Multiple cycles of filtration and cleaning procedures (see in Figure 5.1.8), described as method 1 & 2 below.

The two cleaning methods were performed on membrane NF 270 in this project.

Method 1:

- ✧ In the cleaning process, the fouled membrane was taken out of the stirred cell rig, then soaked and shaken by tweezers in 0.01 M NaOH solution for approximately (120 seconds). Most of the brown layer on the fouled membrane was removed.
- ✧ After cleaning, the membrane NF 270 was rinsed by Milli-Q water outside of the stirred cell. Then the cleaned membrane was soaked in Milli-Q water overnight in a refrigerator.

- ✧ The cleaning cycle was repeated four times after each filtration cycles.

Method 2:

- ✧ In the cleaning process, only the feed solution was replaced with the Milli-Q rinsing water and then the cleaning solution.
- ✧ After each filtration, the feed was replaced by 0.01 M NaOH ($\text{pH} \approx 12$) and without applied pressure, the membrane is soaked in the NaOH solution in the stirred cell for 30 minutes.
- ✧ After cleaning, the whole stirred cell with the fouled membrane is rinsed with Milli-Q water. The filtration and cleaning cycle is repeated 5 or 6 times to test the cleaning effects on membrane flux recovery during filtration of the tea solution.
- ✧ The pressure for filtration tests was 5 bar, and the stirring rate was constant at 450rpm.

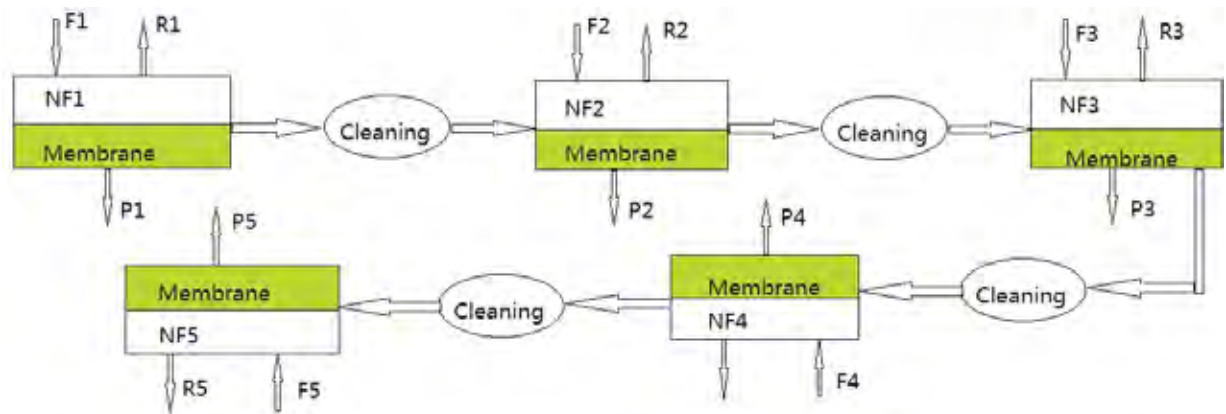


Figure 5.1.8: Multiple cleaning cycles (F: feed, P: permeate, R: retentate)

3) Effect on filtration flux of NTR-7450 membrane after cleaning cycles

In this part of the experiment, the NTR-7450 membrane was used in the cleaning cycles. 0.01 M NaOH was applied as the cleaning solution. Water flux after chemical cleaning was also recorded to compare with the initial water flux.

After the 3rd cycle, the membrane surface of NTR-7450 cannot be cleaned totally with yellow spots on it (see Figure 5.1.9).



Figure 5.1.9: Membrane NTR-7450 with yellow spots after the 3rd cleaning cycle

As shown in Figure 5.1.9, the flux rate declined significantly, approximately 20% in each cleaning cycle, from 14 L/m²h to 9 L/m²h. And after three cleaning cycles, yellow spots appeared on the membrane surface (Figure 5.1.9).

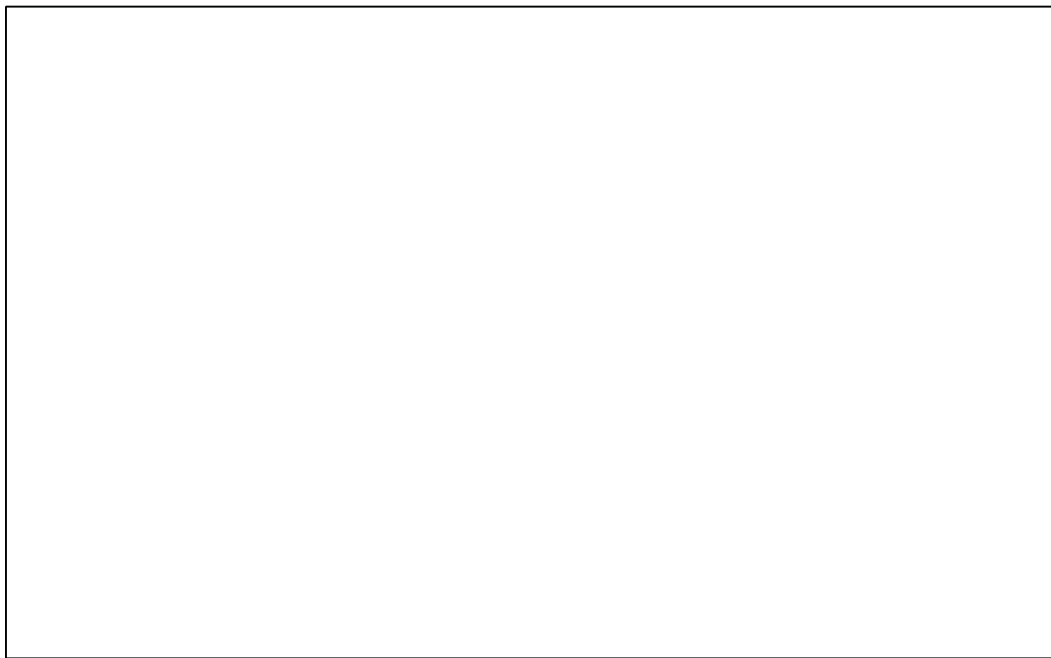


Figure 5.1.10: Black tea flux change of NTR-7450 membrane in 3 cleaning cycles



Figure 5.1.11: Milli-Q water flux change in cleaning cycles

From Figure 5.1.11, it showed again that the membrane performance of NTR 7450 has been affected significantly during the three cleaning cycles with the water flux dropping more than 50% compared to the initial flux. Table 5.1.3 shows that the rejection rates of NTR-7450 membrane in nanofiltration remained high and steady over the three cycles.

	Rejection	Mass balance error
1 st cycle	96.25%	14.80%
2 nd cycle	93.50%	10.90%
3 rd cycle	95.40%	14.40%

Table 5.1.3: Rejection and error in cleaning cycles by mass balance

4) Effect of filtration on flux rates of the NF 270 after cleaning cycles

The same NF 270 membrane was used to carry out the cleaning cycles. The flux changes of the two cleaning methods are shown in Figure 5.1.12 and

5.1.13.

Figure 5.1.12: Effect of flux change after cleaning method 1

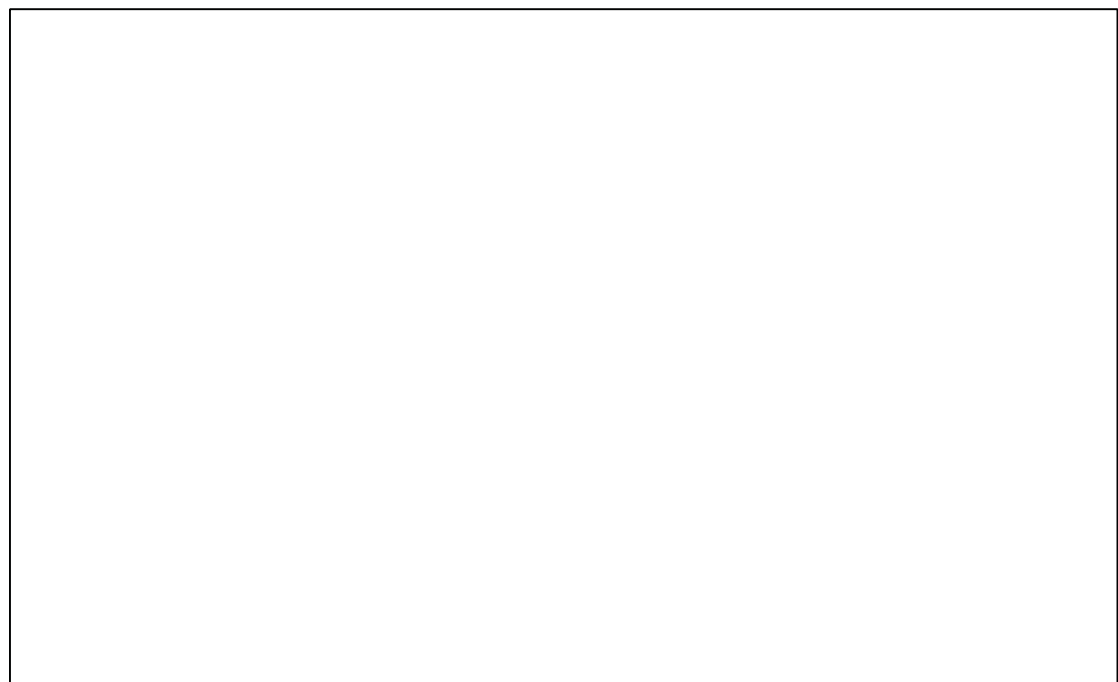


Figure 5.1.13: Effect of flux change after cleaning method 2

In Method 1, Figure 5.1.12 shows that after the 1st cleaning cycle, the flux rate increased from 20 to 25 L/m²h, which was about 25% boost comparing to the initial flux. Applying Cleaning Method 2 (Figure 5.1.13) showed the similar flux boost, which was increased from 30 to 35 L/m²h. From the 2nd to 5th filtration, the flux of membrane showed little obvious change NF 270 during the cleaning cycles in both of

the cleaning methods.

5) Flux recovery of membrane NF 270 by chemical cleaning

In this cleaning cycle test, the water flux of a pre-cleaned virgin membrane was recorded after each cleaning and water rinse and compared with membrane performance after chemical cleaning.

	Virgin NF 270	2 nd cycle	3 rd cycle	4 th cycle	5 th cycle
Filtration	20.6L/m ² h	25.0 L/m ² h	28.1 L/m ² h	26.2 L/m ² h	28.1 L/m ² h
Recovery rate	-----	121.3	112.4%	93.2%	107.2%

Table 5.1.4: Flux recovery after cleaning method 1

Flux(L/m ² h)	Virgin NF 270	2 nd cycle	3 rd cycle	4 th cycle	5 th cycle
Water flux	43.1				
Filtration	34.3-16.9	33.7-18.7	29.3-16.9	30.6-16.9	31.8-18.7
After water rinse	24.3	30.0	30.0	30.0	33.7
After cleaning	65.5	61.8	56.2	60.0	65.5
Recovery rate	151.9%	94.4%	90.9%	106.8%	109.2%

Table 5.1.5: Flux recovery of NF 270 membrane after cleaning method 2

For the virgin membrane NF 270, the water flux recoveries for both of the cleaning methods were 125% and 150% after the 1st cycle of chemical cleaning. Another studies has established that flux rates increased by approximately 50% when the 1st cleaning was performed on a virgin membrane after normal operation [84]. This phenomenon is not reported by many industries using membranes because in most cases the membranes used in a laboratory are virgin membranes. In reality, industrial membranes are cleaned before the first usage. In these experiments, the characteristics of the membrane after pre-cleaning might have changed considerably

from the characteristics of the virgin membranes [45]. Some opening of previously closed pores may have occurred with cleaning.

Considering the higher recovery rate obtained in method 2 and also that there was less damage to the membrane since it did not have to be removed from the rig, cleaning method 2 is recommended as a better choice compared with method 1.

6) Polyphenols concentration during filtration and cleaning cycles

The following figures were drawn from the UV absorbance data collected in cleaning method 2.

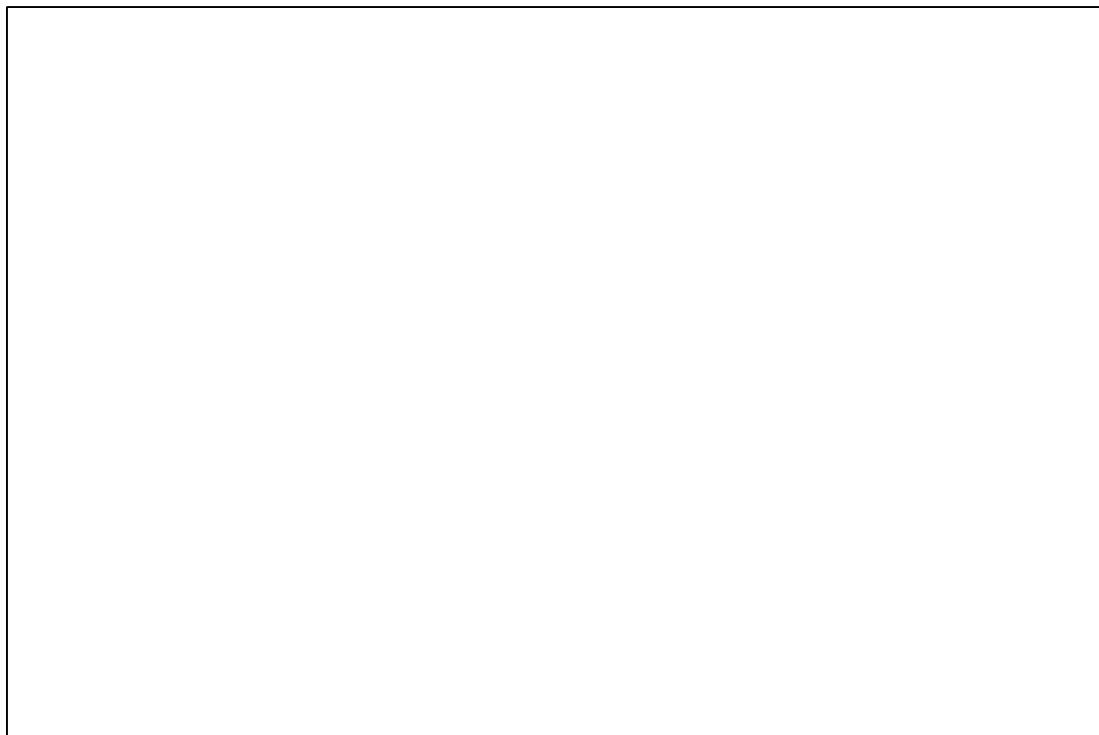


Figure 5.1.14: Concentration ratio of feed, permeate and retentate in cleaning cycles (2nd to 5th cycle)

As shown in Figure 5.1.14, the polyphenols concentration ratio in permeate generally increased from 0.38 to 0.59 with a slight drop after the 1st cleaning, which indicates the increasing amount of polyphenols penetrating through the membrane with additional cleaning.

	1 st cycle	2 nd cycle	3 rd cycle	4 th cycle	5 th cycle
Rejection	88%	94.3%	95%	89.1%	86.7%
Error	8.6%	7.1%	4.4%	2.6%	17.5%

Table 5.1.6: Rejection and error of NF 270 membrane on total polyphenols during cleaning cycles

Table 5.1.6 shows that polyphenols rejection remained at a very high ratio during cleaning cycles. The errors were also under 10% except for the 5th cycle which were calculated by mass balance.

7) Comparison of membrane performance for the NF 270 and NTR-7450 during cleaning cycles

For membrane NF 270, the virgin membrane flux increased by 50% after cleaning. After filtration, each time the flux recovered nearly 100% (ranging from 94.4% to 109%) with cleaning after 30 minutes. In the meantime, polyphenols rejection was also estimated by use of relative concentration of polyphenols in permeate solution measured with UV-Vis spectrophotometer. In the 5 cleaning cycles, the rejection rates were quite high and clustered between 87-95%. Just the error of the 5th cycle was approximately 17%, while the others were controlled under 10%.

As noted above, virgin membranes used in the laboratory are different from the industrial membranes which have been cleaned before usage. Nystrom et al. research [45], also reported a flux boost after cleaning due to the surface coating covering virgin membranes. Although this flux boost is not mentioned specifically in industrial applications, this phenomenon does exist with industrial membranes. Because those membranes are pre-cleaned the phenomenon is ignored.

In the rest of the cleaning cycle experiments in this research, flux recoveries were around 100% which indicates that most of the fouling accumulated on the

membrane surface is removable. The rejection rates during cleaning remained in the range of 87-95%, which means the membrane remained in a good condition to function as a barrier after chemical cleaning.

For membrane NTR-7450, the permeate flux of tea solution decreased over the three cycles from 15 to 11 and 9 L/m²h at the end of each filtration. The water flux recoveries of NTR-7450 were 73%, 74%, and 87% comparing to the previous flux. After the third cleaning, the water flux of NTR-7450 was just 48% of the initial water flux.

Membrane NTR-7450 achieved quite a high rejection rate of 99.25%, 96.50%, and 98.4% in these three cycles. The results show the excellent performance of NTR-7450 in separating and concentrating polyphenols in black tea which is consistent with high retention.

Both the permeate flux and flux recovery of membrane NTR-7450 are significantly lower than membrane NF 270, which limited the overall performance of NTR-7450. In Liikanen et al. [84]paper, membrane NTR-7450 was investigated with other NF membranes, such as NF 70 and NF 255. Their study of flux change during filtration also noted the flux recovery differences between NTR-7450 and NF 70 & 255. Membranes NF 70 and NF 255 demonstrated quite stable flux rates but NTR-7450 experienced an obvious decrease of flux. This results point to higher internal pore fouling which was not easily removed by cleaning.

5.2.8 Streaming potential comparison of NF 270 and NTR-7450

Change of surface property can be used to characterise the degree of membrane fouling for UF and nanofiltration processes. The surface charge of the NF 270 and NTR-7450 membranes were measured after filtrations in different conditions and

compared with their values at virgin status.

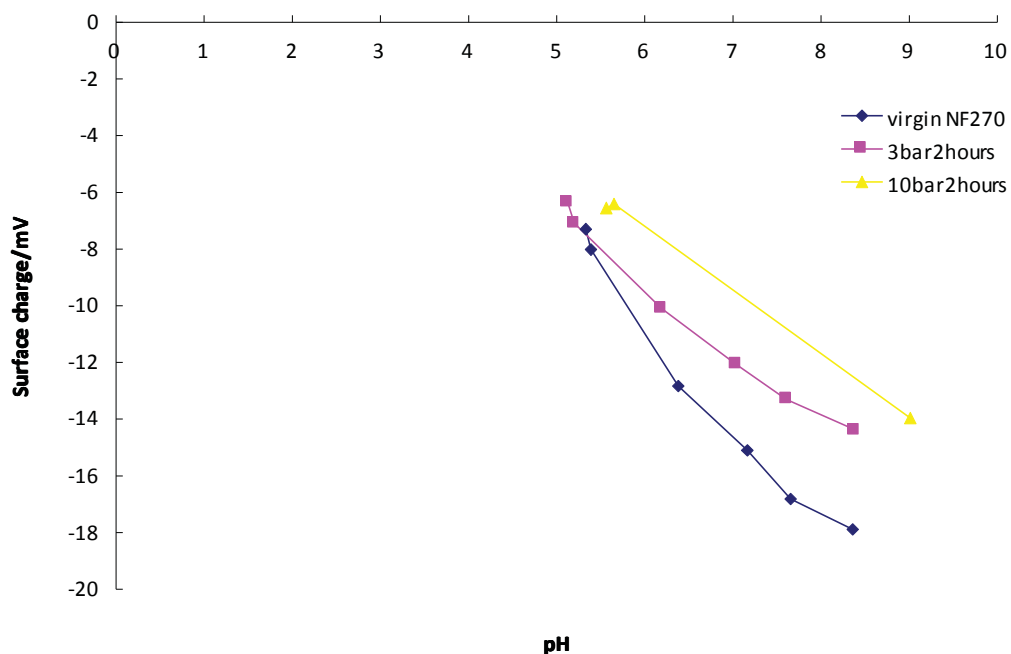


Figure 5.1.15: Surface charge of virgin and fouled NF 270 membranes

The results in Figure 5.1.15 indicated that the surface charge of NF 270 became less negative after membrane fouling compared to the virgin membrane. With the increase in TMP, the negative charge was reduced further.

In Cho et al. study on fouled NF membranes, the polyamide NF membranes presented less negative zeta potential values after fouling of natural organic matter [77].

The results showed that the foulants on membrane NF 270 was more positive charged or neutral which leaded to the less negative zeta potential of NF 270 after fouling. In addition, less negative charged with increasing TMP revealed that more foulants were accumulated on membrane NF 270 when higher pressure was applied on the feed.

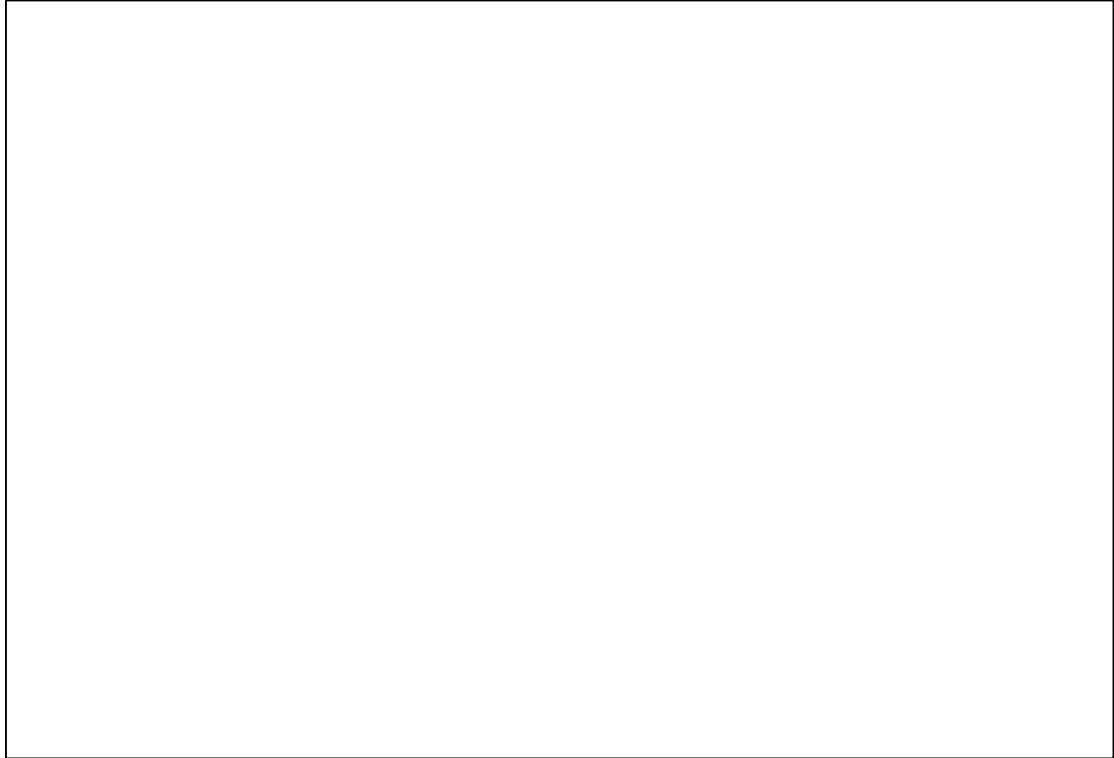


Figure 5.1.16: Surface charge of virgin and fouled NTR-7450

The zeta potential of virgin NTR-7450 membrane measure in this study is slightly more negative then the results of Nystrom et al.. In their study, the zeta potential of virgin NTR-7450 at pH ranging from 5 to 7 was -7.5 mv to -10 mv [80]. The difference could be due to the difference in membrane washing procedure before zeta potential tests. The incomplete removal of surface preservative from the virgin membrane during washing process may have affected the surface charge. However the trend of this study is very similar with the above mentioned report.

It is showed in Figure 5.1.16 that the surface charge of fouled NTR-7450 was less negative than the virgin membrane. Similar with the observation shown in Figure 5.1.15, the foulant on the membrane surface have reduced the membrane surface charge compared to the clean membranes. The surface charges measured on both fouled membranes were dominated by the foulants on the membrane, therefore similar surface charge of both fouled membrane were observed.

5.3 Analysis of polyphenol composition

UV-Spectrophotometry was used to analyze the total polyphenols feed and permeate samples; capillary electrophoresis and GC-MS were applied in order to determine the specific components in those samples.

5.3.1 Capillary Electrophoresis (CE)

To analyze polyphenolic components, capillary electrophoresis was used to study the composition of tea sample as well as the amounts of each component. As shown in Table 2.6.1 in the summary methods for analyze polyphenols by capillary electrophoresis, both 210 nm and 250 nm wavelengths were chosen in the following CE tests.

Capillary 1:

AGIG1600-62211, Capillary Standard HPCE 50 μ m id 72 cm

With capillary1, the organic acid (standard solution purchased with the capillary with known spectrum) was tested to make sure the capillary electrophoresis equipment worked properly.

According to the organic acid graphs, the three peaks were clear and the retention time was stable (from 7 min to 10 min), which means the equipment of capillary electrophoresis worked well (see Figures 5.2.1 & 5.2.2).

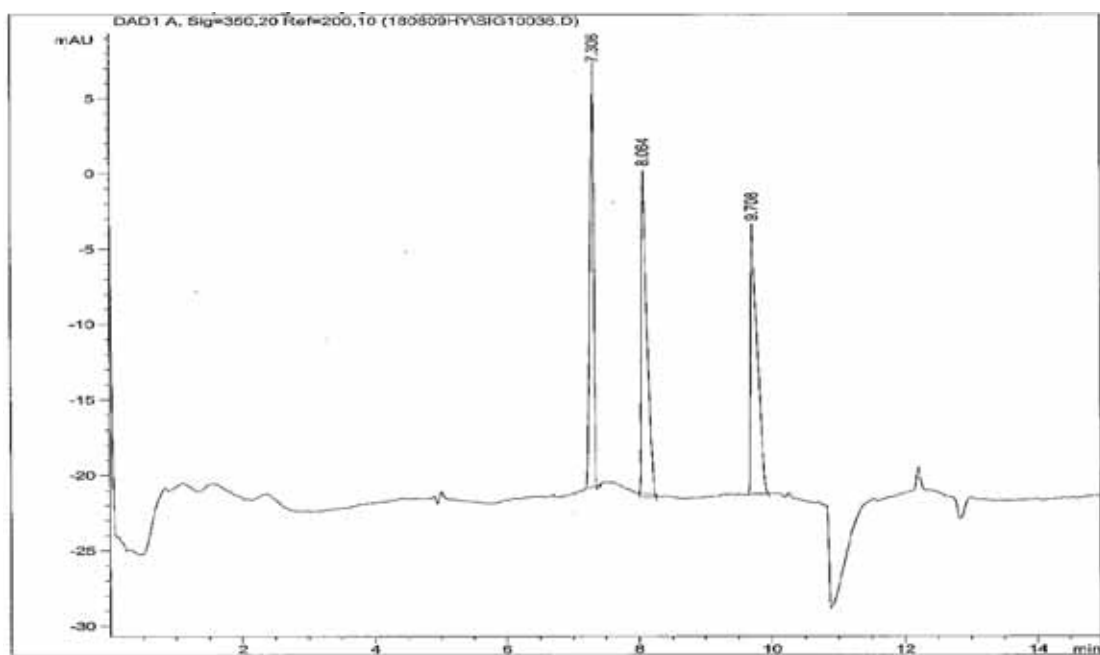


Figure 5.2.1: Electropherogram of organic acid with capillary 1

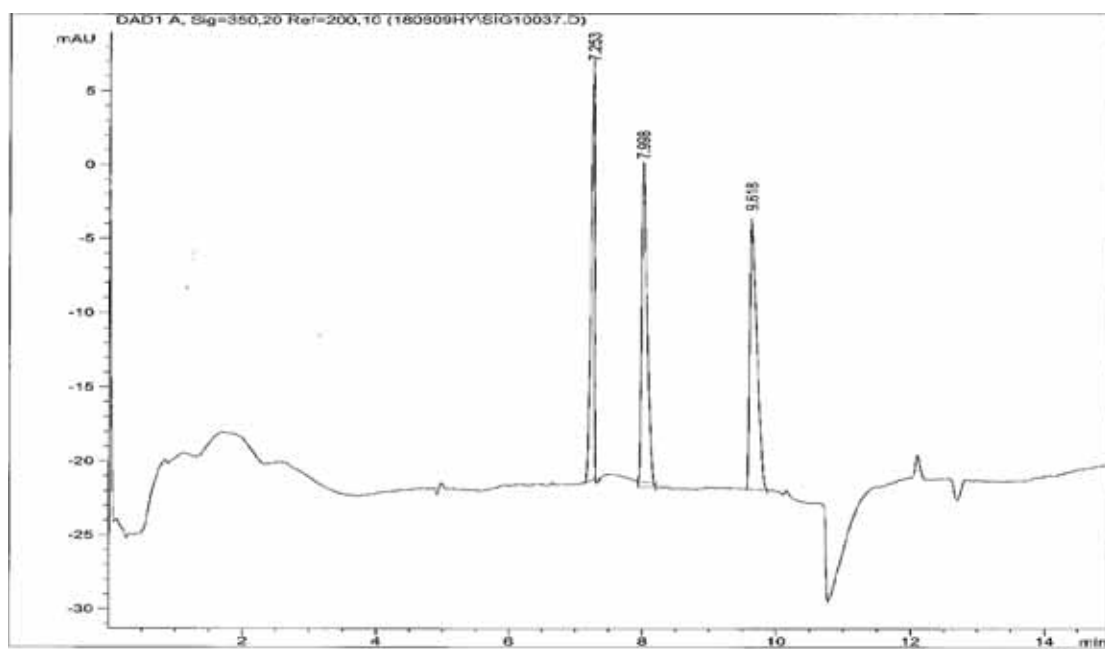


Figure 5.2.2: Electropherogram of organic acid with capillary 1 (repeated)

Capillary 2:

AGIG1600-61239, PVA CE-cap. Ext. light path 50 μ m 56 cm.

Buffer 1 was used in the tests of capillary 2.

Results:

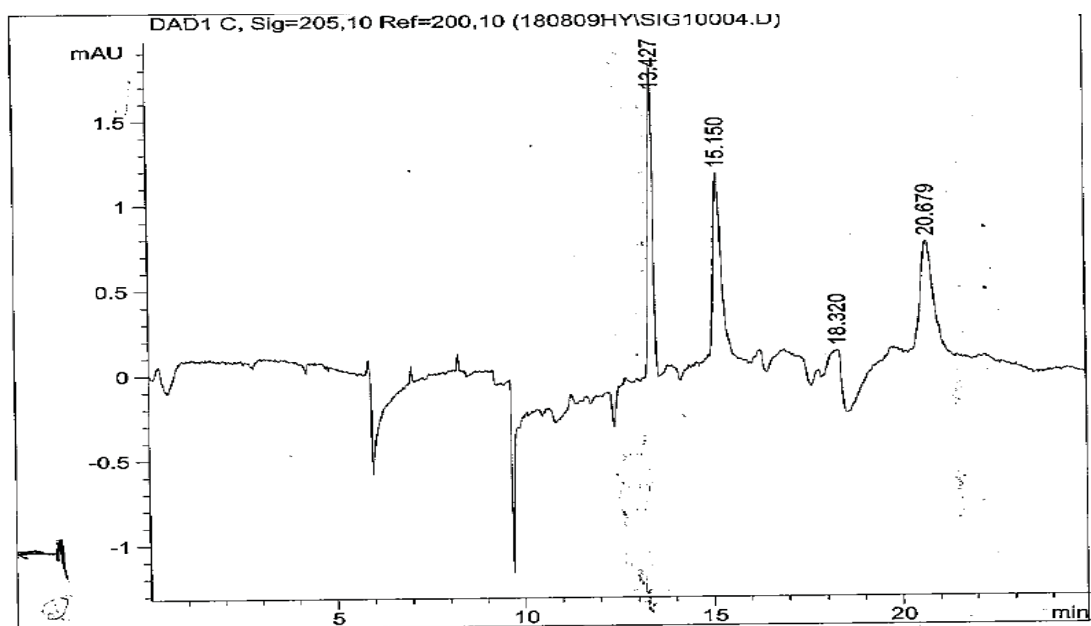


Figure 5.2.3: Electropherogram of 33% black tea sample with capillary 2

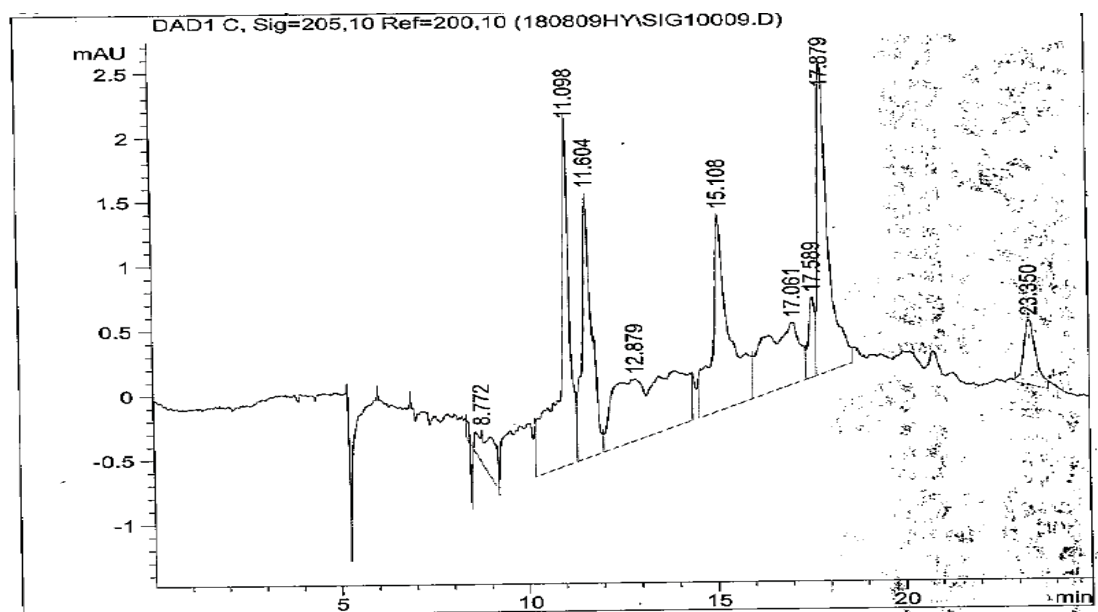


Figure 5.2.4: Electropherogram of 33% black tea sample with capillary 2 (repeat 1)

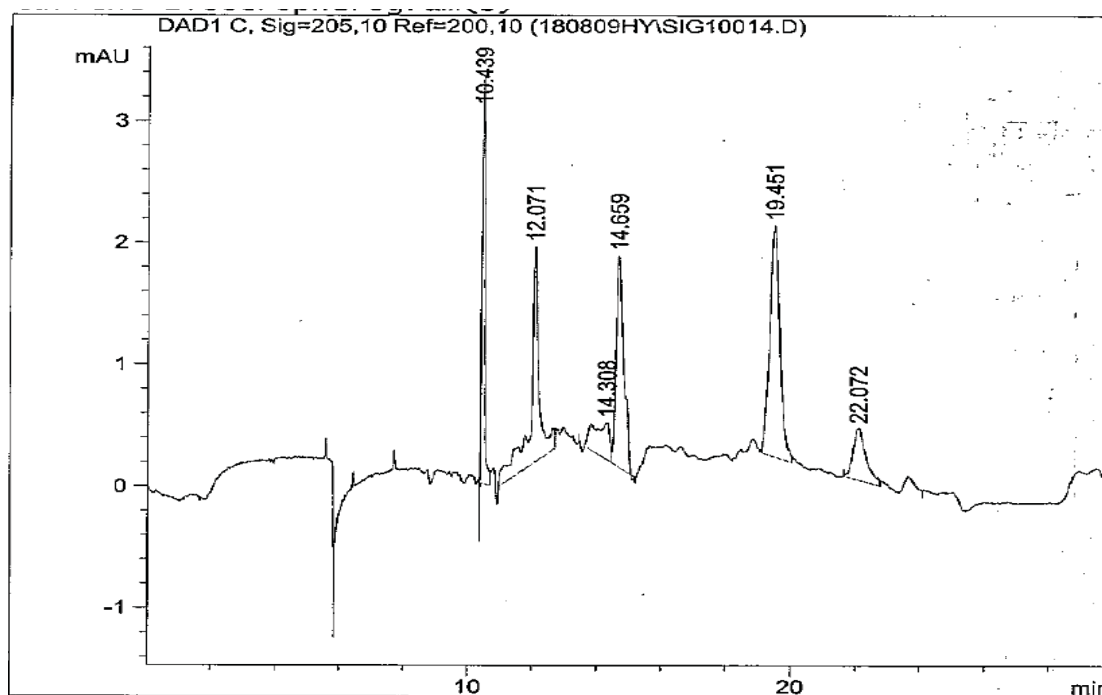


Figure 5.2.5: Electropherogram of 33% black tea sample with capillary 2 (repeat 2)

In the above Figure 5.2.2 to 5.2.5, although several positive peaks were detected, the elution time was not consistent. In this case, the result obtained in this condition cannot be considered as acceptable in analyzing the constituents in tea samples.

Capillary 3:

AGIG1600-61232, Ext. Light Path CE cap. 50 μ m 56 cm

First, this capillary was tested with the standard sample of organic acid. Conducted using the same buffer as Capillary 1, the peaks and retention time were relatively consistent.

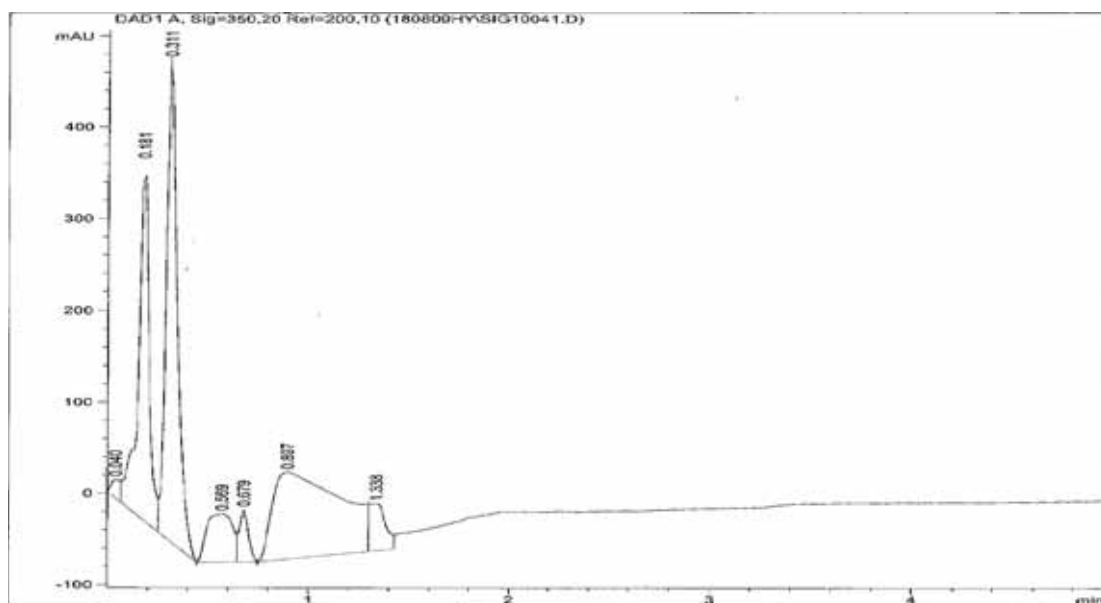


Figure 5.2.6: Electropherogram of organic acid (sample) with capillary 3

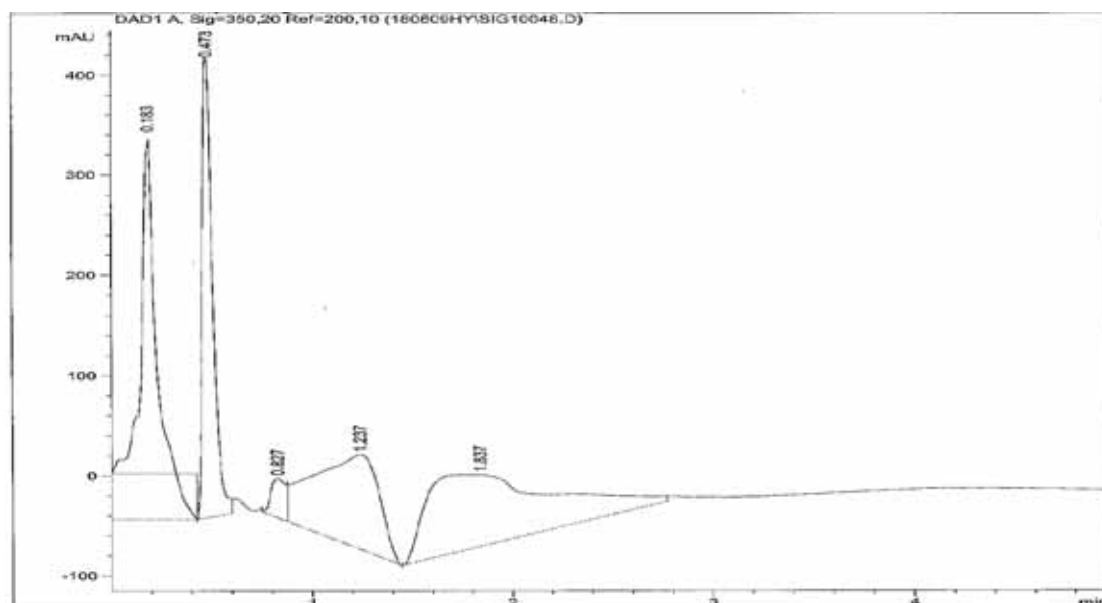


Figure 5.2.7: Electropherogram of organic acid (sample) with capillary 3 (repeat1)

However when the tea sample was tested with buffer 1 on capillary 3, unfortunately, the results of the tests were not reproducible. Consequently, the results displayed in Figures 5.2.6 to 5.2.10 are not reliable and were inconclusive. As both the equipment and the column have been proved reliable in other studies, another factor may have affected the results was the buffer material used.

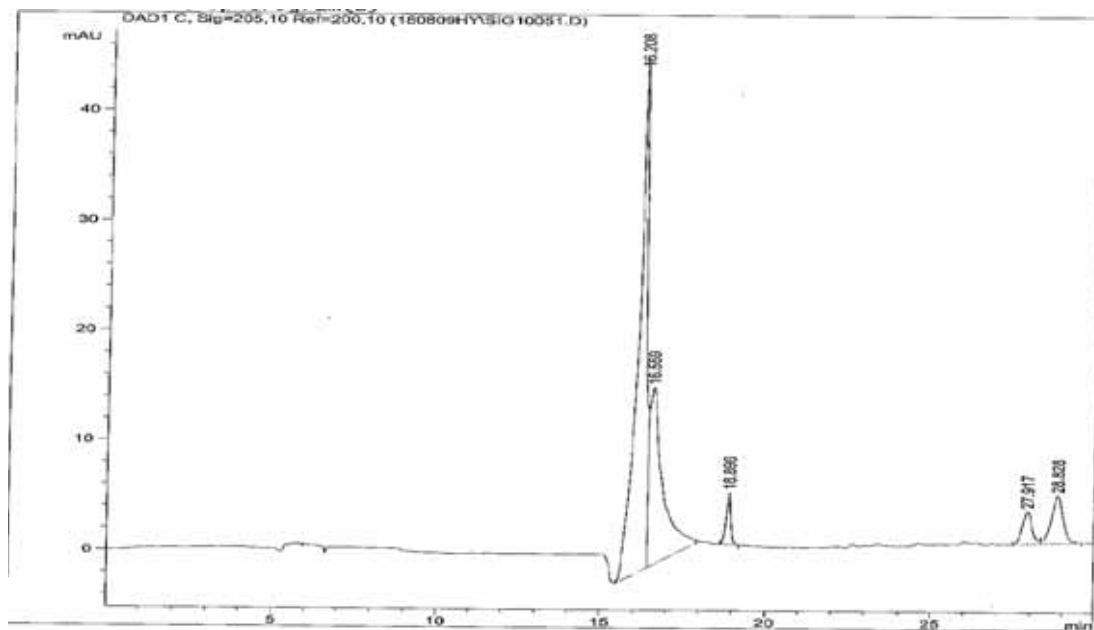


Figure 5.2.8: Electropherogram of 33% black tea sample with buffer 1 and capillary 3

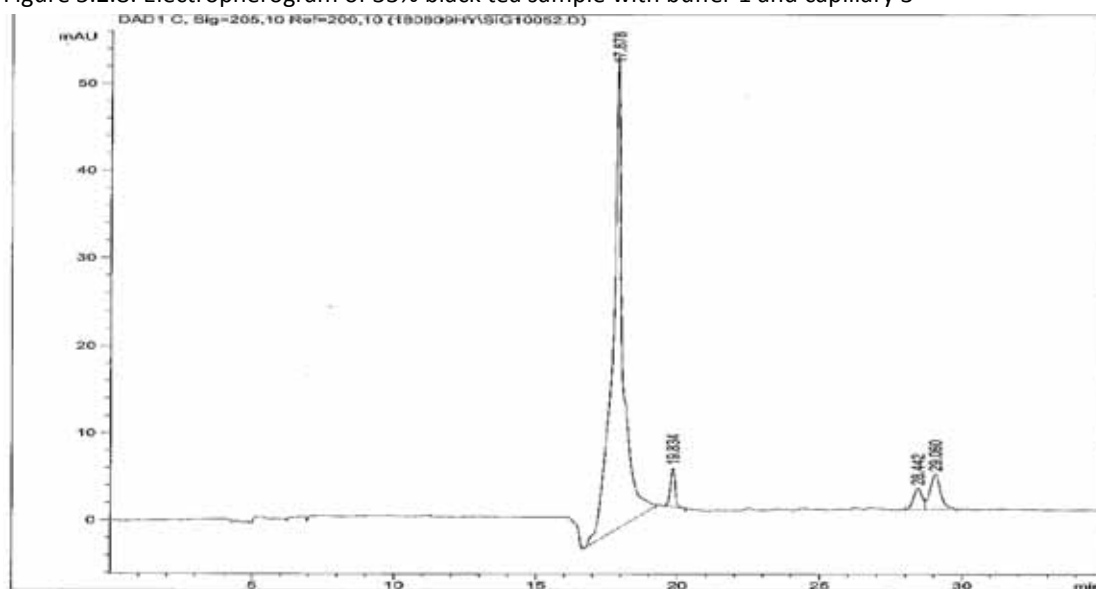


Figure 5.2.9: Electropherogram of 33% black tea sample with buffer 1 and capillary 3 (repeat 1)

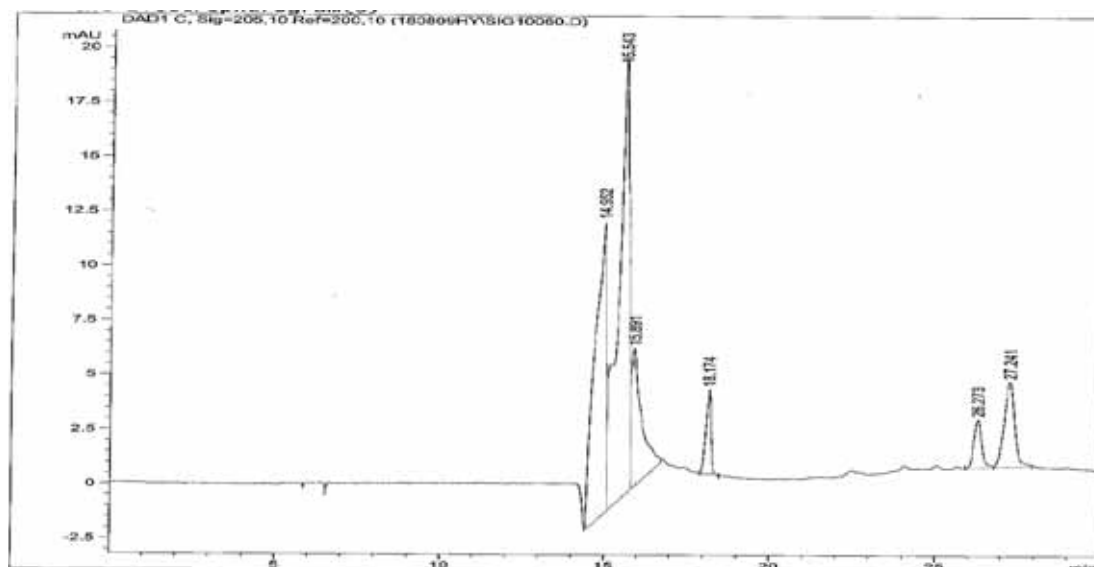


Figure 5.2.10: Electropherogram of 33% black tea sample with buffer 1 and capillary 3 (repeat 2)

Since the current used with Buffer 1 was just around 7 μ A, a new borate buffer (Buffer 2) was tested on capillary3. However, with Buffer 2, only one peak can be detected, even with different tea concentrations. In this case, buffer 2 was unable to separate the different polyphenols in capillary 3.

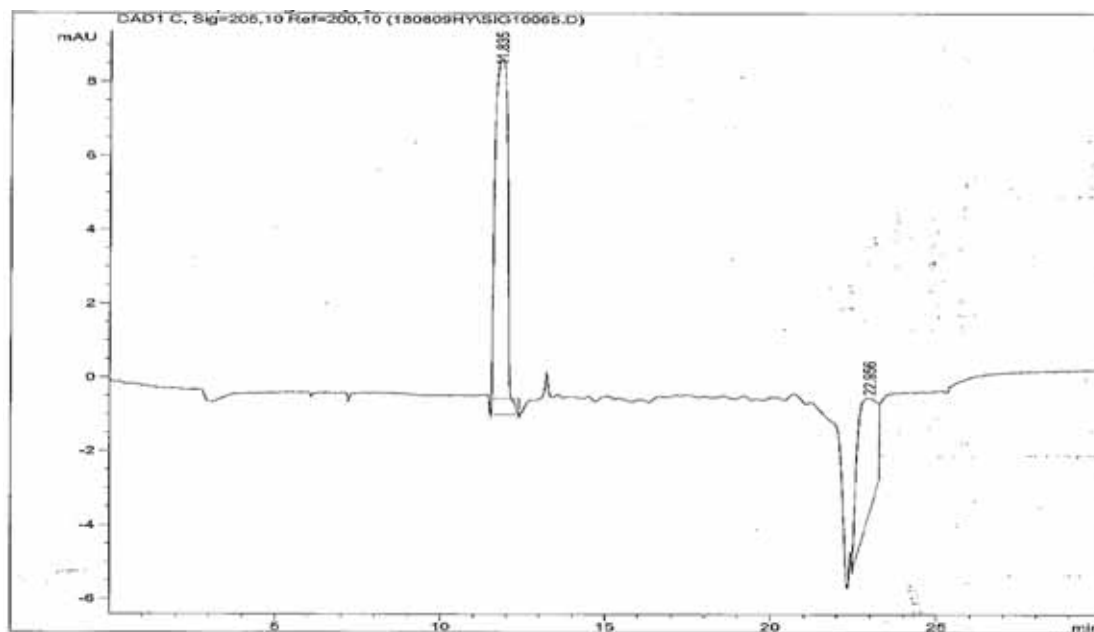


Figure 5.2.11: Electropherogram of 20% black tea sample with buffer 2 and capillary 3

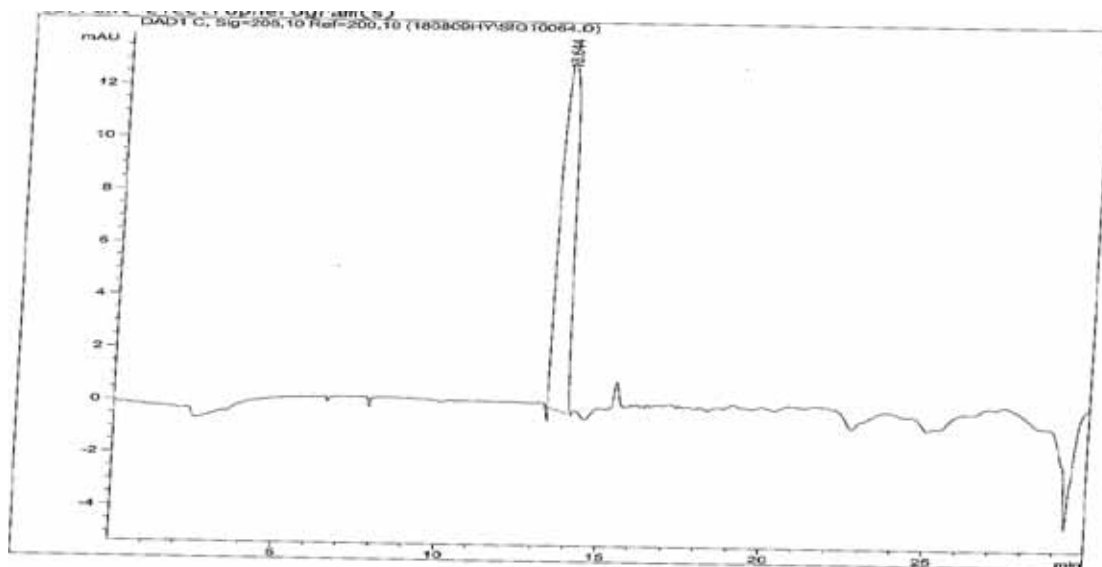


Figure 5.2.11: Electropherogram of 33% black tea sample with buffer 2 and capillary 3

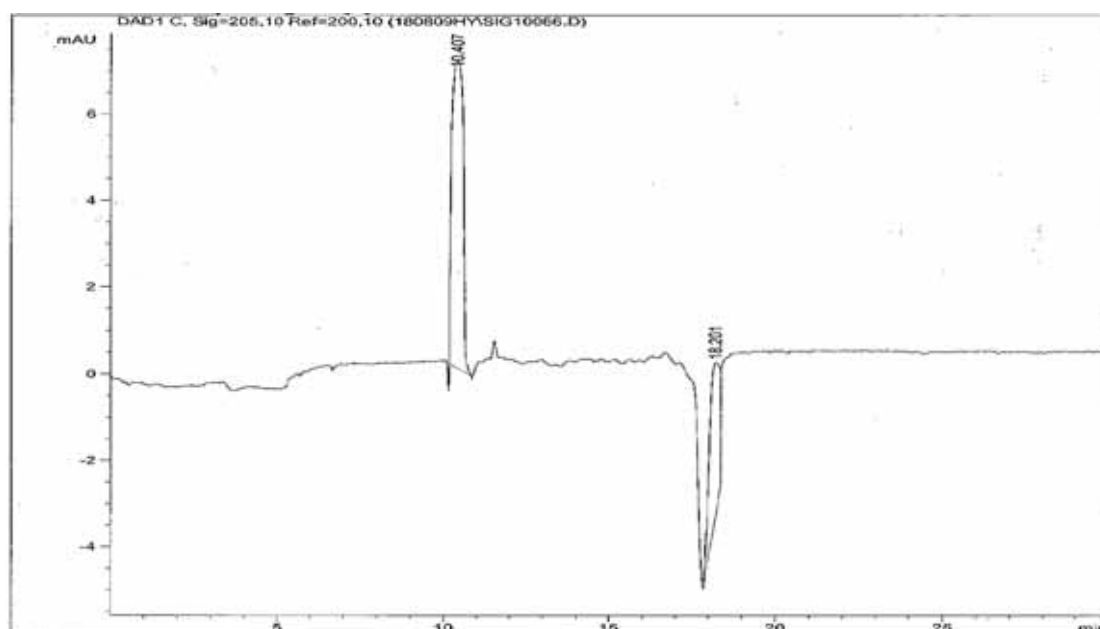


Figure 5.2.12: Electropherogram of 10% black tea sample with buffer 2 and capillary 3

In these laboratory experiments, capillary electrophoresis was tried with three different columns and 2 different buffer solutions. However, the peaks obtained for the tea samples are not stable which makes CE not an appropriate method to analyze polyphenolic components in this research.

5.3.2 Gas Chromatography-Mass Spectrometry (GC-MS)

GC-MS was used to detect the specific components in black tea in feed and retentate. The GC-MS analysis was conducted previously by Milou [58], who also tested tea polyphenols extraction using similar technique. All the components in the sample were identified using mass spectral library provided by the GC-MS supplier. 50 mg 5-Methoxy Salicylic acid was used to the tea sample as an internal standard to quantify the phenolic acids.

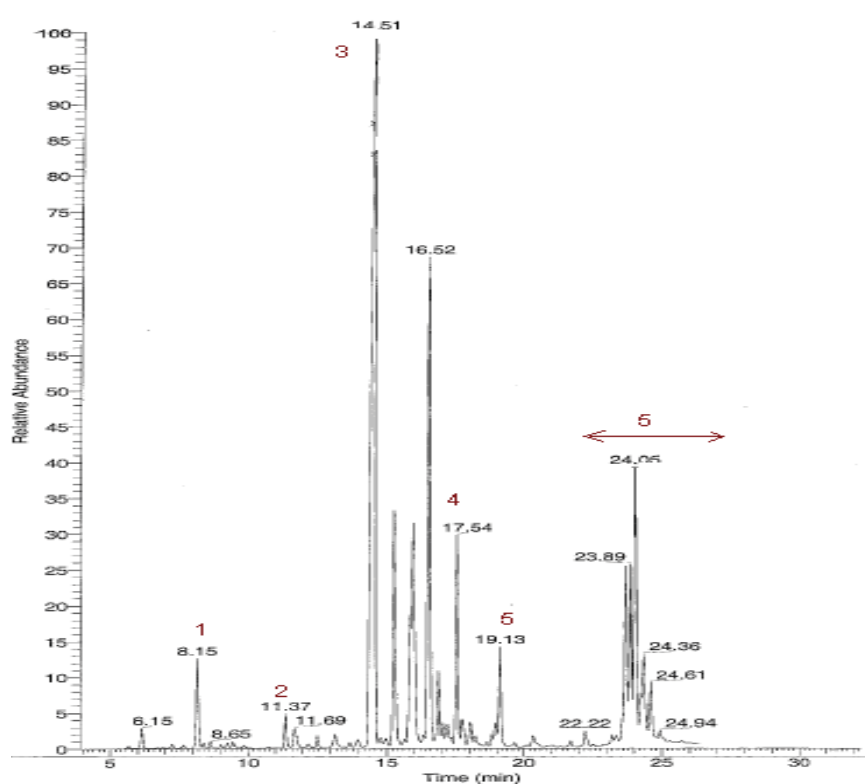


Figure 5.2.13: Feed of black tea solution (1.8 g tea, 5min, infused in 300 ml boiling water), 1: Silanol, 2: Malic acid, 3: I.S., 4: Gallic acid, 5: Sugar

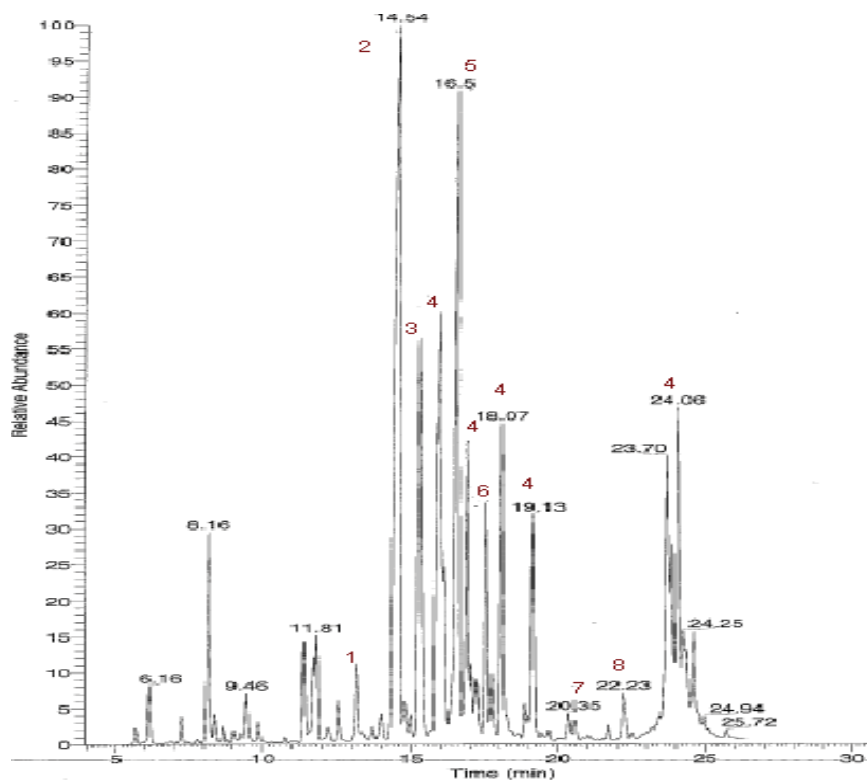


Figure 5.2.14: Feed of black tea solution (3.6 g tea, 5min, infused in 300ml boiling water), 1: Glutamic acid, 2: I.S., 3: Solvent, 4: Sugar, 5: Quinic acid, 6: Gallic acid, 7: Sinapinic acid, 8: Karachine

From Figures 5.2.13 and 5.2.14, we can find the graph with lower tea concentration (Figure 5.2.13) is much clearer while the graph obtained from higher concentration (Figure 5.2.14) has a much higher density of peaks. We can conclude that the chromatography was more defined at a lower concentration for analysing the major components in black tea. In this case, the GC-MS can identify most components clearly which gives a better picture of which components are present in black tea.

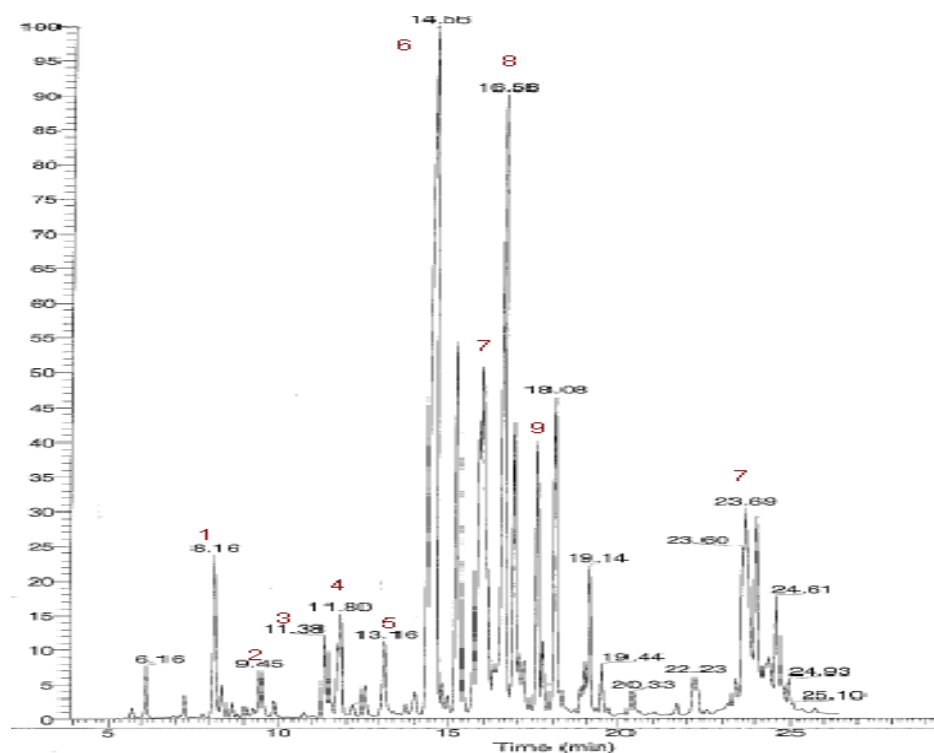


Figure 5.2.15: Permeate of macrofiltration (0.22 μ m) of black tea solution (3.6 g, 5min, 300 ml, infused in 70°C), 1: Silanol, 2: L-Serine, 3: Malic acid, 4: L-Aspartic acid, 5: Glutamine, 6: I.S., 7: Sugar, 8: Quinic acid, 9: Gallic acid

Comparing Figure 5.2.13 and 5.2.15, we can achieve a high concentration of polyphenols from black tea feed with 5 minutes infusion and boiling water which can support the result of extraction method on extracting temperature and infusion time proven by UV-Vis spectrophotometer.

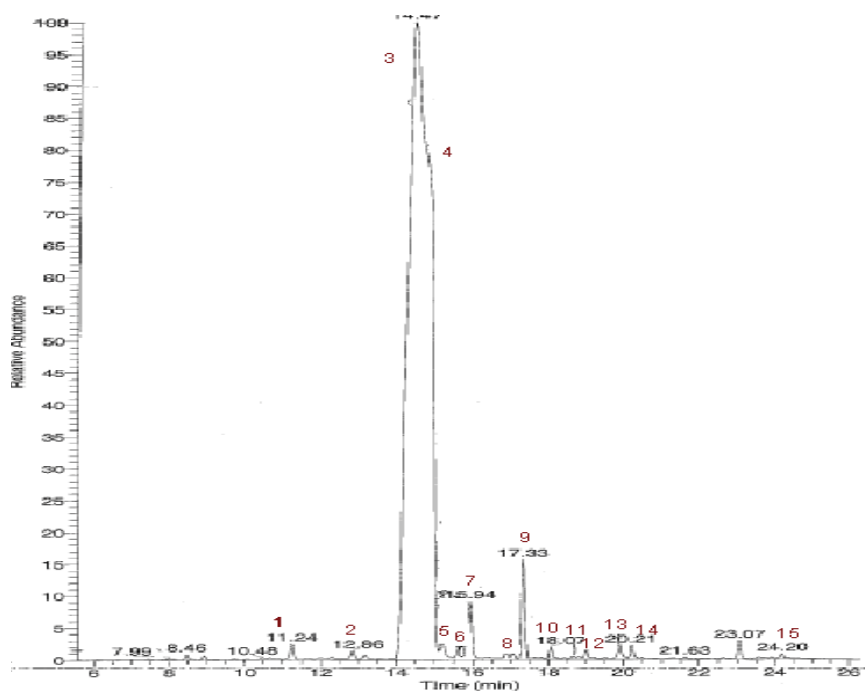


Figure 5.2.16: Retentate of black tea solution after NF 270 filtration, 1: O-Salicylic acid, 2: P.H.B.A., 3: I.S., 4: Vanillic acid, 5: Dihydroxybenzoic acid, 6: Protocatechinic acid, 7: Caffeine, 8: Coumaric acid, 9: Gallic acid, 10: Hexadecanoic acid, 11: Ferulic acid, 12: Caffeic acid, 13: Oleic acid, 14: Octadecanoic acid 15: Sinapinic acid

From Figure 5.2.13 and 5.2.16, we can find the constituents of feed and retentate were almost same, both containing caffeine, gallic acid, and hexadecanoic, but in differing amounts. Which means that the distribution of components are the same even if the concentration is different from feed.

5.4 Conclusion

From the results of flux change in NF process with varied TMP, filtration time and stirring rates, the following conclusion can be made:

When trans-membrane pressure increased from 3 bar to 8 bar, the initial permeate flux rose but not in proportion. The flux/pressure factor at 3 bar was about 5.7, while they were 7 and 6.5 respectively at the pressure of 5 bar and 8 bar. After 2 hours' filtration, the flux/pressure factor declined to 4.8, 4.5 and 3.5. In addition, the

decreasing ratios of flux were separately 15.8%, 35.7% and 46.2% from 3 to 8 bar. However, when the pressure rose to 10 bar, the initial permeate flux did not increase, in contrast, it decreased by almost 29.2% comparing to the flux at 8 bar in the first 20 to 40 minutes with a flux/pressure factor of just 4.8 which was even approximately 15.8% lower than 3 bar. In addition, rejection was also calculated and remained stable (90%, 86%, 86%) in the NF filtration at 3 bar, 5 bar, and 8 bar.

To investigate the effect of filtration time, two trans-membrane pressures (3 bar and 5 bar) were applied in the NF process. At 3 bar, the permeate flux dropped by 14.7% in the first 2 hours and another 20.7% in the following 2 hours. At 5 bar, the decreasing factors of permeate flux were separately 35.7% and 10%. The rejection at 5 bar was 86% for 2 hours, 79% for 4 hours, and 86% for 6 hours.

The study of effect of stirring rates was conducted on membrane NF 270 at 5 bar. Stirring rates of 250 rpm, 450 rpm, and 650 rpm were applied in this NF process. The flux change in 2 hours filtration was from 29.3 to 22.5 L/m²h (250 rpm), 35.0 to 22.5 L/m²h (450 rpm), and 33.7 to 26.0 L/m²h (650 rpm).

Membrane NTR-7450 and NF 270 were both tested and compared on filtration flux and rejection of polyphenols. In the 2 hours' filtration, membrane NTR-7450 experienced a flux drop from 25.8 L/m²h to 17.7 L/m²h. Although the absorbance values of polyphenols in permeate did not show obvious increase or decrease with time, the results obtained by NTR-7450 were higher (approximately 76%) than NF 270 (approximately 65%), which indicated that NF 270 had a better performance in concentrating polyphenols than NTR-7450.

0.01 M NaOH solution (pH $\approx$ 12) was applied as a effective alkaline cleanser which can still restore most of the membrane properties. Multiple cleaning cycles were carried out on the fouled NF membranes (NF 270 and NTR-7450). For membrane

NTR-7450, the permeate flux declined approximately 20% after each cleaning cycle, and after three cycles, obvious yellow spots appeared on the membrane surface. For membrane NF 270, the permeate flux experienced a 25% increasing after the chemical cleaning of virgin membrane, and in the following cleaning cycles (2nd to 5th filtration), the permeate flux of NF 270 did not show significant change which indicated that membrane NF 270 has higher tolerance to alkaline cleaning and better persistence in long-run usage than NTR 7450. The NTR-7450 membrane probably suffers from greater internal fouling compared to NF 270.

In order to investigate the membrane surface property before and after fouling, streaming potential was applied. For membrane NF 270, fouled membrane showed less negative charge than virgin one, and with increased TMP conducted, the negative charge reduced further. For membrane NTR-7450, the fouled membrane also experienced less negative charge than virgin membrane. The change of surface charge of two membranes converged to the similar foulant on membrane.

Capillary electrophoresis was tried to analyse the composition of polyphenols in tea sample. Three different capillaries and two different buffer solutions were tested, however, the results were not conclusive. GC-MS were used to analyse the extraction methods of polyphenols from black tea as well as the permeate and retentate. The graph of black tea feed extracted from 1.8 g tea in 300 ml boiling water had a clearer chromatography than the result from 3.6 g tea. When the other parameters were the same, the chromatography of tea feed with extraction temperature at 70° C showed a clearer peaks than the result obtained at 100° C. These results are consistent with the conclusion in chapter 4 about polyphenols extraction by UV-Vis spectrophotometric method. These results show little or no difference in the distribution of polyphenols species in the permeate and retentate after the NF process.

Chapter 6 CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

The following conclusions were drawn from the experimental results of this research.

Membrane selection

In literature review (chapter 2), it has been investigated that the molecular weights of main polyphenols in black tea below 300 Da (Table 2.3 and 2.4). Because the overall performance in salt rejections, water flux and TMP tolerance are better than ceramic and liquid membranes, traditional synthetic polymer membranes were used in this research. Considering of the membrane pore size in Table 2.6, nanofiltration membrane was chosen to perform the filtration research. Both NF 270 membrane and NTR-7450 membrane were tested and compared. The NF 270 membrane shows higher permeate flux and higher tolerance to chemical cleaning which makes it as a better choice than NTR-7450 membrane. In order to gain a better observation with less fouling, pre-treatment was also investigated and the ultrafiltration membranes of 1k Da and 5k Da were used.

Polyphenols extraction methods

In this part, varied parameters were performed and compared in order to obtain a optimum extraction method of black tea polyphenols. These parameters include brew temperature (70°C, 100°C), infusion time (5 minutes, 10 minutes), solvent constituent (30% ethanol and pure Milli-Q water), and tea concentration (1 g tea, 2 g tea).

In chapter 4, the polyphenols extracted at 100°C was approximately 20% higher than at 70°C, which indicates that the total polyphenols found in high extraction temperature. However, because one main component of tea polyphenols is theaflavins which is sensitive to temperature and it will be oxidized at a temperature over 90°C. In this case, 70°C was chosen as the brew temperature which can preserve the diversity of polyphenols obtained.

The total polyphenols amount analyzed by UV-spectrophotometer after 10 minutes was approximately 30% higher than the amount gained after 5 minutes. In high temperature, more polyphenols such as flavanoids and phenolic acids can be soluted, however, the oxidization can be much severer due to the long exposing at high temperature environment. Considering to keep the variety of tea polyphenols in feed, 5 minutes was chosen as the infusion time in order to decrease the oxidization of polyphenols as much as possible.

In this part of experiments, the total polyphenols extracted from different concentrations did not show obvious consistency. Even the absorption observed at high tea concentration (2 g tea infused) was about 6% lower than the data obtained at lower tea concentration (1 g tea infused). According to the GC-MS, at a lower concentration, the types of polyphenols detected are similar to high concentration, however, the peaks were more clearly defined. Take both the analytical results into account, 1 g tea infused in 300 ml water was chosen.

The polyphenols detected with 30% ethanol mixed in extraction solvent was about 30% higher than pure Milli-Q water, which indicates that polyphenols are more soluble in organic solvents. But, in consideration of safety issues, usage of 30% ethanol solvent can lead to chemical residual, which is harmful to human health. In this case, although the taking pure Milli-Q water as the extraction solvent cannot obtain maximum amount of tea polyphenols, it can supply a results free of chemical

residual interference.

After the extraction of polyphenols from Milli-Q water, vacuum filtration with ashless filter paper 42 (Whatman) was applied to the initial tea feed to remove the impurities and large particles in it.

The filter paper was covered by a layer of brown particles after vacuum filtration. Besides, the UV absorption of extracted polyphenols with vacuum filtration was approximately 8% (average in three repeat tests) higher than without vacuum filtration. This result presents that the removal of the impurities and large brown particles can improve the effectiveness of polyphenols detection in tea feed.

Ultra-filtration as pre-treatment

Both 1k Da and 5k Da UF membranes were used as pre-treatment on tea feed before nanofiltration. Due to the low flux of UF 1k Da membrane (9 L/m²h), which is too low comparing to the normal flux of nanofiltration (25-30 L/m²h), only UF 5k Da membrane was chosen as UF pre-treatment.

The pre-treatment of UF 5k Da membrane was conducted at a TMP of 3 bar. The UV absorptions of polyphenols with/without UF pre-treatment were observed after 2 hours filtration at 5 bar. But the UV absorbance showed no significant different after UF pretreatment, comparing with the feed with vacuum filtration only. In this case, vacuum filtration was the only protocol used as pretreatment before nanofiltration.

Nanofiltration tests

During nanofiltration, varied trans-membrane pressures (3, 5, 8 bar) were applied on nanofiltration membrane NF 270. The initial permeate flux were 17, 35, 53 L/m²h with ascending TMP, while the ending flux were 17, 25, 31 L/m²h respectively. This results shows that the flux change was consistent with the TMP during 3-8 bar.

UV absorption of feed, permeate and retentate were investigated during nanofiltration. According to the results of mass balance, the rejection rates of membrane NF 270 at 3 bar, 5 bar, and 8 bar were quite close, which were 90%, 86%, 86% separately (Table 4.2). The data here presents that during 3-8 bar, the membrane NF 270 remains a good performance of high rejection rate.

With NF 270 membrane, the effect of retention time was also investigated. During this part, permeate flux with 2 hours' and 4 hours' filtration were measured and the result of 4 hours' filtration was quite close to the result of 2 hours' without obvious decline. It implies that the fouling situation did not accumulate significantly in the first 4 hours' filtration.

The UV absorbance in these two situations were also tested, the data with 4 hours' filtration obtained a rejection of 79%, which was approximately 10% lower than the data of 2 hours' filtration. The decrease in rejection may due to the concentration polarization at longer filtration time is more severe. In the filtration performance, the effects of fouling are reflected in the change of flux decline and flux recovery after cleaning.

With NF 270 membrane, the effect of stirring rate was studied as well. Stirring rates of 250, 450, 650 rpm were investigated in this research. The permeate flux at 450 rpm was approximately 17% higher than 250 rpm in the beginning and gradually closed to the result of 250 rpm. At the end of the 2 hours' filtration, both flux were around 25 L/m²h. In nanofiltration, NTR-7450 membrane was performed and compared with NF 270. The permeate flux with NTR-7450 was approximately 25% lower than NF 270.

Membrane cleaning

For cleaning protocol, 0.01 M NaOH solution was used as cleaning solution for 30

min with stirring. The cleaned membrane NF 270 was then recycled in a new cycle of tea feed filtration. This cycle repeated five times. During this phase, the virgin membrane flux got a 50% increase after cleaning. After filtration, each time the flux could recover nearly 100% (distributing from 94.4% to 109%) with 30 min cleaning. In the meantime, polyphenols rejection was also measured by UV-Vis spectrophotometer. In those 5 cleaning cycles, the rejection rates were quite high and close, which were 88%, 95%, 89%, 94%, 87% respectively. Just the error of the 5th cycle was approximately 17%, while the others were controlled under 10%.

Analysis

In analysis part, capillary electrophoresis was tried with three different columns and 2 different buffer solutions, however, the peaks read for the tea samples cannot stay stable.

GC-MS was used to separate the components in tea feed, permeate and retentate, which was proved to distinct most of the constituents clear and stable. With GC-MS, the former research about extraction methods were also investigated by detecting the detailed components and amount in tea solution extracted.

Zeta Potential was used to measure the surface charge of different membranes. Fouled NF 270 membrane has less negative surface charge and the more the fouling is, less surface charge it is. The situation of NTR-7450 membrane is similar, and the fouled membrane has less negative charges comparing to virgin one.

6.2 Recommendations

A comprehensive study in polyphenols extraction on more parameters is recommended. More extraction temperatures could be tested to evaluate the relationship between temperatures and total polyphenols extracted. In addition,

more infusion time choices and different percentages of organic solvent can also be investigated to discover whether these parameters are linear to polyphenols extracted or not.

In addition to RC and PES membranes, more UF membranes with different molecular weight cut off and materials can be studied for the pre-treatment of original black tea feed. Both permeate flux and persistency to cleaning could be compared and evaluated to determine the best choice to conduct the pre-treatment to black tea feed to reduce fouling during nanofiltration.

Nanofiltration period in this study was limited to 2 hours and 4 hours. Long filtration should be done (as the filtration time in industry is more than 4 hours) with cross-flow rig, to study the flux change and fouling mechanism in a cross-flow system which will experience less cake accumulation and concentration polarization in filtration process.

Wider range of trans-membrane pressures can be applied in the nanofiltration process, and more NF membranes with different materials and MWCO can be applied to search for a best determination. Fouling performance is likely to decrease with smoother and more hydrophilic membranes.

Compositions of permeate in NF process at different time can be analyzed by GC/MS or other analytic methods to track the species penetrating from feed with time. Based on the permeate compositions and concentration of polyphenols and other compounds, fouling mechanism in this NF process can be predicted.

Acid chemical cleanser can be applied on the fouled NF membranes to study the flux recovery and membrane persistence in cleaning cycles, comparing to the alkaline chemical cleanser used in this research.

Due to the failure of analysis by capillary electrophoresis, further study can explore analytic methods, such as HPLC or GC, to analyze the composition of tea samples. The contact angle of membranes can also be studied before and after fouling to investigate the hydrophobicity change which may contribute to explain the fouling mechanism and membrane property.

Instead of stirred-cell with flat sheet, it will be interesting to run the experiment in other types of membranes, such as hollow fiber, spiral wound and tubular due to variation in mass transfer regimes.

Study on single polyphenolic compound will be interesting, as comparison to the extraction, concentration and cleaning of the polyphenols mixture. The rate of original single polyphenols can be a standard rate of extraction, concentration and cleaning, so that the process of each polyphenols in the mixture can be well explained by comparing with each standard rate.

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